

Taxing Production Networks: Trade, Misallocation, and Vertical Integration

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Abstract

Taxes on business revenue fall on every intermediate transaction, like tariffs inside the production network. I measure their cost using a large reform that replaced a turnover tax with a value-added tax in China's services. Entry and revenue in treated sectors rise by a fifth, trade involving services by a third, and manufacturers move service tasks to the market. A spatial production-network model with an endogenous make-or-buy margin implies a 1.3 percent gain in real income, 29 percent through that margin. At equal revenue, a value-added tax delivers 1.4 to 2 percent more real income than a turnover tax.

Keywords: Macroeconomic Policy; Misallocation; Firm Growth; Trade; Vertical Integration; Spatial General Equilibrium.

JEL Classification: C23, D22, E60, R12, L60, O25.

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1 Introduction

The past two centuries have witnessed the increasing liberalization in trade of goods and services across locations (Donaldson, 2015). Nonetheless, significant barriers and frictions to trade across locations and sectors remain between and within countries. The literature has documented the impacts of trade barriers, including tariff and non-tariff barriers. This paper applies insights from this literature to within-country markets, examining how a business tax on intermediate inputs distorts sector–region transactions. The business tax functions as a within-country tariff, where firms pay taxes on revenue (therefore intermediate inputs). Not only does this tax hinder trade between sectors and regions by raising input costs, it incentivizes firms to engage in excessive (inefficient) vertical integration in order to evade taxes on intermediate goods. In turn, this further reduces cross-sector trade, particularly depressing demand from upstream suppliers (Coase, 1937; Williamson, 1971). Despite the distortions and inefficiencies created by this tax system, it is still used or has been used in many countries, developing and developed¹ (Best et al., 2015; Hansen et al., 2022; Phillips & Ibaid, 2019).

This paper studies how taxes on intermediate inputs shape production networks and contributes to the literature by being the first paper (to my knowledge) to integrate the endogenous choice of vertical integration into a quantitative spatial general equilibrium model. Embedding the intermediate input tax into a production network captures the distortions of the business tax and incorporating the endogenous choice of vertical integration matters for quantitative exercise. To see why this modeling choice is essential, it is useful to see how the business tax operates and why its cascading nature distorts production networks. When taxes are levied on revenue, they cascade through production networks: every intermediate sale embeds taxes paid at earlier stages. As a result, market-sourced goods appear artificially expensive, nudging firms toward the inefficient vertical integration. This mechanism resembles a tariff but is more pervasive because it operates within the domestic supply chain, compounding across sectors and regions more frequently. In complex production networks, cascading taxes distort trade flows and firm organization:

¹For example, in the United States in fiscal year 2017, \$157 billion of taxes were imposed on business-to-business sales of products, services and equipment, which comprised 42% of total state and local sales tax revenue (Phillips & Ibaid, 2019). Examples from developing countries include Argentina’s provincial gross receipts tax (*Ingresos Brutos*), Brazil’s cumulative federal contributions on revenue (PIS/COFINS) and municipal services tax, India’s regime before the 2017 goods and services tax, the Philippines’ percentage tax on the gross sales of firms outside the value-added tax, Pakistan’s minimum tax on turnover (Best et al., 2015), Russia’s turnover taxes for roads and housing, abolished by 2003, and Vietnam’s turnover tax, replaced by a value-added tax in 1999. Taxing intermediate transactions is a common feature of tax systems, not a Chinese peculiarity.

a firm facing intermediates with multiple layers of embedded tax may resort to cheaper but lower-quality inputs or integrate production in-house to avoid the tax altogether. Neither choice is efficient: the first sacrifices access to better-priced inputs, while the second diverts firms into activities where they could lack a comparative advantage.

A clean measurement of these distortions requires a setting in which a turnover tax is removed on a large scale while the rest of the tax system stays in place. China's replacement of its business tax with a value-added tax provides one. Before the reform, service firms paid the business tax on their total revenue, intermediate inputs included, while manufacturing firms paid a value-added tax on their value added. Any transaction with a service firm at either end therefore carried a tax that the buyer could not credit, so links through service sectors were more expensive than links among manufacturing firms. Between 2012 and 2016 the reform moved every service sector to the value-added tax, in a staggered sequence across sectors and provinces. It removed the tax on intermediate transactions for a third of the economy, and it did so by changing how the tax is levied, which is what allows the turnover tax and the value-added tax to be compared on the same production network.

The reform began in 2012 in nine pilot provinces for seven service sectors, was extended to the whole country a year later, and reached the remaining service sectors by 2016. Using this staggered timeline, I document how the production network responds when the tax on intermediate transactions is removed. The number of registered firms in reformed service sectors rises by about 21 percent relative to manufacturing sectors in the same province and year, and the revenue of service firms in the tax survey, a repeated cross-section that includes the small and newly entered firms that make up most of the sector, rises by about 22 percent. Manufacturers in the pilot provinces, whose own tax regime did not change, raise their purchases of the reformed services from the market relative to manufacturers elsewhere, and the administrative expense that houses their internal service departments falls, though the latter is less precisely estimated. At the region-sector level, gravity regressions on interregional input-output tables before and after the reform show that intermediate trade flows with a service sector at either end rise by 36 percent relative to flows among manufacturing sectors.

Despite these positive outcomes, reduced-form analysis alone is insufficient to evaluate the welfare implications of the reform. The policy's primary objective – lowering the tax burden on service firms – also led to unintended consequences. Local governments experience tighter budget constraints due to declining tax revenues, potentially affecting public service provision. Additionally, sectoral inter-linkages imply that shocks in one sector can propagate throughout the economy, complicating efforts to isolate the reform's

effects. A further challenge is that one of the key pre-reform distortions, excessive vertical integration, leaves no trace in input-output tables, which record purchases between firms and not the tasks a firm performs for itself.

To quantify the general-equilibrium effects, I develop a spatial model of the production network with 31 provinces, the rest of the world, and 41 sectors. Producers are perfectly competitive, trade intermediate and final goods across regions and sectors, and in every sector choose between buying service inputs from the market and producing them with their own labor. The tax system is modeled transaction by transaction. A noncreditable tax enters the buyer's input price and cascades down the chain; a creditable tax is remitted by the seller and refunded to the buyer along the invoice chain, so that the cascading wedge and the statutory remittance are distinct objects. Revenue is collected where the tax is paid and rebated to the households of that province. This is the origin principle that governs the provincial share of China's value-added tax; the model applies it to all revenue. Removing the business tax raises welfare by lowering the effective cost of market inputs and by moving service tasks out of in-house departments. The gains are uneven across space, though not in the way the policy debate expected. Wages rise in every province, yet a province can lose in welfare when its service sectors buy their inputs from the rest of the country, because under origin-based assignment its budget refunds the input credits on goods whose tax other provinces collected.

This paper contributes to the literature on trade, taxation, and firm boundaries by embedding the make-or-buy decision of Bernard et al. (2019) in a quantitative multi-region, multi-sector model of the production network in the tradition of Caliendo et al. (2022a) and using it to evaluate a tax reform. Each sector's intermediate bundle is a constant-elasticity aggregate over origins and sectors in which producing service tasks with the firm's own labor is one more option, so the tax-induced relative price of market inputs determines how much of the bundle is made in-house, and the margin is open in every sector, services included. The model is cast under perfect competition and constant returns and solved in exact hat algebra. An earlier variant added endogenous entry with heterogeneous firms; I set it aside because the entry feedback in a roundabout input-output network is unstable at standard elasticities (Section 6), and I use the entry evidence as motivation rather than as a calibration target.

Two parameters govern the quantitative results. The first is the trade elasticity, which under perfect competition is also the elasticity of bilateral flows with respect to the tax wedge. I set it to 4.5, the midpoint of the range of 3 to 6 in the international trade literature (Caliendo & Parro, 2015; Head & Mayer, 2014; Simonovska & Waugh, 2014), and do not estimate it from the reform window: the 2012–2017 change in input-output flows identifies

the wedge response only jointly with the secular deepening of service inputs, which loads on the same sector pairs. Because the model is solved in exact hat algebra, iceberg trade costs, productivities, and the productivity of in-house tasks are absorbed by the observed 2012 allocation and never need to be estimated.

The second parameter is the share of service tasks that firms produced in-house before the reform, which input-output tables do not record. I measure it directly from the 2011 wave of the tax survey. In-house service production has an accounting location: administrative expense, the income-statement line that houses a firm's accounting, human-resources, legal, information-technology, and planning departments, net of the purchased professional fees booked there. Measured for every sector, from 279,000 manufacturing and 340,000 non-manufacturing firms, in-house service tasks are 5.6 percent of gross cost in the aggregate and absorb 22 percent of core value added. The share is highest in the service industries themselves and in knowledge-intensive manufacturing, where the reform bites hardest, and the same expense line falls after the reform while purchased services rise, in the direction and timing the mechanism requires.

Four results follow. First, removing the turnover tax raises welfare, and the make-or-buy margin is a large part of the reason. The reform raises aggregate real income by 1.30 percent; real wages rise by 3.3 percent and real tax revenue falls by 10.7 percent, so households gain more from higher wages and lower prices than they lose in rebated revenue. In the baseline, in-house tasks are 13 percent of the input bundle in the reformed sectors and 6 percent in the sectors that were always under the value-added tax. Under the reform this share falls by 6.8 percentage points more in the reformed sectors than in the others. A model without the margin delivers a gain of 0.92 percent, so ignoring vertical integration would miss 29 percent of the gain.

Second, a turnover tax is an expensive way to raise revenue. The reform gave up revenue as well as changing the structure of the tax, so I compare it with a cut in business-tax rates that gives up the same revenue. That cut alone raises real income by 0.98 percent: each yuan of business-tax revenue forgone returns about 1.6 yuan to households, because the tax compounds along the chain. The reform delivers a third more, 1.30 percent, for the same revenue. And once purchases are creditable, revenue is cheap to raise: scaling the new rates on services up until real revenue is back at its 2012 level lowers the gain only to 1.1 percent, a cost of about 1.1 yuan per yuan raised. What a tax rate costs depends on the credit chain beneath it. The same increase in service rates lowers welfare by 1.0 percent when buyers cannot credit it and raises welfare when they can. Completing the credit chain in the sectors the legislation left out raises the gain to 1.54 percent at a cost of one

point of revenue, and a uniform 17 percent value-added tax on every sector would raise welfare by 3.9 percent and revenue by 25 percent.

Third, I step outside the reform and compare a uniform value-added tax with a uniform turnover tax at matched fiscal capacity. That the value-added tax dominates in a closed economy is a theorem; the model puts a magnitude on it in a calibrated economy with trade. The crediting architecture is worth 1.4 percent of real income at the revenue China raises after the reform, slightly more than the reform itself delivered, and 2 percent at the revenue it raised in 2012; the value grows more than proportionally with the revenue raised, and trade with the rest of the world does not reverse it.

Fourth, a value-added tax collected at origin redistributes across regions when the credit chain is extended. The seller's region keeps the output tax, and the buyer's region refunds the credit. Extending credits to service sectors, which are concentrated in a few provinces and buy goods from everywhere, moves 17.3 percent of the national tax base between provincial budgets. Welfare rises in 29 of 31 provinces. The two that lose are service-intensive provinces whose service sectors buy their inputs from the rest of the country: Beijing pays out, on net, 31 percent of its baseline revenue through the credit chain. Specialization in services is not enough to produce a loss; Shanghai, whose service sectors buy far less from other provinces, pays out 4 percent on net and gains. Sharing the national change in revenue in proportion to baseline collections turns both losses into gains and halves the dispersion of provincial welfare changes, leaving aggregate real income unchanged.

The results bear on the design of taxes wherever business inputs are taxed, which includes gross receipts taxes, sales taxes that reach business purchases, and value-added taxes with broken credit chains. They say that the cost of such taxes rises more than proportionally with the revenue they raise, that part of the cost takes the form of firms producing for themselves what they would otherwise buy, and that extending a credit chain shifts revenue between regions when the tax is assigned at origin. The framework, which makes the boundary of the firm an equilibrium outcome in a quantitative spatial model, can be applied to other changes in the relative price of market inputs.

The remainder of the paper is organized as follows. Section 2 discusses the related literature. Section 3 documents the policy background, Section 4 works through crediting, remittance, and cascading in a two-firm example, and Section 5 presents the empirical evidence. Sections 6 to 8 set out the model, its calibration, and the counterfactuals. Section 9 concludes.

2 Relation to the Literature

The model builds on quantitative multi-region, multi-sector trade models with input–output linkages (Caliendo & Parro, 2015; Caliendo et al., 2018, 2019; Eaton & Kortum, 2002; Redding & Rossi-Hansberg, 2017) and on models in which tariffs or distortions act on intermediate inputs and shape entry (Baqae & Farhi, 2024; Caliendo et al., 2020, 2022b; Kucheryavyy et al., 2023), and it uses the exact-hat-algebra approach of Caliendo and Parro (2015) to hold unobserved fundamentals fixed. The make-or-buy margin follows Bernard et al. (2019), who allow firms to produce an input in house or buy it from the market, and, as in G. M. Grossman and Helpman (2002), one producer’s organizational choice depends on the industry environment and on the choices of other producers, since the price of market-sourced services is itself an equilibrium object.² Quantitative general-equilibrium models with an endogenous integration margin exist for multinational sourcing (Garetto, 2013) and for domestic outsourcing of labor services (Bilal & Lhuillier, 2025); to my knowledge this is the first to embed the margin in a quantitative spatial trade model and to use it to evaluate a tax reform.

The paper is also closely related to the literature on misallocation with intermediate inputs. Several papers extend Hsieh and Klenow (2009) and Restuccia and Rogerson (2008) to settings in which distortions act on the use of intermediates, e.g., Jones (2011, 2013), Bartelme and Gorodnichenko (2015), Fadinger et al. (2022), Bigio and La’O (2020), E. Liu (2019), Caliendo et al. (2022b), Osotimehin and Popov (2023), Hang et al. (2020), and Baqae & Farhi (2020).³ These papers typically treat the wedge as an unobserved residual inferred from factor or expenditure shares. Closest in mechanism is Boehm and Oberfield (2020), who show that weak contract enforcement pushes Indian plants away from purchased inputs toward in-house production and quantify the resulting productivity loss. This paper differs along three dimensions. First, the wedge is a statutory tax rate on every intermediate transaction, observed directly and removed by a staggered reform, so its effects on entry, revenue, and trade are identified from policy variation rather than inferred from cross-sectional dispersion. Second, the wedge moves the make-or-buy choice of downstream firms, and I measure that margin directly in firm data and embed it in the model; the entry of upstream service providers, which the reform also raised, is

²Endogenous input choice also appears in models of production-network formation (Acemoglu & Azar, 2020; Arkolakis et al., 2026; Dhyne et al., 2023; Oberfield, 2018; Panigrahi, 2021), which let firms choose suppliers but not whether to internalize production.

³See V. M. Carvalho and Tahbaz-Salehi (2019) and Baqae & Farhi (2019) for the propagation of shocks through networks, and Gabaix (2011), Foerster et al. (2011), Acemoglu et al. (2012), V. Carvalho and Gabaix (2013), and di Giovanni et al. (2014, 2018) on granular origins of aggregate fluctuations.

documented as evidence of the same shift in demand but is not part of the quantitative model. Third, the wedge is spatial as well as sectoral, so the counterfactual delivers the regional incidence of removing it.

The paper connects this network view to the public-finance literature on cascading taxes. The case against taxing intermediate transactions goes back to Diamond and Mirrlees (1971); Keen (2014) derives the welfare cost of cascading, and Ebrill et al. (2001), Keen and Lockwood (2010), Brockmeyer et al. (2024), and de Mooij et al. (2025) document why the value-added tax has displaced turnover taxes in most countries (Smart and Bird, 2009 measure the investment response to one such replacement in Canada), though gross-receipts taxes persist in several US states (Hansen et al., 2022; Testa & Mattoon, 2007) and in developing economies with limited enforcement capacity (Best et al., 2015). Empirical work on VAT adoption and VAT chains studies compliance and the segmentation of supplier networks across tax regimes (Agrawal & Zimmermann, 2024; Bachas et al., 2019; de Paula & Scheinkman, 2010; Gadenne et al., 2019; Gerard et al., 2018; Hoseini & Briand, 2020; Naritomi, 2019; Pomeranz, 2015). Two papers share the question. Caprettini and Ciccone (2015) study Brazil's conversion of a turnover tax to a VAT, document a fall in vertical integration, and show in a sequential-production model that the cost of cascading rises with the number of stages; Delalibera et al. (2024) remove cumulative taxation in a multi-sector input-output model of Brazil and find large output gains. Neither has a spatial dimension or a make-or-buy margin, and neither measures that margin in the firm data of the reform it studies. For the Chinese reform itself, Xing et al. (2024) show that removing the turnover tax raised the sales, employment, and research spending of treated service firms, and attribute the response to increased outsourcing by their manufacturing customers and not to changes in tax burdens or prices; a set of papers documents higher specialization and productivity among treated firms and their manufacturing customers (Chen & Wang, 2016; Dai & Zhao, 2021; Z. Fan & Peng, 2017; C. Liu et al., 2026; Peng et al., 2021; Yu & Qi, 2022).⁴ Their reduced-form finding is that the in-house margin of downstream firms is where the reform acts. Relative to this work, the contribution is to measure that margin directly, to embed it in general equilibrium, and to quantify what it and the credit chain are worth: how the reform compares with a cut in the turnover tax that gives up the same revenue, and how a value-added tax compares with a turnover tax at equal revenue. Fajgelbaum et al. (2019) ask a related question for the dispersion of state tax rates in the United States; here the distortion is the base rather than the rate.

⁴The other large Chinese VAT quasi-experiment, the 2009 extension of input credits to capital goods, is studied by Y. Liu and Mao (2019) and Chen et al. (2023). Ghosh et al. (2026) find that India's state-level switch from sales tax to VAT reduced vertical integration among listed firms.

Finally, the paper relates to the literatures on internal trade costs in China and on firm boundaries. Interprovincial barriers in China have been measured and debated since Young (2000) and Holz (2009), with evidence on local protectionism and its fiscal roots in Poncet (2005), Bai et al. (2004), Barwick et al. (2021), E. Liu et al. (2026), and Han and Wu (2024), and quantitative assessments of internal trade and migration costs in Tombe and Zhu (2019), J. Fan (2019), Ma and Tang (2020), and Che et al. (2024); provincial border effects remain large (Jiang et al., 2025), and services are costlier to trade across space than goods (Gervais & Jensen, 2019). These papers treat internal frictions as transport, institutional, or protectionist costs; this paper identifies a friction that is written into the tax code and that falls on cross-sector as well as cross-province transactions, and measures its effect on trade and on firm boundaries from the reform.⁵ On firm boundaries, the transaction-cost and property-rights theories (Antràs, 2003; Antràs & Helpman, 2004; Coase, 1937; S. J. Grossman & Hart, 1986; Hart & Moore, 1990; Lafontaine & Slade, 2007; Legros & Newman, 2013; Williamson, 1971) predict that anything raising the relative price of market inputs pushes production in house, and Alfaro et al. (2016) and Q. Liu et al. (2019) confirm this for tariff-induced price changes, including in China; Atalay et al. (2014), Hortaçsu and Syverson (2007), and Berlingieri (2014) show what integrated firms actually do and how much of service-sector growth reflects outsourcing by manufacturers. By embedding a tax-induced integration margin in a general-equilibrium model, I quantify how much of the welfare gain from the reform a model without the make-or-buy margin would miss.

3 Policy Background

3.1 Dual-tax System in China

The manufacturing sector in China has long been the traditional economic driver. However, in recent years, the service sector has emerged as an increasingly critical component of the economy, playing a pivotal role in innovation-driven growth and contributing significantly to GDP and tax revenue (Zilibotti, 2017). Despite its growing importance, the service sector has historically been subject to a different and more burdensome tax treatment compared to manufacturing firms, facing heavy taxation on intermediate inputs. China's dual tax system created significant economic distortions. Under this system, firms in the service sector were required to pay business taxes on their total revenue — including intermediate

⁵The role of domestic frictions in quantitative trade models is emphasized by Ramondo et al. (2016), Coşar and Fajgelbaum (2016), and Agnosteva et al. (2019); on market access and infrastructure see Donaldson (2015), Donaldson and Hornbeck (2016), and Faber (2014).

Table 1: Timing and Industries of B2V Reform

Reformed industries	Regions	Implementation date
Transportation,	Shanghai	2012.01.01
R&D and technical services, IT services,	Beijing	2012.09.01
cultural and innovation services,	Jiangsu	2012.10.01
logistics auxiliary services,	Anhui	2012.10.01
attestation and consulting services,	Fujian	2012.11.01
and tangible assets leasing services	Guangdong	2012.11.01
	Hubei	2012.12.01
	Tianjin	2012.12.01
	Zhejiang	2012.12.01
	Nationwide	2013.08.01
Postal service, rail transportation	Nationwide	2014.01.01
Telecommunication	Nationwide	2014.06.01
Real estate, construction, finance, and other services	Nationwide	2016.05.01

Notes: This table outlines the waves of the B2V reform across different industries, regions and time.

inputs — while manufacturing firms operated under the value-added tax (VAT) system that only taxed value added.

The distortions operated through multiple channels. First, service firms faced heavy taxation because business taxes applied to their gross revenue rather than just value added. The revenue at the same time included the after-tax intermediate input. This is just one layer inside the production network showing the “taxes on taxes”. Second, when manufacturing firms purchased intermediate inputs from service providers, they bore the multiple layers of taxation incorporated in the prices of the service goods. The same logic applied to service-to-service transactions as well. This tax cascading made cross-sector trade artificially expensive and incentivized manufacturing firms to engage in excessive vertical integration.

This tax system discouraged the development of efficient production networks by making it costly to source inputs from the most productive suppliers across sectors. It particularly hampered the service firm’s growth by increasing operational costs and reducing demand from downstream clients. The resulting misallocation of resources reduced overall economic efficiency. Recognizing these problems, policymakers sought to alleviate the tax burdens on service firms and harmonize the tax structure.

3.2 Business-to-Value Added (“B2V”) Tax Reform

To promote the growth of service firms, the government started to implement the “Business to Value-added Tax” reform by transitioning the service sector to a VAT system. Starting in 2012, this reform was carried out progressively across sectors and provinces, and the timeline is displayed in Table 1. It replaced the business tax with a value-added tax for service sector firms (see Table 2) and, at the same time, made the tax on most intermediate purchases creditable for every sector; a few items, among them catering, loan interest, resident services, entertainment, and, until 2019, passenger transport, remain noncreditable (Appendix C.2; the model does not carve out passenger transport, which the input-output table does not separate from freight). By extending the VAT system to service firms, the reform let these firms deduct the tax paid on their inputs, so that they are taxed only on their value added.

To see how the reform impacted firms in the service sector, I take the Railway Transportation sector in Table 2 for instance. Before the “B2V” reform, Railway Transportation sector would pay 3% on the total revenue of its output (including both intermediate inputs and value added). After the reform, it pays 11% only on the total value added out of its output. The numeric tax rate before and after the reform is not necessarily the same for all sectors. In fact, as seen in Table 2, the statutory rate rises in every reformed sector except entertainment. Therefore, merely seeing from the perspective of tax payment, whether a sector is subject to lower tax burden also depends on its value added margin.

Four features of the pre-reform system matter for the measurement and are taken up in Section 6.7. First, buyers under the value-added tax could deduct a deemed 7 percent of the freight they paid to carriers under the business tax, so the cascade on freight was partly relieved before the reform. Second, Shanghai’s pilot started on 1 January 2012 and left railways, post, and telecommunications outside, so the 2012 data record a system that was already partly reformed in one province. Third, exported services under the pilot were zero-rated only in part; most were exempt, which removes the output tax and denies the credit on inputs (Appendix C.3). Fourth, the value-added tax is shared between the central and provincial governments, and the provincial share is assigned to the province where the tax is paid; the business tax accrued almost entirely to the provinces.

How the switch changes what a service firm remits depends on its rate, its exports, and how much of its input bill becomes creditable. Figure 1 reports a mechanical prediction for each of the 341 reformed service region–sector pairs in the 2012 interregional table, holding quantities and prices fixed. Before the reform the pair remits its business-tax rate times gross output. After it, the pair remits the value-added tax on its domestic sales minus the

Table 2: Change of Tax Scheme and Rate

Industry Name	Industry Code	BT Rate	VAT Rate
Railway transportation	G53	3%	11%
Road transportation	G54	3%	11%
Water transportation	G55	3%	11%
Air transportation	G56	3%	11%
Handling and transportation agency	G58	3% (handling) or 5% (agency)	6%
Warehousing	G59	5%	6%
Postal services	G60	3%	11%
Telecommunications, broadcast and satellite transmission services	I63	3%	11% (basic) or 6% (value-added)
Internet services	I64	5%	6%
Software and information technology services	I65	5%	6%
Leasing	L71	5%	11% or 17%
Business services	L72	5%	6%
Research and experimental development	M73	5%	6%
Professional technical services	M74	5%	6%
News and publication	R85	3%	6%
Radio, television, film and recording production	R86	3%	6%
Culture and art; sports	R87, R88	3%	6%
Entertainment	R89	5–20%	6%
Accommodation and catering	H61, H62	5%	6%
Finance and insurance	J66–J69	5%	6%
Resident services	O79	5%	6%
Construction	E47	3%	11%
Real Estate	K70	5%	11%

Notes: Statutory business-tax rate before the reform and statutory value-added-tax rate after it, by industry (GB/T 4754-2011 codes). The business tax was levied on gross revenue, intermediate inputs included; the value-added tax allows input tax to be credited, so a higher statutory rate does not imply a higher burden.

tax on its creditable domestic purchases, at the statutory rates and with the carve-outs as legislated. The predicted payment falls by more than 10% in 77% of pairs and rises by more than 10% in 16%; in the aggregate it falls by 47%, and 11% of pairs move to a net refund position. The pairs whose payment rises are concentrated in real estate, where the rate goes from 5 to 11 percent and purchased inputs are a small share of output, and in transport. The majority of service sectors therefore remit less under the new system at unchanged behavior, and a minority remit more.

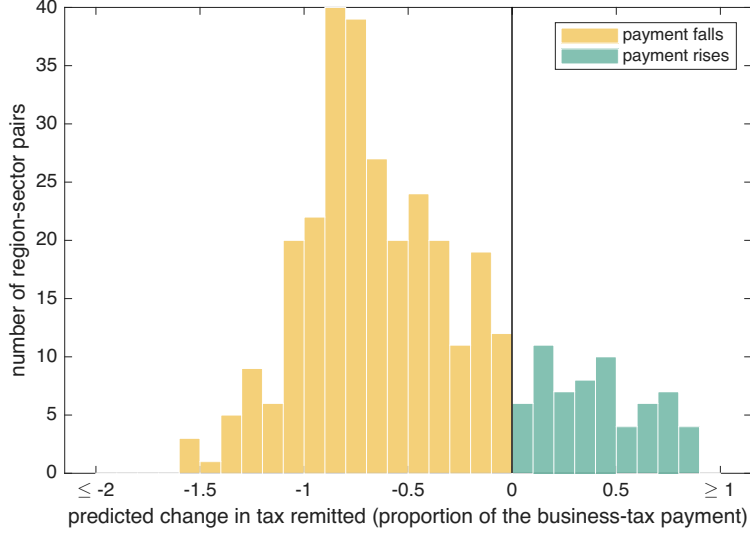


Figure 1: Distribution of Predicted Tax Payment Change

Notes: Each observation is one of the 341 region–sector pairs in the eleven reformed service sectors of the 2012 interregional input-output table. The horizontal axis is the predicted change in the tax the pair remits, as a proportion of its business-tax payment: $(T_1 - T_0)/T_0$, with $T_0 = \tau^{BT}Y$ the statutory business-tax rate times gross output and $T_1 = v(Y - E) - \sum_k \delta_k v_k X_k$ the value-added tax on domestic sales (E are exports, zero-rated) minus the tax on creditable domestic purchases X_k ; $v = t/(1 + t)$ is the tax share of an invoice at the statutory rate t , and δ_k is the creditable share of purchases from sector k under the 2016 rules. Quantities and prices are held at their 2012 values. A change below -1 is a net refund position.

4 A Numerical Illustration: Crediting, Remittance, and Cascading

A two-firm example shows how the tax works, and in particular separates *where* a tax is remitted from *who* bears it. An upstream firm contributes \$50 of value added and sells to a downstream firm that contributes a further \$100. I work through four cases: full crediting, excess credits, and the two ways in which a business tax breaks the credit chain. Rates are tax-exclusive throughout. The model of Section 6 carries the same objects: a tax that the buyer can credit does not enter the buyer’s input price, and a tax that the buyer cannot credit does.

Case 1: Full crediting (both firms under the value-added tax). Let the upstream statutory rate be 13% and the downstream rate 17%. The upstream firm invoices the tax-inclusive amount $1.13 \times 50 = 56.5$ and remits

$$T_1 = 50 \times 0.13 = 6.5$$

to its own tax authority. The downstream firm credits the \$6.5 invoiced upstream, so the input costs it 50 net of the recoverable tax, and it remits

$$T_2 = \underbrace{(100 + 50)}_{\text{cumulative value added}} \times 0.17 - \underbrace{50 \times 0.13}_{\text{credit for tax paid upstream}} = 25.5 - 6.5 = 19.$$

Total tax collected along the chain is $T_1 + T_2 = 25.5 = 150 \times 0.17$: the *downstream* rate applied to the cumulative value added, whatever the upstream rate. The tax remitted upstream is rebated through the buyer's credit, so it never enters the buyer's cost. The tax *collected* is positive even though no tax is embedded in any input price: the wedge on a transaction and the remittance of a firm are distinct objects, a distinction made precise in Section 6.5.

Remittance versus incidence. The upstream rate still decides *where* revenue is collected. Raise it from 13% to 17% with the downstream rate at 17%: the upstream firm now remits $T_1 = 50 \times 0.17 = 8.5$ and the downstream firm remits $T_2 = 25.5 - 8.5 = 17$; the total is unchanged at 25.5. A change in the upstream rate is a *pure transfer* between the two tax authorities, equal to the change in the rate times the upstream sale, here $0.04 \times 50 = 2$. The downstream firm bears nothing in either configuration: its larger remittance is offset by a smaller tax-inclusive payment to its supplier (56.5 rather than 58.5; input payment plus remittance is 75.5 in both), and the final consumer pays the downstream rate on the cumulative value added regardless. The split governs which *government* books the revenue. It matters when the two firms are in different regions and revenue is assigned at origin, to the region where the tax is remitted. This is the rule that governs the provincial share of China's value-added tax, and the model applies it to all revenue (6.27). Under it a creditable sale across regions moves revenue between them, since the seller's region collects the output tax and the buyer's region refunds the credit, even though the two entries net to zero nationally. A destination rule would assign the whole chain's tax to the region of final consumption.

Case 2: Excess credits and refunds. When a firm's output tax falls below the credit on its inputs, its remittance is negative. Here that requires a downstream rate below 5.67% when the upstream rate is 17%. Suppose the upstream rate is 17% (so $T_1 = 8.5$) and the downstream rate is 5%. The downstream firm owes $150 \times 0.05 = 7.5$ and credits \$8.5, so

$$T_2 = 150 \times 0.05 - 50 \times 0.17 = 7.5 - 8.5 = -1 :$$

the downstream tax authority *refunds* \$1. Total collections are again $T_1 + T_2 = 7.5 = 150 \times 0.05$. The refund is what enforces taxation of the chain at the downstream rate. Exporters under zero-rating are in this position, with a downstream rate of zero. I assume throughout that

refunds are paid in full. This is the clean credit-invoice benchmark, and it is more generous than Chinese practice before 2019, when excess credits of domestic sellers were carried forward and not refunded (Section 6.7). If refunds are rationed, the excess credit is a cost to the firm and cascading re-emerges, as in Case 3.

Case 3: A business-tax seller and a value-added-tax buyer. Now the upstream firm is a service producer under the pre-reform business tax, and the buyer cannot credit the tax. To keep the comparison with Case 1 exact I keep the upstream rate at 13%; the business tax itself was 3 to 5 percent of the tax-inclusive price, and the last sentence of this case gives the numbers at 5 percent. The invoice is unchanged at $1.13 \times 50 = 56.5$ and the upstream firm still remits $T_1 = 6.5$. But the whole \$56.5 is now a cost to the downstream firm and enters its tax base:

$$T_2^{BT} = (100 + 56.5) \times 0.17 = 26.605.$$

Total tax collected rises to $T_1 + T_2^{BT} = 33.105$, which is \$7.605 above Case 1. The excess equals $T_1 \times 1.17 = 7.605$: the \$6.5 of upstream tax is lost as a credit and is then taxed again at the downstream rate (a further \$1.105). This is tax on tax, or *cascading*. Unlike the transfer in Case 1 it raises the *total* tax collected, and because it sits in the buyer's input price it raises the buyer's cost and every price downstream of it; with competitive firms it is paid by final consumers and by factors, not out of profits. The wedge compounds with the number of noncreditable links in the chain. It also bears on what the buyer does for itself. A buyer that produces the input with its own labor pays no tax on it, so the market supplier must be more than 15 percent cheaper ($7.605/50$) than in-house production to win the business. This is the make-or-buy wedge of the model. At a business tax of 5 percent of the tax-inclusive price (5.26 percent tax-exclusive), the same steps give $T_1 = 2.63$, $T_2^{BT} = 25.95$, and an excess of 3.08, or 6 percent of the input's value.

Case 4: A value-added-tax seller and a business-tax buyer. The chain also breaks when the buyer is the one under the business tax. Let the upstream firm be a goods producer under the value-added tax at 17%, remitting $T_1 = 8.5$ on an invoice of 58.5, and let the downstream firm be a service producer under a business tax of 5 percent of its tax-inclusive revenue. It has no value-added-tax liability to credit against, so the \$8.5 is a cost. Its costs are $58.5 + 100 = 158.5$, its price is $158.5/0.95 = 166.84$, and it remits

$$T_2 = 0.05 \times 166.84 = 8.34,$$

for a total of 16.84. The reform moves the service firm to the value-added tax at, say, 6%. It now credits the \$8.5 and remits $T_2 = 150 \times 0.06 - 8.5 = 0.5$, and the total falls to $9.0 = 150 \times 0.06$. The upstream authority collects 8.5 before and after. The whole fall in

revenue, from 8.34 to 0.5, is at the downstream authority, because that is where the new credit is claimed. When the goods producer and the service firm are in different regions, the reform therefore lowers revenue in the region that hosts the service firm, on account of a tax that the other region collected and keeps. This is the mechanism behind the regional results of Section 8.1.

5 Empirical Evidence

In this section, I present the empirical evidence on how the “B2V” tax reform affects firm dynamics (i.e., firm entry and growth) and interregional trade. The empirical investigation motivates my development of the quantitative general equilibrium model in section 6. I unfold this section with the dataset I use, empirical strategy, and main findings.

5.1 Data

Firm Entry. To document how “B2V” tax reform affects firm entry, I measure the number of firms at the province-by-sector-by-year level from 2008 to 2021 by aggregating the individual records of the administrative firm registry, collected from Qichacha.⁶ The registry covers every firm that has ever registered in China.

Firm Activities. To investigate the impact on firm activities following the “B2V” reform, I use the National Tax Survey Database of the Chinese State Administration of Tax (SAT) from 2008 to 2016. The database includes information on the firms’ income, expenditure, and all tax-related outcomes. The sampling of this dataset ensures that large firms are included every year and that smaller firms are sampled as repeated cross-sectional data. The data surveys around 600,000 firms per year, accounting for 65% of total enterprise tax revenue. The analysis uses the waves from 2010 onward.

Trade. To study the impact of the tax reform on interregional trade and cross sector-regional intermediate input use, I use the sector-by-region level trade value generated from the China inter-regional Input-Output Tables (IRIO). The IRIO is available every 5 years, I use data from 2012 and 2017 (Li et al., 2023) that cover the B2V reform period. The IRIO reports bilateral trade flows between 31 provinces and 41 sectors.

⁶Qichacha <https://www.qcc.com/> is a comprehensive business information platform based in China, providing up-to-date detailed and reliable data about companies.

Table 3: Summary Statistics of the Repeated Cross Sections

Year	Avg Revenue	Avg Asset	Obs.	#Service Firms
2010	145,804	424,514	746,772	155,387
2011	191,562	553,581	691,353	176,515
2012	210,576	635,929	689,395	191,414
2013	215,808	661,738	699,369	218,240
2014	197,115	697,199	716,765	236,810
2015	200,727	740,333	700,525	221,667
2016	159,509	766,051	608,799	205,182

Notes: This table presents summary statistics for the repeated cross sections of the National Tax Survey used in the analysis. All monetary values are reported in thousands of Chinese Yuan (CNY). The repeated cross sections include a broad range of firms each year, including smaller and newer entries, with sample sizes ranging from approximately 600,000 to over 740,000 and service firm counts increasing from 155,387 in 2010 to 205,182 in 2016.

5.2 Empirical Strategy

The “B2V” reform was implemented gradually across regions (for a subset of sectors) between 2012 and 2013, and gradually across sectors (nationwide) between 2014 and 2016. I exploit this policy variation using an event-study DID framework. While the staggered implementation of the reform across provinces and sectors provides a valuable source of variation for my empirical specification, potential identification concerns remain. These include the possibility of anticipatory effects in provinces that implemented the reform later, endogenous timing of reform implementation based on unobserved local economic conditions, and the challenge of establishing a clear control group given the interconnected nature of the economy. Furthermore, the parallel trends assumption crucial for my identification strategy may be violated if pre-reform trends in outcomes differed systematically between early and late-adopting provinces or sectors, potentially due to unobserved factors correlated with both the timing of reform implementation and our outcomes of interest.

To address these concerns, every specification includes sector-by-province and province-by-year fixed effects, so that the comparison is between reformed and unreformed sectors within the same province and year. The event-study estimates in the next section show no differential pre-trends, which supports the parallel-trends assumption without proving it. Since economic linkages between sectors make any control group imperfect, the results, especially the comparison between service and manufacturing firms, should be interpreted as relative effects: comparing more heavily impacted sectors to less impacted sectors rather than absolute treatment effects. To quantify the general equilibrium impacts and obtain

welfare estimates that account for these cross-sector linkages over space, I develop a spatial general equilibrium model that captures the interconnected nature of the economy that allows me to evaluate the reform's total effects in [Section 6](#).

5.2.1 Firm Entry

I begin by investigating the impact of the reform on extensive firm entry using the event study difference-in-differences as follows:

$$\log(M_{s,p,t}) = \alpha + \sum_{k \neq -1} \beta_k \times \text{Treat}_{s,p,k} + \eta_{s,p}^1 + \eta_{p,t}^2 + \epsilon_{s,p,t}, \quad (5.1)$$

where $M_{s,p,t}$ is the number of firms in sector s , province p and year t . $\text{Treat}_{s,p,k}$ is equal to 1 if sector s in province p was reformed at $t - k$, and 0 otherwise (and 0 for all manufacturing sectors). $\eta_{s,p}^1$ and $\eta_{p,t}^2$ are sector-by-province and province-by-year fixed effects, respectively. I cluster standard errors $\epsilon_{s,p,t}$ at the sector-by-province level, as this represents the level of the tax reform variation. First, I did not include the sector-by-year fixed effects as they are directly correlated with the treatment variable, which varies by sector and year. Second, the empirical analysis faces an additional identification challenge from a concurrent policy implemented in 2013 that aimed to stimulate small firm entry nationwide.⁷ Since service sectors are characterized by a higher proportion of small firms compared to manufacturing sectors, the policy would disproportionately impact service firms, potentially confounding the estimated effects of the tax reform.

To address this endogeneity concern, I employ two complementary strategies. First, I include province-by-year fixed effects to account for time-varying provincial-level variables, which could account for the impact of the 2013 small firm policy across provinces. Second, I use the methodology of Callaway and Sant'Anna, [2021](#), using the not-yet-treated service sectors as control units rather than manufacturing sectors, to help identify the absolute treatment effect of the reform more accurately.

5.2.2 Firm Revenue

I next examine the intensive margin using the tax survey, which reports each firm's total revenue. The survey is a repeated cross-section that includes the small and newly entered firms that make up most of the service sector ([Table 3](#)), and I estimate on the cross-sections

⁷For more information on the policy, see https://www.gov.cn/zwggk/2013-07/25/content_2455259.htm.

with the same two-way fixed-effects structure as for entry:

$$\log(\text{Revenue}_{f,s,p,t}) = \alpha + \sum_{k \neq -1} \beta_k \times \text{Treat}_{s,p,k} + \eta_{s,p}^1 + \eta_{p,t}^2 + \epsilon_{f,s,p,t}, \quad (5.2)$$

where $\text{Revenue}_{f,s,p,t}$ is the sales of firm f in sector s , province p , and year t , $\text{Treat}_{s,p,k}$ is defined as above, and $\eta_{s,p}^1$ and $\eta_{p,t}^2$ are sector-by-province and province-by-year fixed effects. Standard errors are clustered at the sector-by-province level.

The investment variable is largely missing in the tax survey dataset. To provide additional evidence of the reform's impact on firm activities and complement the revenue analysis, I examine land transactions as an alternative measure of firm expansion and investment. Land transactions are particularly informative in the Chinese context because land use rights represent a major input for business operations, and land purchases often signal firm growth and expansion plans. Moreover, since land sales constitute an important source of fiscal revenue for local governments⁸, analyzing land transactions gives insights about both firm behavior and local government finances. Using comprehensive data that includes every land transaction record in China's primary land market, I find that firms in reformed sectors and regions pay higher land prices and acquire larger land areas compared to their counterparts in non-reformed regions or sectors (see [Appendix B](#)). These results corroborate the firm-level revenue analysis by providing independent evidence that the tax reform stimulated firm expansion and investment activities.

5.2.3 Outsourcing and the Make-or-Buy Margin

A central mechanism in this paper is that the business tax distorted firms' *make-or-buy* decisions: purchased service inputs carried a noncreditable tax, while producing the same service tasks in-house with own labor carried none, so the tax favored vertical integration. I test for this margin among manufacturing firms. Their own tax regime never changes; the only thing the reform changes for them is whether the services they buy carry an input credit. This isolates the make-or-buy margin from the changes in output tax, filing regime, and accounting definitions that a service firm experiences at its own switch date.

The variation is across provinces. In a comparison across sectors by reform date, every manufacturing firm is exposed, because its service suppliers switch regardless of its own sector, so a sector-level design does not exclude spillovers from the reformed sectors to the control group. The pilot provides a comparison that does: manufacturers in the nine pilot

⁸Land transaction data is obtained from the Ministry of Natural Resources (formerly the Ministry of Land and Resources), which is available at www.landchina.com.

provinces could credit their purchases of the pilot services from 2012, and manufacturers elsewhere only from August 2013. I estimate

$$y_{it} = \sum_{\tau \neq 2011} \beta_{\tau} \text{Pilot}_{p(i)} \mathbf{1}[t = \tau] + \alpha_i + \delta_{j(i),t} + \varepsilon_{it}, \quad (5.3)$$

where $\text{Pilot}_{p(i)}$ indicates that firm i is located in a pilot province, α_i is a firm fixed effect for the firms observed in more than one wave, $\delta_{j(i),t}$ is a two-digit-industry-by-year effect, and standard errors are clustered by province. The base year is 2011, the last year before any adoption. Because the rest of the country adopted in August 2013, β_{τ} measures the effect of the pilot provinces' head start; it is a relative difference and bounds the total response from below.

Two outcomes measure the two sides of the margin. The *in-house cost share* is administrative expense over operating cost plus administrative expense; administrative expense is the income-statement line that houses a firm's own accounting, legal, information-technology, and planning departments. The *purchased-service share* is the sum of seven itemized purchased service categories reformed by 2015 (freight transport, research and technical services, information technology, cultural and creative services, logistics support, leasing of movable property, and attestation and consulting) over operating cost, available from 2011. The level of the in-house share used in the model is measured separately from the pre-reform cross-section (Section 7), so the response estimated here is a check on the mechanism and not a calibration target.

5.2.4 Cross-region-sector Trade

The firm-level evidence demonstrates that replacing the business tax with the value added tax stimulated individual firm growth and expansion. However, a key insight from the reform is that it should also affect how firms trade with each other across sectors and regions by reducing the frictions of cross-sector trade. To examine this mechanism directly, I analyze changes in interregional trade patterns using input-output data.

I estimate the impact of the tax reform on interregional trade flows using the Inter-Regional Input-Output (IRIO) tables for 2012 and 2017, which captures the trade flow before and after reform's completion. This analysis allows me to test whether the removal of the artificial tax burdens increased trade flow involving service sectors. I employ a gravity-style specification to test this hypothesis:

$$\text{IntUse}_{ok,dj,t} = \alpha + \beta \cdot \text{Treat}_{k,j,t} + \eta_{o,t}^1 + \eta_{d,t}^2 + \eta_{o,k,d,j}^3 + \epsilon_{o,k,d,j,t}, \quad (5.4)$$

where $\text{IntUse}_{o,k,d,j,t}$ is the intermediate trade flows from sector k of origin province o to sector j of destination province d in year t if it's the intensive margin; or equal to 1 if it's positive and 0 otherwise for the extensive margin. $\text{Treat}_{k,j,t}$ is defined as one if the year is in 2017 and the trade flow is imported from the service sector or exported to the service sector.⁹ $\eta_{o,t}^1$, $\eta_{d,t}^2$, and $\eta_{o,k,d,j}^3$ are origin-province-by-year, destination-province-by-year, and origin-cell-by-destination-cell fixed effects. I use the two-way clustered standard errors at origin-sector and destination-sector level. I use the Poisson pseudo-maximum-likelihood (PPML) estimator for estimations to deal with the issue of having many zeros and since the estimation is based on the levels of the dependent variable rather than its logarithm, both small and large values contribute to the estimation process in a balanced manner (Silva & Tenreyro, 2006).

5.3 Main Findings

5.3.1 Firm Entry

I begin with the extensive margin. Panel (A) of Figure 2 plots the event-study coefficients for the log number of registered firms in reformed service sectors relative to manufacturing sectors in the same province and year. Before the reform, the coefficients are small, negative, and statistically indistinguishable from zero, so service and manufacturing sectors were on similar paths within a province. The number of service firms then rises steadily relative to manufacturing: by about 6 log points in the reform year, 16 in the first year after, 32 in the second, and 43 in the third. Pooling the post-reform periods, service sectors see about 21% more firms after the reform than their manufacturing counterparts.

Two features of the design matter for reading this estimate. First, the province-by-year fixed effects absorb any province-wide shock to firm registration, including the 2013 nationwide policy that eased small-firm registration, so the coefficients compare reformed service sectors only with manufacturing sectors in the same province and year. Second, because the comparison group is manufacturing, the estimate captures the differential effect of removing the business tax on the sectors that received it. Entry rising only after the reform, and continuing to rise over the following years, is what one expects when demand for market-supplied services rises: once service output carries a creditable VAT invoice, serving business customers becomes more profitable, and more firms find it worthwhile to register and operate in the formal sector.

⁹Shanghai is dropped to avoid the contamination because it was treated in 2012, January.

5.3.2 Firm Revenue

I next turn to the intensive margin. Panel (B) of [Figure 2](#) reports the event-study estimates for log firm revenue in the repeated cross-sections of the tax survey, which include the smaller and newly entered firms that make up most of the service sector. The pre-reform coefficients are close to zero and statistically insignificant, which supports the parallel trends assumption behind the comparison with manufacturing firms. After the reform, service firms experience a substantial and statistically significant increase in revenue relative to their manufacturing counterparts, approximately 22% on average over the post-reform years. The response is largest in the reform year, dips in the following year, and recovers to between 24 and 30 log points in years two and three, so the gain is not a one-time jump but a persistent shift in the relative size of service firms.

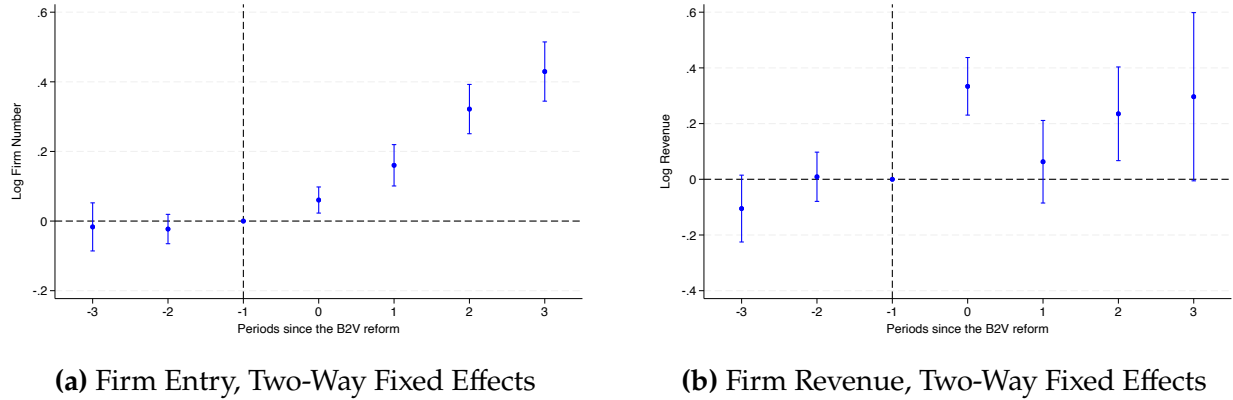
Because the survey is a repeated cross-section, the revenue estimate combines the growth of incumbent service firms with the arrival of new ones. Read together with the entry result in Panel (A), the two panels describe the same adjustment from two sides: more firms operate in the reformed service sectors, and the firms that are observed sell more. Both are consistent with the mechanism in the model: when the business tax, which taxed service sales in full and left business customers no credit, is replaced by a VAT that lets those customers deduct the tax on their service purchases, demand for market-supplied services rises. The model itself has no entry margin, so the entry response is motivating evidence for that shift in demand rather than a prediction of the model ([Section 6](#)). The reform lowered the effective cost of buying services from the market, and the service sectors that gained this crediting grew along both margins.

5.3.3 Outsourcing and the Make-or-Buy Margin

[Figure 3](#) reports the estimates. The purchased-service share of manufacturers in the pilot provinces rises relative to manufacturers elsewhere, by 0.25 to 0.44 share points of operating cost, and the difference is statistically significant (at the 1 percent level in 2015). The in-house cost share moves the other way: it is flat in 2010, before the pilot, falls by about 0.1 share points in 2012 and 2013, and by 0.4 to 0.5 share points from 2014. The in-house estimates are less precise than the purchased-service ones; they are significant at the 10 percent level in 2014 and 2015 and not at the 5 percent level in any year.

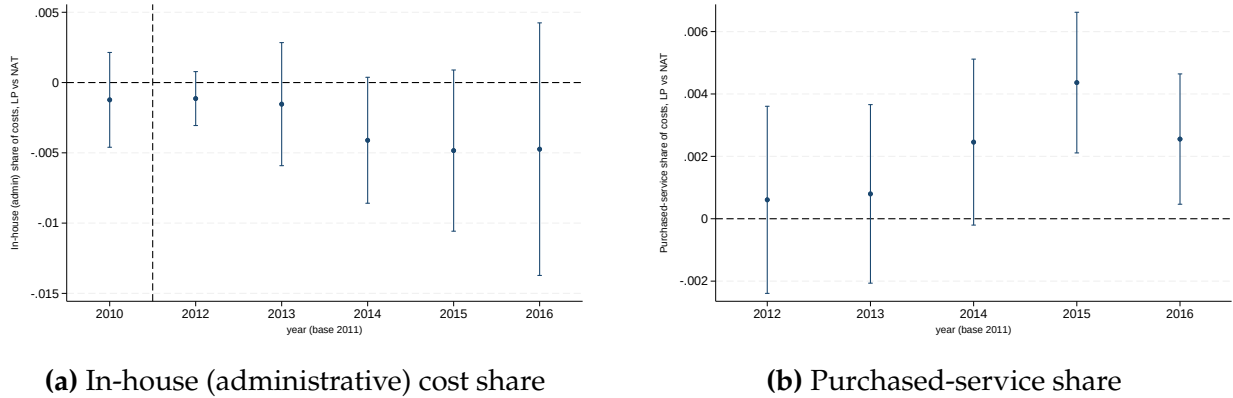
The two outcomes are measured in different blocks of the survey and move in opposite directions by comparable amounts. A reporting response to newly creditable invoices would raise the recorded purchases but has no reason to lower recorded internal costs,

Figure 2: Effect of the Reform on Firm Growth in Service Sectors



Notes: This figure presents two-way fixed-effects event-study estimates of the reform's effect on service-sector firms relative to manufacturing firms. Panel (A) shows the log number of registered firms; Panel (B) shows log firm revenue in the repeated cross sections of the National Tax Survey. The x-axis is time relative to the reform implementation, with -1 the omitted reference period and 0 the reform year. Vertical bars are 95 percent confidence intervals.

Figure 3: The Make-or-Buy Margin: De-Integration on Both Sides of the Ledger



Notes: Event-study coefficients on pilot-province × year interactions, base year 2011, with firm and two-digit-industry-by-year fixed effects; standard errors clustered by province, 95 percent confidence intervals. Sample: manufacturing firms in the raw tax-survey waves 2010–2016; the purchased-service outcome begins in 2011 with the itemized fields. Panel (A) is administrative expense over operating cost plus administrative expense; Panel (B) sums the seven itemized purchased modern-service categories reformed by 2015 as a share of operating cost. Pilot provinces adopted the crediting in late 2012 and all others in August 2013, so the coefficients are head-start effects and bound the total response from below.

and the cost of complying with the value-added tax, if anything, raises administrative expense. I therefore read the pattern as manufacturers moving service tasks from their own departments to the market once the tax on purchased services became creditable. Since both groups are treated by August 2013, the estimates are differences due to a head start of about one year and understate the full response.

5.3.4 Cross-region-sector Trade

The trade flow analysis provides direct evidence that the tax reform enhanced economic integration by reducing trade frictions in intermediate goods transactions. I find substantial increases in trade involving service sectors both within and across regions following the reform.

Panel A of Table 4 presents the estimated effects on intermediate input trade flows. The reform raised trade flows involving service sectors — either as suppliers or purchasers of intermediate inputs — by 0.309 log points, a 36% increase, relative to purely manufacturing-to-manufacturing transactions, on flows converted to basic prices. Within provinces the coefficient is 0.407 (a 50% increase) and across regions 0.232 (26%). Because the 2017 table sits only one year after the reform’s completion for the largest cohorts, these are short-run responses; the wedge-based specification and its interpretation are in Appendix C.1. The three columns are separate regressions, each with its own fixed effects, so the pooled coefficient is not a weighted average of the other two. The response is positive both within and across provinces, so the effect of the reform is not confined to local markets.

These results indicate that the tax reform facilitated more efficient allocation of resources across sectors and regions, allowing firms to source inputs from potentially more productive suppliers and enhanced the interconnectedness of the production networks.

The pooled treatment coefficient in Panel A of Table 4 averages over flow types that shed different numbers of tax wedges, and the theory makes a sharper prediction than a single positive sign. Classifying each flow by the tax status of its two endpoints yields four types: service-to-service flows shed two wedges (the seller’s cascading tax and the buyer’s non-credibility), service-to-manufacturing flows shed the seller-side wedge and gained buyer crediting, manufacturing-to-service flows gained only the buyer’s input credit, and manufacturing-to-manufacturing flows — creditable on both sides since 1994 — serve as the untreated benchmark. The response should therefore be ordered: largest for service-to-service, intermediate for service-to-manufacturing, smallest but positive for manufacturing-to-service. Panel B of Table 4 shows this ordering of the point estimates in all three samples. As in Panel A, the columns are separate regressions: within a flow type the pooled coefficient can lie below both the within-province and the cross-province ones, because the province-by-year fixed effects absorb different variation in each sample, and the within- and cross-province coefficients are not statistically distinguishable from each other. One further pattern stands out. The manufacturing-to-service coefficient, identified purely off the buyer-credit margin with no change on the seller side, is small and only marginally significant: most of the reform’s trade response operated through the removal

Table 4: Effects of the Reform on Intermediate Trade Flows

	All (1)	Intra-Province (2)	Inter-Province (3)
<i>Panel A. Pooled: flows with at least one service endpoint</i>			
B2V	0.309** (0.029)	0.407** (0.042)	0.232** (0.072)
Cells (incl. zeros)	3,230,882	104,222	3,126,660
<i>Panel B. By flow type (manufacturing → manufacturing omitted)</i>			
Service → Service	0.643** (0.057)	0.752** (0.066)	0.834** (0.163)
Service → Manufacturing	0.158** (0.055)	0.255** (0.074)	0.318** (0.137)
Manufacturing → Service	0.066* (0.035)	0.046 (0.041)	0.027 (0.064)
Cells (incl. zeros)	2,206,520	86,126	2,120,394
Origin cell × destination cell FE	✓	✓	✓
Origin province × year FE	✓	✓	✓
Destination province × year FE	✓	✓	✓

Notes: PPML estimates of equation (5.4) on intermediate input flows reconstructed from the raw 2012 and 2017 MRIO tables and converted to basic prices before estimation (creditable cells deflated by the invoice factor); on published invoice values the coefficients are approximately 0.05 larger, the difference being the mechanical revaluation of newly creditable cells. A cell is a province-by-sector pair. Column (1) pools all flows; columns (2) and (3) split them into intra-province and inter-province flows. Panel A reports the coefficient on B2V, the differential log change between 2012 and 2017 of flows with a service sector at either endpoint relative to manufacturing-to-manufacturing flows. Panel B classifies each flow by the pre-reform tax status of its endpoints: service sectors are those moved from the business tax to VAT by the reform; manufacturing comprises the always-VAT sectors (including wholesale and retail); flows touching exempt or non-market sectors (education, health, public administration) are excluded, which is why Panel B has fewer cells. Manufacturing-to-manufacturing flows are the omitted benchmark, so each coefficient is the differential log change of that flow type relative to always-creditable flows. The predicted wedge ordering — service-to-service (two wedges removed) > service-to-manufacturing (seller wedge plus buyer credit) > manufacturing-to-service (buyer credit only) > 0 — holds in every column. Standard errors two-way clustered by origin and destination cell in parentheses. ***, **, * indicate significance at the 1%, 5%, and 10% levels. Appendix C.1 reports the wedge-based continuous specification and its short-run interpretation.

of the cascading wedge on service suppliers rather than through the credit margin alone. One caveat disciplines the magnitudes: the 2012–2017 window coincides with China’s secular service-sector expansion, which loads on the same ordering (flows with two service endpoints most, one endpoint less, none not at all), so the service-to-service magnitude in particular should be read as an upper bound on the reform component.

The empirical analysis demonstrates that eliminating the tax on intermediate inputs generated substantial positive effects on firm entry, growth and trade flows. Building on these empirical insights, a quantitative general equilibrium model can extend the analysis in several important dimensions to provide a more comprehensive understanding of the reform’s economic impact.

First, while the empirical estimates effectively identify the relative effects between service and manufacturing sectors, a structural model can quantify the absolute welfare changes

by accounting for general equilibrium effects that propagate through the interconnected production network. This allows for a complete assessment of the reform’s net benefits.

Second, a quantitative model can explicitly identify mechanisms that reduced form analysis suggest but cannot directly examine. These include: firms’ vertical integration decisions, a key channel through which the business tax distorted production choices — the outsourcing evidence above measures a sliver of this margin directly and disciplines its strength, while the model aggregates it over the full input bundle; the distributional consequences across regions, particularly how provinces with different economic structures were affected; and the fiscal implications for local governments that experienced a varying degree of tax revenue changes.

Third, a structural model can decompose the sources of observed improvements by distinguishing two potential channels: simple tax burden reduction versus the elimination of cross-sector trade frictions that distorted production networks in every stage. This decomposition can provide valuable insights for optimal tax policy design by identifying which mechanisms drive the largest welfare gains.

6 The Model

I develop a quantitative spatial general equilibrium model of production networks in the tradition of Caliendo et al., 2022a, cast under perfect competition and constant returns to scale, in which producers choose how much of their service inputs to buy from the market and how much to produce in-house (Bernard et al., 2019).¹⁰ There are $N = 32$ (31 Provinces + 1 Rest of the World) regions and $J = 41$ sectors (agriculture; 25 mining, manufacturing, and utility sectors; and 15 construction and service sectors, of which eleven were moved from the business tax to the VAT by the reform), with regions denoted by o and d and sectors by k and j . Labor is the sole factor of production supplied inelastically and mobile across sectors within a region but immobile across regions.

The structure of the economy is as follows. Each region-sector pair (o, k) supplies a distinct composite good under perfect competition, produced with labor and a composite intermediate input that bundles market-purchased inputs from all region-sectors with in-house service tasks. Goods are differentiated by origin à la Armington, and prices

¹⁰An earlier variant added a firm-entry margin with selection, as in Melitz, 2003 and Krugman, 1980. That margin is shelved for a deliberate, measured reason: with love-of-variety entry in a roundabout input-output network, the entry feedback loop is supercritical at standard elasticities (its measured gain exceeds one), so the entry-augmented equilibrium is *tâtonnement*-unstable and would require an equilibrium-selection caveat. The entry event study of Section 5 is accordingly used as motivating evidence, not as a calibration target; the selection stack is retained for companion work.

equal marginal delivered cost. Ad valorem taxes apply to sales revenue: buyers pay the tax-inclusive price and producers receive the net-of-tax price, following Caliendo et al., 2020. Households consume final goods, supply labor, and receive tax revenue as a lump-sum rebate.

6.1 Preference

In each region, L_d measures the representative households who maximize utility from consuming the region-sector composite goods $Y_{ok,d}^C$:

$$u(Y_d^C) = \left[\sum_{o=1}^N \sum_{k=1}^J s_{ok,d}^{\frac{1}{1+\theta}} (Y_{ok,d}^C)^{\frac{\theta}{1+\theta}} \right]^{\frac{1+\theta}{\theta}}, \quad (6.1)$$

subject to the budget constraint $\sum_{o=1}^N \sum_{k=1}^J p_{ok,d} Y_{ok,d}^C \leq I_d$, where $p_{ok,d}$ is the tax-inclusive price paid by consumers. I denote households' income by $I_d \equiv \left(\sum_{j=1}^J w_d L_{dj} \right) + D_d + T_d$, where income is the sum of labor wages, transfer payments, and tax revenue T_d collected in region d (defined in Section 6.5). $s_{ok,d}$ is the preference weight residents in region d place on goods from region o and sector k ; $\theta + 1$ is the elasticity of substitution between any two origin-sector composites, so that θ is the trade elasticity.

Utility maximization yields the CES price index

$$P_d = \left(\sum_{o=1}^N \sum_{k=1}^J s_{ok,d} p_{ok,d}^{-\theta} \right)^{-\frac{1}{\theta}}, \quad (6.2)$$

with $U(Y_d^C) = I_d/P_d$, and the demand

$$Y_{ok,d}^C = s_{ok,d} p_{ok,d}^{-(1+\theta)} P_d^\theta I_d, \quad (6.3)$$

which depends on the tax-inclusive price: consumers respond to the price they actually pay.

6.2 Production

6.2.1 Final Goods

Each pair (d, j) hosts a continuum of identical competitive producers with constant-returns technology¹¹

$$y_{dj} = [\ell_{dj}]^{\alpha_{dj}} [Y_{dj}^M]^{1-\alpha_{dj}}, \quad (6.4)$$

where ℓ_{dj} is labor and the intermediate input Y_{dj}^M is a CES aggregate of intermediate goods from all sectors and regions, including *in-house production goods* (vertical integration). The parameter α_{dj} captures the relative importance of labor versus intermediate inputs in production and can vary across regions and sectors.

The outer nest is Cobb–Douglas, so the split between direct labor and the input bundle is fixed. The response to taxes comes from inside the bundle, which is CES: when taxes make market-purchased inputs more expensive, firms substitute towards other inputs and towards in-house provision of services, and back when the taxes are removed. The marginal cost under value-added taxation (VAT) or business taxation (BT) implies the following unit cost bundle for good j produced in region d :

$$c_{dj} = \mathbf{C}_1(\alpha_{dj}) (w_d)^{\alpha_{dj}} (P_{dj}^M)^{1-\alpha_{dj}}, \quad (6.5)$$

where α_{dj} denotes the labor share, which is allowed to vary flexibly across regions and sectors.¹²

This cost formulation embeds the key tax distortions discussed above. In particular, P_{dj}^M denotes the composite price of intermediate inputs faced by firms and is measured at purchasers' prices. As a result, all noncreditable taxes paid on intermediate inputs—such as business taxes or noncreditable VAT—are already incorporated into P_{dj}^M and therefore enter firms' marginal costs directly. In contrast, creditable VAT paid on intermediate inputs between VAT-registered firms is assumed to be fully rebated through the input tax credit mechanism and thus does not affect P_{dj}^M .

¹¹The continuum matters for interpreting the *diagonal* input cells: a producer's purchase from its own region-sector is a purchase from the cell's composite, of which its own output is of measure zero, so literal self-trade never occurs. The diagonal flow is the aggregate of arm's-length transactions between distinct legal persons within the cell — construction subcontracting being the canonical case — which is precisely what the input-output table records and what the tax law reaches: the business tax and VAT attach to transactions between legal persons, intra-firm transfers are neither taxed nor recorded, and the model's in-house row sits exactly on that boundary.

¹²The associated constant is $\mathbf{C}_1(\alpha_{dj}) = \left(\alpha_{dj}^{\alpha_{dj}} (1 - \alpha_{dj})^{1-\alpha_{dj}} \right)^{-1}$; it varies across (d, j) through the labor share α_{dj} .

Consequently, under a neutral VAT system with full crediting, intermediate input choices and marginal costs are unaffected by taxation, and VAT operates purely as a wedge between producer prices and final consumer prices. By contrast, under the pre-reform dual-tax system, transactions involving business-tax sectors give rise to noncreditable taxes that become embedded in intermediate input prices, raising marginal costs and generating tax cascading along production networks. These distortions are summarized in the cross-sector tax wedges, which will be introduced later and operate entirely through P_{dj}^M in the cost function.

This formulation is cast in the valuation convention of the SNA basic-price accounts: recoverable VAT is excluded from intermediate inputs and value added, while noncreditable taxes are embedded in purchasers' prices and therefore enter firms' marginal costs. The published Chinese benchmark input–output table follows a different convention. Its transaction cells are recorded at invoice values on all domestic links—including fully creditable ones—so that a cell on a creditable VAT link exceeds the buyer's true economic cost by the input VAT that is subsequently refunded. Columns nonetheless balance because the net-taxes-on-production entry (生产税净额) in the value-added block records each sector's *net* remittance—output tax invoiced less input credits reclaimed—rather than its gross output tax, so the input VAT embedded in the intermediate cells is exactly offset by the credit subtracted in the tax entry. Neither the published cells nor the published value-added block can therefore be read directly as the model's objects; Section 6.7 constructs the model-consistent table by a link-by-link valuation conversion.

6.2.2 (Composite) Intermediate Input

The intermediate input of pair (d, j) is a CES aggregate of the composite goods of all origins and sectors and of service tasks produced in-house:

$$Y_{dj}^M = \left[\sum_{o=1}^N \sum_{k=1}^J \iota_{ok,dj}^{\frac{1}{1+\theta}} \left(Y_{ok,dj}^m \right)^{\frac{\theta}{1+\theta}} + \iota_{H,dj}^{\frac{1}{1+\theta}} \left(Y_{H,dj}^m \right)^{\frac{\theta}{1+\theta}} \right]^{\frac{1+\theta}{\theta}}. \quad (6.6)$$

The weight $\iota_{ok,dj}$ captures the importance of the good of sector k and region o in the input bundle of sector j in region d , and $\iota_{H,dj}$ that of in-house tasks.

In-house Production. In-house service tasks are produced with the firm's own labor at constant returns,

$$Y_{H,dj}^m = z_{dj}^H \ell_{dj}^{IH}, \quad (6.7)$$

where z_{dj}^H is the productivity of in-house tasks in region d and sector j . No tax falls on them: they are not a transaction between legal persons.

Cost minimization gives the price index of the bundle,

$$P_{dj}^M = \left\{ \sum_{o=1}^N \sum_{k=1}^J \iota_{ok,dj} p_{ok,dj}^{-\theta} + \iota_{H,dj} \left(\frac{w_d}{z_{dj}^H} \right)^{-\theta} \right\}^{-\frac{1}{\theta}}, \quad (6.8)$$

where $p_{ok,dj}$ is the purchaser's price, so that noncreditable taxes are embedded in P_{dj}^M , and the demand for the good of (o, k) ,

$$Y_{ok,dj}^m = \iota_{ok,dj} \left[\frac{p_{ok,dj}}{P_{dj}^M} \right]^{-(1+\theta)} Y_{dj}^M. \quad (6.9)$$

A firm in (d, j) therefore employs two kinds of labor: labor for its primary product, with cost share α_{dj} , and labor for in-house service tasks, which belongs to the input bundle and competes with market inputs. Writing $\gamma_{H,dj}^P$ for the in-house share of the bundle, the two sum to the value-added share observed in the input-output table, $\alpha_{dj} + \gamma_{H,dj}^P(1 - \alpha_{dj}) = \zeta_{dj}^0$.

One elasticity. The bundle has a single nest, so one parameter, $1 + \theta$, governs substitution between origins of the same good, between goods of different sectors, and between market inputs and in-house tasks. This is a strong restriction. It is what allows the model to be solved in exact hat algebra from the input-output table alone, without separate estimates of cross-sector and make-or-buy elasticities, which the reform window does not identify (Section 7). It has two consequences that the reader should keep in mind. Substitution across sectors is as elastic as substitution across origins, which makes sectoral reallocation in the counterfactuals larger than a nested structure with a lower upper-level elasticity would deliver. And the make-or-buy response is tied to the trade elasticity: the contribution of the margin reported in Section 8 is its contribution at that elasticity, and Appendix C.2 reports how it varies with θ .

Competitive Pricing. Under perfect competition and constant returns, the good of pair (o, k) is supplied at marginal delivered cost. Following Caliendo et al., 2020, the ad valorem tax applies to sales revenue: the buyer pays the tax-inclusive price p , while the producer receives $p/(1 + t)$. Zero profits then pin down the tax-inclusive prices to producers in (d, j)

and to consumers in d :

$$p_{ok,dj} = (1 + t_{ok,dj}) c_{ok} \kappa_{ok,dj}, \quad (6.10)$$

$$p_{ok,d} = (1 + t_{ok,d}) c_{ok} \kappa_{ok,d}. \quad (6.11)$$

The net-of-tax price received by the producer is marginal delivery cost, and the tax wedge $(1 + t)$ is passed through one-for-one to the buyer. Because the wedge is passed through in full, the tax is borne by the buyer—and ultimately by the final consumer—not by the selling producer; what the credit mechanism $t_{ok,dj}$ governs is instead *where* along the chain each dollar of revenue is remitted, and whether it cascades. The numerical illustration of Section 4 makes this distinction between the *location of remittance* and the *incidence of the tax* precise.

The cross-sector tax rate $t_{ok,dj}$ captures the VAT credit mechanism that generates differential pricing across transaction types. Under China's pre-reform dual-tax system, manufacturing firms operating under the value-added tax (VAT) could claim input tax credits only when transacting with other VAT-registered firms. Transactions involving service firms subject to the business tax, as well as sales to final consumers, did not allow for input tax credits.

Formally, before the reform,

$$t_{ok,dj} = 0$$

if both k and j are manufacturing sectors operating under the VAT system. In this case, although VAT is charged on the invoice, the buyer can fully claim input tax credits, so intermediate inputs are effectively traded at VAT-exclusive prices and tax cascading is avoided. This price neutrality, however, is not the same as *fiscal* neutrality. Regardless of whether the input credit is ultimately deducted, the upstream rate t_k still governs how much tax is remitted at the upstream stage versus the downstream stage—and hence, when revenue is assigned at origin, which jurisdiction collects and rebates the revenue. Full crediting pins down the aggregate tax and the consumer price at the downstream rate, but it leaves the *spatial incidence* of the revenue open: a lower t_k shifts collection toward the downstream production stage. The upstream rate is thus neutral for prices and marginal costs but not for the geography of tax revenue, as the numerical illustration in Section 4 makes precise.

For any transaction involving a service sector subject to the business tax—either as seller or buyer—input tax credits are unavailable. The VAT charged on inputs is therefore

noncreditable and becomes embedded in the buyer's input cost, implying

$$t_{ok,dj} = t_k,$$

where t_k is the statutory (net) ad valorem rate of sector k .

A remark on rate conventions is needed here, because China's two pre-reform taxes are stated on different bases. The VAT is *price-exclusive* (价外税): the statutory rate τ_k applies on top of the seller's net price, so it maps directly into the model's convention, $t_k = \tau_k$. The business tax is *price-internal* (价内税): the statutory rate τ_k^{BT} applies to gross receipts, tax included, so the seller of a transaction valued at X keeps $(1 - \tau_k^{BT})X$. Matching this to the model, in which the seller keeps $X/(1 + t_k)$, requires

$$\frac{1}{1 + t_k} = 1 - \tau_k^{BT} \quad \Longleftrightarrow \quad t_k = \frac{\tau_k^{BT}}{1 - \tau_k^{BT}}, \quad (6.12)$$

so that statutory business-tax rates of 3% and 5% correspond to $t_k \approx 3.09\%$ and $t_k \approx 5.26\%$ in the model's net-rate convention. Under (6.12) the implied remittance on a business-tax sale, $\frac{t_k}{1+t_k}X = \tau_k^{BT}X$, equals the statutory tax on gross receipts—exactly the amount recorded in the service sector's net-taxes-on-production entry—so the convention is what allows the remittance equation (6.26) to reproduce the tax row of the published table for business-tax sectors.

After the reform, the reformed sectors operate under the VAT system, and input tax credits are available on most intermediate transactions, where

$$t_{ok,dj} = 0.$$

Not on all of them. Purchases of catering, loan services, resident services, and entertainment remain noncreditable; exempt buyers (agriculture, education, health) and public administration credit nothing; and in 2012 buyers under the VAT could credit a deemed share of two inputs whose sellers issued no VAT invoice, farm products and freight. The general form of the wedge covers all of these cases. Let $\delta_{ok,dj} \in [0, 1]$ be the share of the tax on a purchase that the buyer credits. Then

$$1 + t_{ok,dj} = 1 + (1 - \delta_{ok,dj}) t_k, \quad (6.13)$$

which is zero-wedge at $\delta = 1$ and the full rate at $\delta = 0$; the values of δ before and after the reform are given in Section 6.7 and Appendix C.2. The reform is a change in $\{\delta_{ok,dj}\}$ and in $\{t_k\}$.

For transactions with final consumers, the tax treatment differs. Consumers cannot claim input tax credits, and the consumer-facing tax rate is therefore given by

$$t_{ok,d} = t_k.$$

In all cases, consumers bear the full statutory tax burden.

6.3 Price Indices

Substituting the competitive prices (6.10)–(6.11) into the CES aggregators, the composite intermediate input price in (6.8) becomes

$$P_{dj}^M = \left[\sum_{o=1}^N \sum_{k=1}^J \iota_{ok,dj} [(1 + t_{ok,dj}) c_{ok} \kappa_{ok,dj}]^{-\theta} + \sum_{k=1}^J \iota_{Hk,dj} \left(\frac{w_d}{z_{dj}^H} \right)^{-\theta} \right]^{-\frac{1}{\theta}}, \quad (6.14)$$

and the consumer price index (6.2) becomes

$$P_d = \left[\sum_{o=1}^N \sum_{k=1}^J s_{ok,d} [(1 + t_{ok,d}) c_{ok} \kappa_{ok,d}]^{-\theta} \right]^{-\frac{1}{\theta}}. \quad (6.15)$$

Noncreditable wedges enter both indices exactly as trade costs do, which is the formal sense in which cascading taxation acts as an artificial trade friction on the production network.

6.4 Gravity and Trade Shares

Combining the CES demands with the competitive prices yields bilateral expenditures in gravity form:

$$X_{ok,dj}^P = \iota_{ok,dj} \left[\frac{(1 + t_{ok,dj}) c_{ok} \kappa_{ok,dj}}{P_{dj}^M} \right]^{-\theta} E_{dj}^M, \quad (6.16)$$

$$X_{ok,d}^C = s_{ok,d} \left[\frac{(1 + t_{ok,d}) c_{ok} \kappa_{ok,d}}{P_d} \right]^{-\theta} I_d, \quad (6.17)$$

where E_{dj}^M is total intermediate expenditure of pair (d, j) defined below. The elasticity of bilateral trade with respect to the gross tax wedge $(1 + t)$ is $-\theta$: under perfect competition the trade elasticity and the wedge elasticity coincide, and θ is the single elasticity the calibration must take a stand on (Section 7). Total *net-of-tax* sales of region-sector (o, k) are

$$R_{ok} \equiv \sum_{d=1}^N \sum_{j=1}^J \frac{X_{ok,dj}^P}{1 + t_{ok,dj}} + \sum_{d=1}^N \frac{X_{ok,d}^C}{1 + t_{ok,d}}. \quad (6.18)$$

The bilateral trade shares, which are the objects observed in the input-output data and used to calibrate the model, follow directly:

$$\gamma_{ok,dj}^P \equiv \frac{X_{ok,dj}^P}{P_{dj}^M Y_{dj}^M}, \quad \gamma_{ok,d}^C \equiv \frac{X_{ok,d}^C}{I_d}, \quad (6.19)$$

and the implied factor share of intermediate input (o, k) in (d, j) production is $\xi_{ok,dj} \equiv (1 - \alpha_{dj}) \gamma_{ok,dj}^P$. Both $X_{ok,dj}^P$ and $X_{ok,d}^C$ are measured at purchasers' prices, inclusive of *noncreditable* tax only: on a creditable link the wedge is zero and the model's purchaser price coincides with the seller's net price, whereas on a noncreditable link the wedge $(1 + t_{ok,dj})$ separates the two, and only $X/(1 + t)$ reaches the producer, with the difference collected as tax revenue. These objects correspond to the published input-output cells only after the valuation conversion of Section 6.7, which removes the invoiced-and-refunded VAT from creditable cells.

6.5 Market Clearing and Trade (Im)balance

The circular flow of the economy closes as follows: producers receive net-of-tax sales R_{ok} and pay them out to labor and intermediate inputs; households earn wages, rebated tax revenue, and transfers, and spend their income I_d on final varieties; and firms in (d, j) spend $(1 - \alpha_{dj})R_{dj}$ on intermediate inputs. Under constant returns and zero profits, total payments by (o, k) split into $\alpha_{ok}R_{ok}$ for production labor and $(1 - \alpha_{ok})R_{ok}$ for the composite intermediate input, where R_{ok} is net-of-tax sales defined in (6.18). Within intermediate expenditure, the in-house portion $\gamma_{H,ok}^P(1 - \alpha_{ok})R_{ok}$ also accrues to local labor. Total intermediate expenditure (at purchasers' prices) of pair (d, j) is therefore

$$E_{dj}^M \equiv P_{dj}^M Y_{dj}^M = (1 - \alpha_{dj}) R_{dj}. \quad (6.20)$$

Goods market clearing. It is convenient to keep R for *net-of-tax* sales and to denote the bilateral expenditures at purchaser (tax-inclusive) prices by

$$X_{ok,dj}^P \equiv \gamma_{ok,dj}^P (1 - \alpha_{dj}) R_{dj}, \quad X_{ok,d}^C \equiv \gamma_{ok,d}^C I_d, \quad (6.21)$$

as in (6.19); only $X/(1 + t)$ reaches the producer. Net-of-tax sales of (o, k) then equal the sum over destinations of these expenditures net of tax:

$$R_{ok} = \sum_{d=1}^N \sum_{j=1}^J \frac{X_{ok,dj}^P}{1 + t_{ok,dj}} + \sum_{d=1}^N \frac{X_{ok,d}^C}{1 + t_{ok,d}}. \quad (6.22)$$

Two distinct tax objects. Before writing the revenue equations it is essential to distinguish two objects that both go under the name “tax” but play entirely different roles:

- (i) the *cascading wedge* $t_{ok,dj}$ (and $t_{ok,d}$), which enters *prices and costs*—the competitive prices (6.10)–(6.11), the input price index P_{dj}^M , the cost bundle c_{ok} , and the purchaser-price expenditures X^P, X^C . It is 0 on every fully creditable link and positive where the credit chain breaks (business-tax sectors, exempt buyers, the legislated carve-outs, final consumers);
- (ii) the *statutory remittance* T_{dk}^P , which enters *government revenue*. It is built from the statutory output rates t_k applied to a net base, less the input credits reclaimed, and—as the numerical illustration showed—is generically nonzero *even when every wedge* $t_{ok,dj} = 0$. Full crediting collapses the wedge (price/cost neutrality) but not the remittance (fiscal non-neutrality).

The wedge is a function of *creditability*; the remittance is a function of the *statutory rates* $\{t_k\}$ applied on a value-added base through the credit-invoice mechanism. These two maps coincide only after aggregating across the whole economy (shown in the Walras step of Appendix A); sector-by-sector, and for an open region, they differ.

Taxes, revenue, and the input–output accounts. It is worth being explicit about what R_{dj} represents and how it maps to the input–output (IO) data, because the ad valorem tax plays an unusual role: it *generates government revenue without being a cost to any firm*. On any transaction the buyer pays the purchaser (tax-inclusive) price $X_{ok,dj}^P$; of this the seller (o, k) keeps the net-of-tax amount, and the residual is the *non-creditable wedge* embedded in the

buyer's price,

$$X_{ok,dj}^P = \underbrace{\frac{X_{ok,dj}^P}{1 + t_{ok,dj}}}_{\text{net price kept by } (o,k)} + \underbrace{\frac{t_{ok,dj}}{1 + t_{ok,dj}} X_{ok,dj}^P}_{\text{non-creditable wedge in the price}}. \quad (6.23)$$

This decomposition concerns the *price*, not the government's collection. The wedge term is the tax that fails to be reclaimed and therefore sticks in the purchaser price; it is a transfer, not a factor payment, entering neither the marginal cost, the value added, nor the input bill of the selling firm. It must *not* be confused with (o, k) 's remittance, which is output tax *net of input credits*,

$$T_{ok}^P = t_k R_{ok}^{\text{dom}} - \sum_{o',k'} \delta_{o'k',ok} t_{k'} X_{o'k',ok'}^P$$

defined in (6.26). The two differ sharply on a *creditable* link: there $t_{ok,dj} = 0$, so the wedge term in (6.23) is zero—the buyer reclaims the invoiced VAT, and nothing sticks in the price—yet the seller has still *invoiced and remitted* output tax t_k on the sale. Full crediting collapses the price *wedge*, not the *remittance*: by the basic-price convention the recoverable VAT is excluded from measured intermediate use and value added, so the firm acts as a pure *collector* whose invoiced tax appears only in its buyers' prices and never in its own costs, while the revenue still flows to the government through T_{ok}^P . The wedge in (6.23) is strictly positive only where the credit chain terminates—a final consumer, or a business-tax sector—and only there does the invoiced tax coincide with tax that actually sticks in the purchaser price, and hence in the *model-consistent* (basic-price) input–output accounts. The *published* Chinese table instead records invoice values on all domestic links, creditable or not, with the offsetting credit netted inside the tax row of the value-added block. The expenditure shares $\gamma_{ok,dj}^P = X_{ok,dj}^P / E_{dj}^M$ are therefore computed from the *converted* table of Section 6.7, in which creditable cells are deflated by the statutory VAT factor and noncreditable cells retain their published values; reading the shares off the raw table would overstate creditable input shares by the embedded credit and, more importantly, would erase the creditable–noncreditable asymmetry that drives both cascading and the make-or-buy margin. The direct labor share α_{dj} is measured from the core value-added block—labor compensation, depreciation, and operating surplus—*excluding* net taxes on production, which in the model are neither a factor payment nor a cost but the rebated revenue term $T_{dk'}^P$, and *net of* the in-house service cost H_{dj} , which the model relabels as part of the intermediate bundle (Section 7).

Consequently R_{dj} in (6.18) is the gross output of sector (d, j) valued at basic (net-of-tax) prices—the value the firm actually receives and has available to spend. By zero profits it is exhausted by factor and intermediate payments, so it splits into value added and an intermediate bill valued at purchaser prices,

$$R_{dj} = \underbrace{\alpha_{dj} R_{dj}}_{\text{value added}} + \underbrace{(1 - \alpha_{dj}) R_{dj}}_{\text{intermediate use} = \sum_{o,k} X_{ok,dj}^P + H_{dj} = E_{dj}^M}, \quad (6.24)$$

where the second term is total intermediate use: market purchases at purchaser prices (6.21) plus the in-house service cost $H_{dj} \equiv \gamma_{H,dj}^P E_{dj}^M$. (The in-house portion is relabeled local labor, so measured IO value added is $[\alpha_{dj} + \gamma_{H,dj}^P (1 - \alpha_{dj})] R_{dj}$, as in (6.25).) The output tax that sector (d, j) invoices on its *own* sales appears *nowhere* in (6.24): it is neither part of value added nor, when creditable, part of any buyer's cost, since the buyer reclaims it. It re-enters the model only as government revenue—the net remittance T_{dk}^P , output tax less input credits—collected and rebated lump-sum through (6.26)–(6.27). What is invariant to the seller's own output tax is the *technology*: neither α_{dj} nor $\gamma_{ok,dj}^P$ contains the tax the selling sector remits, which enters the equilibrium only through the price wedge (6.10)–(6.11) and the separate, rebated revenue term T_{dk}^P .

Labor market clearing. With labor mobile across sectors within a region, labor demand has two components: production labor, $\alpha_{ok} R_{ok}$, and *in-house* labor, which is paid out of intermediate expenditure with share $\gamma_{H,ok}^P$:

$$w_o L_o = \sum_{k=1}^J \left[\alpha_{ok} + \gamma_{H,ok}^P (1 - \alpha_{ok}) \right] R_{ok}, \quad L_o = \sum_{k=1}^J L_{ok}. \quad (6.25)$$

Tax revenue. Revenue is assigned at origin: the tax on each transaction is remitted by the *seller* to the government of its own region. This is the rule that governs the provincial share of China's VAT; the model applies it to all revenue and abstracts from the central share. Because every region is itself a producer, the revenue collected in region d is the tax its own firms remit on their sales to all destinations, net of the input credits those firms reclaim—so the subscript d here labels the *selling* region, and no revenue accrues to a region merely for buying a good. Writing t_k for the statutory output rate of sector k , and $zr_{dk} \in [0, 1]$ for the zero-rating intensity of its exports (one under the VAT, where the output rate on the export link is zero and input credits remain reclaimable; zero under the business tax, which gave

no relief; Section 6.6), the remittance of sector (d, k) to region d 's government is

$$T_{dk}^P = \underbrace{t_k R_{dk}^{\text{dom}}}_{\text{output tax on domestic sales}} - \underbrace{\sum_{o=1}^{N-1} \sum_{k'=1}^J \delta_{ok',dk} t_{k'} X_{ok',dk}^P}_{\text{input credits it reclaims}} \quad R_{dk}^{\text{dom}} \equiv R_{dk} - z r_{dk} R_{dk}^{\text{exp}}, \quad (6.26)$$

where R_{dk} is net-of-tax sales (6.18), R_{dk}^{exp} is its export component, of which the zero-rated part $z r_{dk} R_{dk}^{\text{exp}}$ leaves the output-tax base, and $X_{ok',dk}^P$ is the purchaser-price expenditure of (d, k) on inputs from (o, k') (6.21). Because credits remain reclaimable while export sales generate no output tax, a sufficiently export-oriented sector has $T_{dk}^P < 0$: its government refunds the accumulated input credit, the position described in Case 2 of the numerical illustration (Section 4). I assume throughout that refunds are honored in full. Chinese practice before 2018 refunded excess credits on exports but carried excess domestic credits forward against future liabilities rather than refunding them; a carry-forward that is never recovered would enter here as a reclaim rate below $t_{k'}$, leaving the wedge structure unchanged but raising the tax actually borne by sectors in a net-refund position (construction and six reformed services in the counterfactual). The full-refund benchmark therefore bounds those sectors' gain, and the revenue loss, from above. Crucially, the rates entering (6.26) are the *statutory output rates* $t_k, t_{k'}$, not the cascading wedge $t_{ok',dk}$; the wedge enters only through the creditable fraction $\delta_{ok',dk} \in [0, 1]$ of each input. The first term is the output tax charged on sector (d, k) 's domestic sales at its statutory rate t_k ; the second is the credit that (d, k) reclaims on its inputs, equal to the seller's rate $t_{k'}$ applied to the creditable fraction of the input expenditure $X_{ok',dk}^P$. The fraction $\delta_{ok',dk}$ is one on every fully creditable link, where the purchaser price is VAT-exclusive and the wedge is zero; zero whenever a pre-reform business-tax service sector is involved, so the embedded tax is not reclaimed and cascades through P_{dj}^M ; and, on the links the legislation left partly creditable, the statutory carve-out fraction of Appendix C.2, the wedge being $1 + (1 - \delta_{ok',dk}) t_{k'}$ in general. The credit sum runs over domestic origins only: VAT on business imports is collected at customs and credited to the registered buyer, so the two entries cancel inside the buyer's provincial budget and neither appears in T_{dk}^P . As Case 1 of the numerical illustration made clear, T_{dk}^P is generically nonzero even when all wedges vanish, because the output-tax term collects the statutory rate on a value-added base. Summing over its sectors, the tax revenue collected in—and rebated lump-sum to—region d is

$$T_d = \sum_{k=1}^J T_{dk}^P + T_d^{\text{imp}}, \quad T_d^{\text{imp}} \equiv \sum_{k=1}^J \frac{t_k}{1 + t_k} X_{Nk,d}^C \quad (6.27)$$

where T_d^{imp} is the VAT collected at the border on region d 's consumer imports, which has no domestic seller and is booked to the consuming province; household income is $I_d = \sum_{k=1}^J w_d L_{dk} + T_d + D_d$. This books each tax dollar where value is *produced*: in a chain made in one region and consumed in another, each production stage's tax stays with its own region and the region of final consumption receives nothing. A *destination*-based rule would instead assign the whole chain's tax to the consuming region; the two raise the same national total, $\sum_d T_d = \sum_d T_d^{\text{dest}}$, and differ only in the cross-region allocation, which is what the regional results of Section 8.1 measure.

Other taxes inside prices. The VAT and the business tax are not the only taxes in the price of a good. Excises and surcharges are levied on the seller and included in the price the buyer pays; tobacco is the leading case. I carry them as a seller-side factor $\chi_{ok} \in (0, 1]$: of each yuan a buyer pays for the good of (o, k) , net of the wedge, the seller keeps χ_{ok} , so the receipts in (6.22) are scaled by χ_{ok} , and the remainder is revenue of region o . The factor is calibrated as the ratio of each sector's output to its receipts in the converted table, so that rows and columns balance at baseline; its median is 0.995, and it is held fixed in the counterfactuals. The corresponding revenue, together with production taxes other than the VAT and the business tax and net of subsidies, forms the residual account of Section 6.7.

Trade (Im)balance Condition. Region o 's imports, net of its exports—both valued at purchaser (tax-inclusive) prices—equal the transfer D_o :

$$\sum_{o' \neq o} \sum_{k'=1}^J \left[\sum_{k=1}^J X_{o'k',ok}^P + X_{o'k',o}^C \right] - \sum_{o' \neq o} \sum_{k=1}^J \left[\sum_{k'=1}^J X_{ok,o'k'}^P + X_{ok,o'}^C \right] = D_o. \quad (6.28)$$

Both sides are valued at invoice prices, inclusive of all tax invoiced whether creditable or not, which is how the published interregional table records trade; in that valuation the interprovincial credit flows cancel and only the transfer D_o remains. In the model's own valuation, net of creditable tax, the same balance carries the net credit routing of the origin rule: imports less exports equal $D_o + N_o$, where N_o is the output tax region o collects on creditable sales to other provinces less the credits it refunds on creditable purchases from them, the quantity measured in Section 8.1. The baseline transfers D_o are calibrated from the household budget, and the published trade table reproduces them to within 0.13 percent of provincial income (derivation in Appendix A).

6.6 The Rest of the World

The foreign region N (ROW) is treated asymmetrically on its supply and demand sides, and the asymmetry is deliberate.

(1) ROW as a supplier (imports). ROW varieties enter every Chinese buyer's CES aggregators as ordinary Armington origins with the anchored cost $\hat{c}_{Nk} \equiv 1$: the foreign cost is the numeraire, which pins the level of domestic wages and prices. Import demand is therefore fully endogenous through the shares, $\hat{\gamma}_{Nk,dj}^P = (\hat{t}_{Nk,dj} / \hat{P}_{dj}^M)^{-\theta}$. At the border the destination principle applies: VAT on business imports is collected at customs and credited to registered buyers, so the net wedge on creditable import links is one, while consumer imports carry the consumer-facing rate.

(2) ROW as a buyer (exports): the export ledger. Baseline export revenue in ROW (numeraire) units is $\bar{R}_{dj}^x$, and the seller-side receipt entering goods-market clearing is

$$R_{dj}^{x,\text{net}} = \bar{R}_{dj}^x \left[zr_j + \frac{1 - zr_j}{1 + t_j} \right], \quad (6.29)$$

where $zr_j \in [0, 1]$ is export zero-rating, carried cell by cell as zr_{dj} where one province differs (Shanghai's pilot cells, whose exports were partly exempt; Appendix C.3): one for VAT sectors — the export-rebate institution, with refundable input credits delivering the negative-remittance position of Case 2 — and zero for pre-reform business-tax services, which had no rebate mechanism.

(3) Export demand. Foreign demand responds with the trade elasticity at fixed foreign expenditure and small Chinese shares: $R_{dj}^{x'} = \bar{R}_{dj}^x (\hat{p}_{dj}^x)^{-\theta}$ with $\hat{p}_{dj}^x$ the FOB price hat (seller cost hat times the export-tax factor relative to its baseline value), so export revenue falls with θ and quantity with $\theta + 1$, consistent with the domestic share system. This is the closure used throughout the paper. The alternative that holds export revenue fixed in ROW units, $q^x = \bar{R}^x / p^x$ (unit quantity elasticity and zero value elasticity, the fixed-foreign-expenditure closure common in regional quantitative models), makes any tax that falls on exports a costless transfer from foreigners (Lemma 1); results under it are reported in Appendix E.

(4) No foreign budget constraint. ROW's own budget is the Walras equation dropped from the system: net imports of each Chinese region are financed by the frozen nominal transfer D_o in (6.28), and ROW income is not modeled. The consequences of the frozen anchors $(\bar{R}^x, D_d)$ for counterfactuals — and the certification exercises that bound them — are documented in Appendix E.

6.7 From the Published Input–Output Table to Model Objects

The quantitative analysis is calibrated to the 2012 Chinese multi-regional input–output table (31 provinces, 41 sectors). I read the published table as valued at invoice prices: every domestic cell records the full amount paid by the buyer, and the value-added block’s net-taxes-on-production row records each sector’s net remittance. The statistical authority describes its producer price as including the VAT payable, output tax less input tax, and does not state how creditable tax is treated cell by cell, so this reading is an assumption, and the checks below are what support it: the tax row reproduces the statutory rate on value added in manufacturing, and the rows and columns of the converted table balance to within the closure factor χ , whose median is 0.995. A sharper test is available for the fixed-capital-formation cells, where the two readings differ most: if those cells excluded the tax that investors credit, the equipment-producing sectors would be out of balance by 6 percent, against 2.5 percent under the invoice reading. The model, by contrast, is cast at basic prices with explicit wedges at credit-chain breaks. This subsection defines the conversion between the two and the consistency checks it must satisfy.

What a published cell contains. Although all domestic cells look uniformly “gross,” their composition differs by the seller’s tax regime, and so does the conversion applied to them:

- (i) *VAT seller, creditable buyer.* The cell equals the seller’s net price scaled by the invoice factor, $X^{\text{IO}} = (1 + \tau_k) \times (\text{net value})$. The invoiced VAT is refunded to the buyer through the input credit, so the net value is simultaneously the seller’s receipts and the buyer’s true economic cost. *Conversion: deflate both sides by $1/(1 + \tau_k)$.*
- (ii) *VAT seller, noncreditable buyer* (final consumers; business-tax, exempt, or small-scale buyers). The same invoice factor applies, but nothing is refunded: the gross value is the buyer’s true cost, while the seller receives only the net value. *Conversion: keep the gross value on the buyer (column/demand) side; deflate by $1/(1 + \tau_k)$ on the seller (row/revenue) side.*
- (iii) *Business-tax seller, any buyer.* The tax is price-internal: the cell equals gross receipts with the BT embedded, there is no separately invoiced tax and no credit. The full cell is the buyer’s cost; the seller keeps $(1 - \tau_k^{\text{BT}})$ of it. *Conversion: keep the gross value on the buyer side; multiply by $(1 - \tau_k^{\text{BT}})$ —equivalently, deflate by $1/(1 + t_k)$ under (6.12)—on the seller side.*

- (iv) *Export column.* Exports are recorded free on board. For VAT sectors they are zero-rated, hence already at basic prices: *conversion: none*. For business-tax sectors, which had no rebate, the seller's receipt is the FOB value net of the business tax: *conversion: none on the buyer side; deflate by $1/(1 + t_k)$ on the seller side*, as in (iii).

Cases (ii) and (iii) are the credit-chain breaks: there, and only there, the buyer-side value exceeds the seller-side value, and the difference is the model's wedge revenue. In case (i) the two sides coincide after deflation and the wedge is zero.

The conversion map. Let $X_{ok,dj}^{\text{IO}}$ denote a published intermediate cell, τ_k the statutory VAT rate of sector k (zero for business-tax sectors), and $t_{ok,dj}$ the pre-reform wedge defined above. The model's purchaser-price expenditure is

$$X_{ok,dj}^P = X_{ok,dj}^{\text{IO}} \frac{1 + (1 - \delta_{ok,dj}) \tau_k}{1 + \tau_k}, \quad (6.30)$$

which strips the invoiced-and-refunded VAT from the creditable share $\delta_{ok,dj}$ of each cell and leaves the rest at its published value: a fully creditable cell is deflated by $1/(1 + \tau_k)$, a noncreditable cell is unchanged. In the 2012 baseline δ is zero or one except on three kinds of links: purchases of farm products and of freight by VAT buyers, which earned deemed credits, and Shanghai's pilot cells, which were under the VAT for part of their sales (see the paragraph on the institutions of 2012 below). Final-demand cells map to $X_{ok,d}^C$ at their published values, since final buyers do not credit; the export column maps at its published value, with the seller's receipt given by (6.29). Seller-received revenue then follows the model identity (6.18) applied to the converted objects,

$$R_{ok} = \sum_{d=1}^N \sum_{j=1}^J \frac{X_{ok,dj}^P}{1 + t_{ok,dj}} + \sum_{d=1}^N \frac{X_{ok,d}^C}{1 + t_{ok,d}} + R_{ok}^{\text{exp}}, \quad (6.31)$$

which, cell by cell, deflates VAT rows by $1/(1 + \tau_k)$ on all domestic sales and multiplies business-tax rows by $(1 - \tau_k^{BT})$, consistent with the seller-side conversions in (i)–(iii). Value added is measured from the core block only,

$$\text{VA}_{dj} = \text{labor compensation} + \text{depreciation} + \text{operating surplus}, \quad (6.32)$$

excluding net taxes on production. The calibrated objects follow: $\alpha_{dj} = (\text{VA}_{dj} - H_{dj})/R_{dj}$, where $H_{dj} = \iota_j R_{dj}$ is the in-house service cost carved out of core value added (Section 7); the shares $\gamma_{ok,dj}^P = X_{ok,dj}^P / (\sum_{o,k} X_{ok,dj}^P + H_{dj})$ and $\gamma_{H,dj}^P = H_{dj} / (\sum_{o,k} X_{ok,dj}^P + H_{dj})$, so that the

market and in-house shares of the bundle sum to one and unit hats reproduce the baseline exactly; $\gamma_{ok,d}^C = X_{ok,d}^C / I_d$ from the converted table, and the final-expenditure weights from gross final-demand cells with the consumer-facing wedge $t_{ok,d} = t_k$ carried explicitly—the tax-inclusive cells and the explicit wedge are two representations of the same object and must not be applied simultaneously.

A worked example: how one cell converts. A single pair of sectors makes the mechanics of (6.30)—and of the published table itself—concrete. Let an upstream sector U generate 50 of value added and sell to a downstream sector S that adds a further 100 and sells to final demand. Impose a uniform rate $\tau = 17\%$ on all links, so that every difference between the two panels of Table 5 is attributable to *credibility* alone. In both panels the published table records each domestic cell at its invoice-inclusive value, and the net-taxes-on-production row (proTax) records the seller’s output tax *less* the credits it reclaims.

Table 5: One intermediate cell under the two credit regimes (uniform $\tau = 17\%$)

	<u>U’s account</u>	<u>S’s account</u>	
	column / row	column (buys, owes)	row (sells)
<i>Panel A. Creditable link (U, S both VAT; $t_{US,S} = 0$)</i>			
intermediate input from U (gross cell)	—	58.5	
core value added (labor + dep. + surplus)	50	100	
net VAT (proTax) = output – credit	8.5	$25.5 - 8.5 = 17.0$	
sale (invoice-inclusive)	58.5		175.5
column total = row total	58.5	175.5	
<i>Panel B. Noncreditable link (U a business-tax/exempt seller; $t_{US,S} = \tau$)</i>			
intermediate input from U (gross cell)	—	58.5	
core value added (labor + dep. + surplus)	50	100	
net VAT (proTax) = output – credit	8.5	$26.945 - 0 = 26.945$	
sale (invoice-inclusive)	58.5		185.445
column total = row total	58.5	185.445	

Panel A (creditable). S ’s row sums to $(50 + 100) \times 1.17 = 175.5$. The crucial feature is that the 8.5 of input VAT enters the accounts *twice with opposite signs*: once embedded inside the gross input cell ($58.5 = 50 + 8.5$), and once subtracted inside the proTax row (net VAT = $25.5 - 8.5 = 17$). The two offset, the column balances at 175.5, and S remits only $100 \times 0.17 = 17$ —the tax on its *own* value added. This is why the conversion (6.30) deflates the creditable input cell, $58.5 \rightarrow 50$: the embedded 8.5 is refunded plumbing, not an economic cost, and removing it recovers both U ’s receipts and S ’s true input cost.

Panel B (noncreditable). Now the 8.5 cannot be reclaimed, so 58.5 is S 's true input cost, its net price rises to 158.5, and the row becomes $(58.5 + 100) \times 1.17 = 185.445$. The proTax row rises to 26.945, because S now remits $(58.5 + 100) \times 0.17 = 26.945$ rather than 17. The escalation decomposes cleanly:

$$\underbrace{100 \times 0.17}_{17 \text{ (own VA)}} \xrightarrow{+8.5} \underbrace{150 \times 0.17}_{25.5 \text{ (lost credit)}} \xrightarrow{+1.445} \underbrace{158.5 \times 0.17}_{26.945 \text{ (tax on tax)}}, \quad (6.33)$$

where the first step is the forgone credit (8.5) and the second is genuine cascading— S charging 17% on the 8.5 of embedded tax it folded into its price, $8.5 \times 0.17 = 1.445$. The total excess burden, $8.5 + 1.445 = 9.945$, sits entirely in the proTax row, and the conversion (6.30) correctly leaves this cell gross (58.5 stays), because here the embedded tax is a real wedge.

Two lessons carry to the calibration. First, the input cell is 58.5 in *both* panels and core value added is 100 in both; every consequence of creditability lives in the proTax row and, through the price, the output row. The upstream sector U is untouched—it remits 8.5 either way—so the cascade is entirely a property of the *break* at S . Second, the published *value-added row total* is 126.945 in Panel B (core 100 plus proTax 26.945) against 117 in Panel A, even though core value added is 100 in both. Reading α_{dj} off published value added would therefore load the tax regime into a technology parameter; measuring it from the core block (6.32) keeps α_{dj} invariant at $100/R_{dj}$, as required.

Why the converted table balances without a tax row. On the converted table the sector identity (6.24) holds in the data by construction, given a residual account: the published tax entry is recovered by (6.34) only up to production taxes other than the VAT and the business tax, subsidies, and cell-level discrepancies, which the model carries as a sector-specific residual that scales with the sector's revenue (capped at ± 50 percent of it, the excess entering income as a lump sum). That residual is 24 percent of the published entry for the VAT sectors in aggregate and 26 percent for the economy, and it is what makes the baseline exact. With it, seller-received output equals core value added plus the intermediate bill at (noncreditable-tax-inclusive) purchaser prices, with no separate tax entry. The published table's net-taxes-on-production row does not disappear from the model—it reappears, exactly, as the government revenue T_{dk}^P of (6.26), collected at the credit-chain breaks and rebated lump-sum. The conversion thus separates the two roles that the published table's tax row conflates: the portion that distorts relative prices (the wedge, surviving only at

chain breaks) and the portion that is pure collection on creditable links (the remittance, price-neutral but fiscally allocative).

Overidentification check. The conversion is disciplined by an identity it does not impose. Because the published tax entry records output VAT invoiced less input credits reclaimed, plus business and other production taxes, net of subsidies, it must satisfy, sector by sector,

$$\text{proTax}_{dk} \approx \underbrace{\tau_k \sum_{\text{dom}} \frac{X_{dk,\cdot}^{\text{IO}}}{1 + \tau_k}}_{\text{output VAT invoiced}} - \underbrace{\sum_{o \leq N-1, k'} \delta_{ok',dk} \tau_{k'} X_{ok',dk}^P}_{\text{input credits reclaimed}} + \underbrace{\tau_k^{BT} \sum_{\text{all}} X_{dk,\cdot}^{\text{IO}}}_{\text{business tax}} + \text{other}_{dk}, \quad (6.34)$$

the empirical counterpart of (6.26), where the domestic sums of the VAT terms exclude the zero-rated export column, whereas the business-tax base includes exports, which carried no rebate, and other_{dk} collects consumption and miscellaneous production taxes net of subsidies. Three features of the published table anticipate this identity. For VAT-intensive manufacturing sectors, in which other_{dk} is small, the net-VAT term implies $\text{proTax}_{dk} \approx \tau_k \times \text{VA}_{dk}$; in the 2012 table the ratio of the tax entry to value added clusters at 16–19 percent across manufacturing sectors, bracketing the statutory 17 percent. Agriculture, which is VAT-exempt and receives production subsidies, shows a slightly negative entry. Service sectors show ratios of tax to gross receipts at or below the statutory business-tax rates of 3–5 percent: construction, finance, and real estate are at or above the statutory rate, while six service sectors pay between a third and three-quarters of it, which is why their pre-reform wedges are calibrated to the effective rate (Appendix C.2). Residuals from (6.34) discipline the assignment of statutory rates $\{\tau_k, \tau_k^{BT}\}$ and the creditability classification underlying the wedge matrix; a decomposition using tax-yearbook revenue by sector and province is reported in Appendix A.

Small-scale taxpayers. The conversion in (6.30) deflates every cell from a VAT sector into another VAT sector, which treats all such transactions as creditable. In practice a fraction of buyers are small-scale taxpayers who cannot reclaim input VAT. Because special VAT invoices are required for crediting and dominate recorded inter-firm transactions, the creditable-majority assumption is the natural baseline for the intermediate cells. The exposure is not confined to the post-reform side: purchases by small-scale taxpayers from VAT sellers are booked as creditable in the baseline as well, so the always-VAT block carries the same approximation in both years, whereas every service transaction before the reform is noncreditable regardless of taxpayer status and is booked exactly; after the reform,

small-scale taxpayers remit a low levy without input credits, and their sales are creditable to buyers only at that levy rate, so the conversion overstates the crediting expansion on the small-scale slice of treated flows, biasing the measured gains upward on that slice. Two features contain the bias: the slice is small (special VAT invoices dominate recorded inter-firm transactions, and small-scale firms lie below the registration threshold by definition), and the effective-rate calibration scales post-reform rates to observed collections, which absorbs part of the small-scale remittance gap on the rate side but none of it on the crediting side. A quantitative bound—rescaling creditability on every link by one minus the small-scale share of business purchases, which the fractional crediting mask supports directly—is left to the robustness analysis.

Institutions of 2012 in the baseline. Three features of the 2012 system enter the baseline through the creditability masks rather than the rates. First, until a transport provider switched to the VAT, general VAT taxpayers could deduct a deemed 7 percent of the freight they paid it, although the provider paid the business tax and issued no VAT invoice. The deduction applied to freight on goods, not to passenger fares, postage, storage, or handling charges, so the link from business-tax transport to a crediting buyer carries, in 2012, a wedge of $(1 + t_{29})(1 - 0.07w)$, with $w = 0.71$ the freight share of the transport, storage, and postal sector's sales to crediting buyers, measured from the transport subsectors of the national input-output tables; the buyer's effective price was about 2 percent below the carrier's receipt. The credit is booked to the buyer's province. It is netted inside the published tax entry, and the table's cells are at invoice value, so the conversion lowers the buyer's input bill by the credit. The reform replaced the deemed credit by ordinary crediting at 11 percent. Second, Shanghai moved its transport and modern-service sectors to the VAT on 1 January 2012, so the published table is the equilibrium of a mixed system. I read Shanghai's pilot cells as the statutes prescribe: transport (88 percent of the sector's sales, since railways and post stayed under the business tax), information services (41 percent, since telecommunications stayed out), leasing and business services, and research and technical services sell and buy under the VAT at their statutory rates, and their exports are zero-rated or exempt as described in Appendix C.3. I then solve the model backward from this table to the economy in which these cells are back under the business tax, with the in-house shares of the unreformed economy set to their measured 2011 values, and I use that economy as the baseline. Applying the pilot to the rebuilt baseline returns the published table to within 10^{-6} , and the welfare effect of the reform computed from the rebuilt baseline equals the ratio of two solves from the published table. The pilot itself raises Shanghai's welfare by less than 0.1 percent. The other eight pilot provinces adopted

between September and December 2012 and are treated as unreformed for the year. Third, public administration has no taxable output and earns no input credits before or after the reform. Final demand, including fixed capital formation, is taxed as consumption in both periods: the model has no capital, and a uniform tax on final demand is the static counterpart of a consumption-type VAT. If investment purchases that the law makes creditable are instead credited in the ledger, the welfare gain is 1.49 percent and the real revenue ratio falls from 0.89 to 0.75.

6.8 Equilibrium

I now formally define the equilibrium under policies $\{t_k, t_{ok,dj}\}$ in this model.

Equilibrium in Levels: Definition 1. Given tax rates $(t_k, t_{ok,dj})$, L_d , trade costs $(\kappa_{ok,dj}, \kappa_{ok,d})$, and foreign costs c_{Nk} , which are the numeraire (Section 6.6), an equilibrium is a wage vector $\mathbf{w}$ and price indices $\mathbf{P}$ that satisfy producers' cost minimization and zero-profit pricing, households' optimization, goods and labor market clearing, and the trade (im)balance condition.

Equilibrium in Relative Changes: Definition 2 . Instead of solving for an equilibrium under the policy before the tax reform, I solve for changes in prices and wages after the tax reform is completed, which I define as an equilibrium in relative changes. In trade literature, "exact hat algebra" is used as an approach to analyze equilibrium adjustments in response to policy changes, such as tax reforms, by focusing on relative changes from a baseline. Instead of recalculating absolute values, variables are expressed in terms of percentage changes or "hats" denoted as $\hat{x} = x'/x$, where x' represents the variable post-reform and x the baseline. This simplifies complex model computations and isolates the impact of specific policy changes on trade flows, costs, and productivity. Specifically, I can match exactly the model to the data in a base year. Then I can identify the effect on equilibrium outcomes from a pure change in cross-sector tax change $t_{ok,dj}$ and consumer-facing tax change $t_{ok,d}$ and can solve for the general equilibrium of the model without needing to estimate parameters that are difficult to identify in the data, such as the in-house productivities z_{dj}^H and the iceberg trade costs $\kappa_{ok,dj}, \kappa_{ok,d}$.

6.9 Counterfactual Equilibrium under the New Tax Structure

I now define the equilibrium of the model in the new tax structure relative to the old one. For the tax wedges I adopt the convention

$$\hat{t}_{ok,dj} \equiv \frac{1 + t'_{ok,dj}}{1 + t_{ok,dj}}, \quad \hat{t}_{ok,d} \equiv \frac{1 + t'_{ok,d}}{1 + t_{ok,d}},$$

so that $\hat{t}$ denotes the relative change in the *gross* tax wedge.

Cost Bundle of Inputs. The cost bundle of inputs becomes:

$$\hat{c}_{dj} = (\hat{w}_d)^{\alpha_{dj}} \left[\hat{P}_{dj}^M \right]^{1-\alpha_{dj}},$$

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$$\hat{p}_{ok,dj} = \hat{c}_{ok} \hat{t}_{ok,dj} \quad (6.35)$$

and

$$\hat{p}_{ok,d} = \hat{c}_{ok} \hat{t}_{ok,d} \quad (6.36)$$

Price Index of Intermediate Inputs. The price index of the intermediate input (based on Equation 6.14) becomes:

$$\hat{P}_{dj}^M = \left[\sum_{o=1}^N \sum_{k=1}^J \gamma_{ok,dj}^P (\hat{c}_{ok} \hat{t}_{ok,dj})^{-\theta} + \gamma_{H,dj}^P (\hat{w}_d)^{-\theta} \right]^{-\frac{1}{\theta}}. \quad (6.37)$$

Assuming that in-house productivities remain unchanged before and after the reform, only the in-house share $\gamma_{H,dj}^P$ is required; its measurement is discussed in Section 7.

Final Consumer Price Index.

$$\hat{P}_d = \left[\sum_{o=1}^N \sum_{k=1}^J \gamma_{ok,d}^C (\hat{t}_{ok,d} \hat{c}_{ok})^{-\theta} \right]^{-\frac{1}{\theta}}. \quad (6.38)$$

The change in the price index works through the baseline shares and the wedge-inclusive cost terms alone: this is the standard hat-algebra system of Caliendo et al., 2022a augmented with the in-house margin and the tax ledger.

Trade Shares. Bilateral trade share in intermediate good expenditure:

$$\hat{\gamma}_{ok,dj}^P = \left[\frac{\hat{t}_{ok,dj} \hat{c}_{ok}}{\hat{P}_{dj}^M} \right]^{-\theta}. \quad (6.39)$$

Bilateral trade share in final consumption expenditure:

$$\hat{\gamma}_{ok,d}^C = \left[\frac{\hat{t}_{ok,d} \hat{c}_{ok}}{\hat{P}_d} \right]^{-\theta}. \quad (6.40)$$

Market Clearing. Denote the counterfactual purchaser-price expenditures by $(X_{ok,dj}^P)' \equiv (\gamma_{ok,dj}^P)'(1 - \alpha_{dj})R'_{dj}$ and $(X_{ok,d}^C)' \equiv (\gamma_{ok,d}^C)'I'_d$, with $(\gamma_{ok,dj}^P)' = \gamma_{ok,dj}^P \hat{\gamma}_{ok,dj}^P$ and $(\gamma_{ok,d}^C)' = \gamma_{ok,d}^C \hat{\gamma}_{ok,d}^C$. The counterfactual goods market clearing, labor market clearing, tax revenue and income are:

$$R'_{ok} = \sum_{d=1}^{N-1} \sum_{j=1}^J \frac{(X_{ok,dj}^P)'}{1 + t'_{ok,dj}} + \sum_{d=1}^{N-1} \frac{(X_{ok,d}^C)'}{1 + t'_{ok,d}} + R_{ok}^{x,net'}, \quad (6.41)$$

where the domestic sums run over the Chinese provinces and the last term is the counterfactual export receipt of Section 6.6, equation (6.29) evaluated at the priced foreign demand: the rest of the world has no budget in the model, so its purchases are not derived from an income.

$$\hat{w}_o w_o L_o = \sum_{k=1}^J \left[\alpha_{ok} + (\gamma_{H,ok}^P)'(1 - \alpha_{ok}) \right] R'_{ok}, \quad (\gamma_{H,ok}^P)' = \gamma_{H,ok}^P \left(\frac{\hat{w}_o}{\hat{P}_{ok}^M} \right)^{-\theta}, \quad (6.42)$$

The tax revenue rebated to region d follows the origin rule (6.26)–(6.27) evaluated at the new tax rates, and income is

$$I'_d = \hat{w}_d \sum_{k=1}^J w_d L_{dk} + T'_d + D_d, \quad T'_d = \sum_{k=1}^J \left[t'_k (R_{dk}^{\text{dom}})' - \sum_{o=1}^{N-1} \sum_{k'=1}^J \delta'_{ok',dk} t'_{k'} (X_{ok',dk}^P)' \right] + (T_d^{\text{imp}})', \quad (6.43)$$

which is (6.26)–(6.27) at the new rates t' , creditable shares δ' , and zero-rating intensities zr' : $(R_{dk}^{\text{dom}})' = R'_{dk} - zr'_{dk} (R_{dk}^{\text{exp}})'$ is counterfactual net-of-tax sales less zero-rated exports, the credit sum runs over domestic origins, and $(T_d^{\text{imp}})'$ is the border tax on consumer imports. The residual account and the factor χ are held at their baseline values. T'_d can be negative for a region in a net-refund position, in which case its government pays out the excess credit.

Welfare. The change in welfare of region d is $\hat{U}_d = \hat{I}_d / \hat{P}_d$.

7 Estimation and Calibration of Model Parameters

7.1 Trade Elasticity

The trade elasticity θ — under perfect competition, both the elasticity of substitution minus one and the elasticity of bilateral flows with respect to the gross tax wedge (Section 6) — is calibrated to $\theta = 4.5$, the midpoint of the 3–6 range established by the international trade literature: Simonovska and Waugh (2014) estimate 4.1, the meta-analysis of Head and Mayer (2014) reports a median of 3.8 across structural gravity estimates and prefers 5.0 from tariff variation, and the aggregate estimate of Caliendo and Parro (2015) is 4.5. I deliberately do not estimate θ from the reform window itself: the 2012–2017 input–output variation identifies the wedge response only jointly with the secular deepening of service inputs, which loads onto precisely the treated sector pairs, and the 2017 table sits at a short-run horizon. Appendix C.1 reports the reform-window wedge specification — directionally consistent, an order of magnitude below the calibrated long-run elasticity — together with this interpretation; robustness of the quantitative results across $\theta \in [3, 6]$ is reported in Appendix C.2.

7.2 Calibration of Other Model Parameters

I use the 2012 Inter-regional Input-Output Table (IRIO) of Li et al., 2023 to calibrate the model. The table is available every five years, and 2012 is the last one before the reform. It is converted to the model’s valuation as described in Section 6.7, and the baseline of every counterfactual is the unreformed 2012 economy obtained from it by undoing Shanghai’s pilot (Section 6.7). As in Caliendo et al., 2020, conditioning on the observed baseline allocation (trade shares, expenditures, and value added) and solving the model in relative changes means that iceberg trade costs, productivities, and the productivity of in-house service tasks never need to be estimated: they are absorbed by the baseline shares. Table 6 provides a summary of the parameters calibrated, their definitions and sources in the model.

The intermediate good trade share ($\gamma_{ok,dj}^P$) and final consumption good trade share ($\gamma_{ok,d}^C$) are based on bilateral trade flows reported in the IRIO 2012. The intermediate input expenditure share ($\xi_{ok,dj}$) and the region-sector specific value-added share (ζ_{dj}^0) are derived

Table 6: Summary of Model Parameter Calibration

Parameter	Definition	Source
$\gamma_{ok,dj}^P$	Intermediate good trade share in 2012	IRIO 2012
$\gamma_{ok,d}^C$	Final consumption good trade share in 2012	IRIO 2012
$\xi_{ok,dj}$	Intermediate input expenditure share in 2012	IRIO 2012
ζ_{dj}^0	Region-sector specific value-added share in 2012	IRIO 2012
$X_{ok,dj}^P, X_{ok,d}^C$	Expenditure on intermediate good and final good	IRIO 2012
$w_d L_{dj}$	Region-sector specific value-added in 2012	IRIO 2012
t_k	Statutory rates, 2012 and 2017	State Council and Ministry of Finance circulars
	Effective business-tax rates, 2012	Tax row of IRIO 2012 over net output
$\delta_{ok,dj}$	Creditable share of each link, 2012 and 2017	Statutes; sub-industry shares from the national table
zr_{dj}	Zero-rating intensity of exports	Statutes (Appendix C.3)
D_d	Transfers	Residual of the household budget, IRIO 2012
χ_{ok}	Other taxes inside prices	Row-column balance of the converted table
θ	Trade elasticity	Calibrated to 4.5 (Section 7)
ι_j	In-house service share of gross cost, 2011	Tax survey (below)

Notes: Parameters and their sources. The baseline is the unreformed 2012 economy: the published table with Shanghai's 2012 pilot undone by a backward solve of the model (Section 6.7). Sub-industry weights (the freight share of transport, the coverage of Shanghai's pilot, and the shares that enter blended sector rates and creditable shares) are measured from the national input-output tables with 139 sectors (2012) and 149 sectors (2017).

from the IRIO 2012, capturing the share of expenditure on intermediate goods and the value-added component in each region-sector pair. The cross-sector tax rates ($t_{k,j}$) are net ad valorem rates reflecting the differential tax treatment observed before and after the Business to Value-Added Tax (B2V) reform, obtained from official government data; the corresponding gross wedges ($1 + t_{k,j}$) enter the model equations. The same official rate schedule (www.gov.cn) also provides the statutory output rate t_k of each sector—the VAT rate for manufacturing (13% or 17% in 2012) and the business-tax rate for services before the reform, and the harmonized VAT schedule afterward. This statutory rate is the primitive underlying both objects: the cross-sector wedge is derived from it as $t_{ok,dj} = (1 - \delta_{ok,dj}) t_k$, zero when the transaction is fully creditable and t_k when it is not creditable at all (6.13), and t_k is applied to sellers' sales to compute the production-stage remittances T_{dk}^P in (6.26), which the origin-based revenue rule aggregates spatially into each region's rebated tax revenue (6.27). The expenditure on intermediate ($X_{ok,dj}^P$) and final goods ($X_{ok,d}^C$) in the base year are used to match the observed trade flows in the model. The wage and labor input ($w_d L_{dj}$) are calibrated based on region-sector specific data, accounting for value-added

contributions from labor across different regions and sectors. These objects replicate the baseline exactly.

The central experiment does not use the statutory rates as they stand. In six service sectors the tax actually paid in 2012 is between a third and three-quarters of the statutory business-tax rate times net output, because of exemption thresholds, exempt items, and incomplete compliance. A tax that was never charged did not cascade, so for these sectors I set the pre-reform wedge to the effective rate (the lower of the statutory and the effective rate in every business-tax sector), and I scale each sector's post-reform rate by the same ratio, which assumes that compliance did not change with the reform. The alternative with statutory rates after the reform is reported in Appendix C.2.

The in-house share: measured, not calibrated. One parameter cannot be read from the input-output table: the share of service tasks that firms produce inside the boundary of the firm before the reform, ι , which governs the make-or-buy margin at the center of the model. I do not calibrate it to a moment of the response to the reform: the responses that are robust to the secular growth of the service sector vary little with ι in the model, and those that vary with ι are contaminated by that growth. I *measure* it directly from the pre-reform cross-section of the tax survey. The in-house production of service tasks has an accounting location: administrative expense, the income-statement line covering the departments that organize and manage production, net of the purchased professional fees booked there (identified by the survey's own itemized outsourcing schedule, Section 5.2.3; firms that fold administrative expense into operating expense and report zero are 1.5 percent of firms, the purchased-fee correction is 6.4 percent of the aggregate, and the income-statement and tax-filing versions of the line agree with correlation 0.976). The measured object is a sector *vector*: for each of the model's sectors, in-house service cost over gross cost (operating cost plus the in-house cost itself), from 279,000 manufacturing and 340,000 non-manufacturing firm-year observations in 2011, the last pre-reform year.

The conceptual bridge deserves to be stated plainly: the administrative departments *are* the in-house service divisions. What sits inside a manufacturer's administrative apparatus — accounting, payroll and human resources, legal and compliance, information technology, planning, procurement, training — is task-for-task the output of the business-service industries the reform de-cascaded, and when a firm dissolves an internal function and contracts it out, nothing about the task changes: the expense migrates from administrative expense (its staff's compensation) to a purchased-service fee. The make-or-buy decision therefore has a ledger address, which is what makes the margin measurable at all. This is not a measurement convenience but the Coase, 1937 definition of the firm — the domain

where coordination by management supersedes coordination by market prices — with the administrative department as that coordination’s physical embodiment (Williamson, 1971); and it is what integrated firms are observed to actually transmit across their boundaries: not goods but intangible, managerial, and organizational inputs (Atalay et al., 2014). In the model the mapping is then literal — in-house provision is labor producing service tasks, carved out of value added, and administrative expense is overwhelmingly the compensation of exactly that labor. A direct content audit supports this: the survey itemizes enough of the envelope to decompose it, and in 2011 manufacturing the identifiable non-task residue — entertainment expense (4.6 percent of the line) and the four small taxes booked there under the accounting standards of the period (≤ 3.4 percent) — totals about 8 percent, purchased professional fees (6.4 percent) are netted out by construction, and expensed research and development (10.7 percent) is itself in-house production of a service task, leaving roughly three-quarters of the envelope as administrative personnel and office operations; consistently, the firm-level correlation between the administrative-expense share and the employee-compensation share is 0.72 (audit in `06j_GlfyContent_Audit.do`). The known edges of the mapping all cut in one direction: some admin content is not outsourceable-task labor (office depreciation, board expenses), some in-house service tasks sit outside the line (a factory’s own drivers and repair crews are booked in operating cost), and selling expense — which contains outsourceable marketing tasks — is excluded; the second and third make the measured ι a lower bound on the in-house margin.

Figure 4 reports the distribution. In-house service intensity is highest exactly where the model’s mechanism predicts the reform bites hardest: service industries themselves (accommodation and catering 0.33, culture 0.17, scientific research 0.16, information 0.13, leasing and business services 0.12, finance 0.11) and knowledge-intensive manufacturing (instruments 0.079, special machinery 0.078), and lowest in commodity processing (metal smelting 0.019, petroleum 0.023, wholesale and retail 0.016). The model takes the full vector — the in-house row weight is sector-specific — and the aggregate, weighted by the model’s intermediate spending, is $\bar{\iota} = 0.056$. A units bridge connects the measured object to the model’s accounting: in-house provision is labor carved out of the sector’s core value added, so the measured cost share ι_j corresponds to the product $\zeta_{dj}^0 \times$ (the in-house fraction of value added), and the model assigns each cell the in-house value-added fraction ι_j / ζ_{dj}^0 ; equivalently, with $H_{dj} = \iota_j R_{dj}$, the direct labor share is $\alpha_{dj} = \zeta_{dj}^0 - \iota_j$ and the in-house share of the intermediate bundle is $\gamma_{H,dj}^P = \iota_j / (1 - \zeta_{dj}^0 + \iota_j)$, the market shares being renormalized to $1 - \gamma_{H,dj}^P$ (Section 6.7) — in aggregate, in-house service tasks absorb 22 percent of core value added, consistent with the 25–60 percent of manufacturing employment found in service support functions in OECD economies (Miroudot & Cadestin, 2017). Two independent

checks corroborate the measurement. First, a benchmark-gap calculation sharing no data or assumptions with it — the shortfall of China’s 2012 market service-input shares relative to the United States’ at China’s industry weights, the de-cascaded allocation the model’s counterfactual converges toward — yields an aggregate of 0.049, and correlates at 0.53 with the measured vector across manufacturing industries once electronics, whose United States counterpart is structurally different, is excluded. Second, the same administrative-expense field *responds* to the reform in the direction and timing the margin requires: it falls on pilot-province crediting timing while itemized purchases rise in near-equal magnitude (Figure 3). Two summary statistics deserve emphasis (Figure 5 plots the firm-level distributions): the unweighted firm-level mean of the measured distribution is 0.10, and recorded in-house provision dwarfs recorded market procurement — 9.7 percent of costs against 1.4 percent in the itemized purchased-service categories, a seven-to-one ratio at reported values. The firm-level make share should be read more cautiously than the in-house share itself: half of manufacturers report no purchases in the itemized categories in 2011, but that mass mixes genuinely internal provision with unrecorded purchases — the schedule covers seven categories rather than all services, it is filed with no tax incentive to document purchases before crediting existed, and the zeros concentrate among the smallest firms, where item non-response is most plausible. The value-weighted contrast, anchored by large firms that do file the schedule, is the robust statement: before the reform removed the cascading penalty on market procurement, in-house provision was the dominant organizational form for service tasks.

The quantitative results below use the measured vector. The welfare consequences of the make-or-buy margin, including the counterfactual in which the margin is shut down altogether and the sensitivity of the headline result to the level of the in-house share, are reported in Section 8.1 (Figure 9).

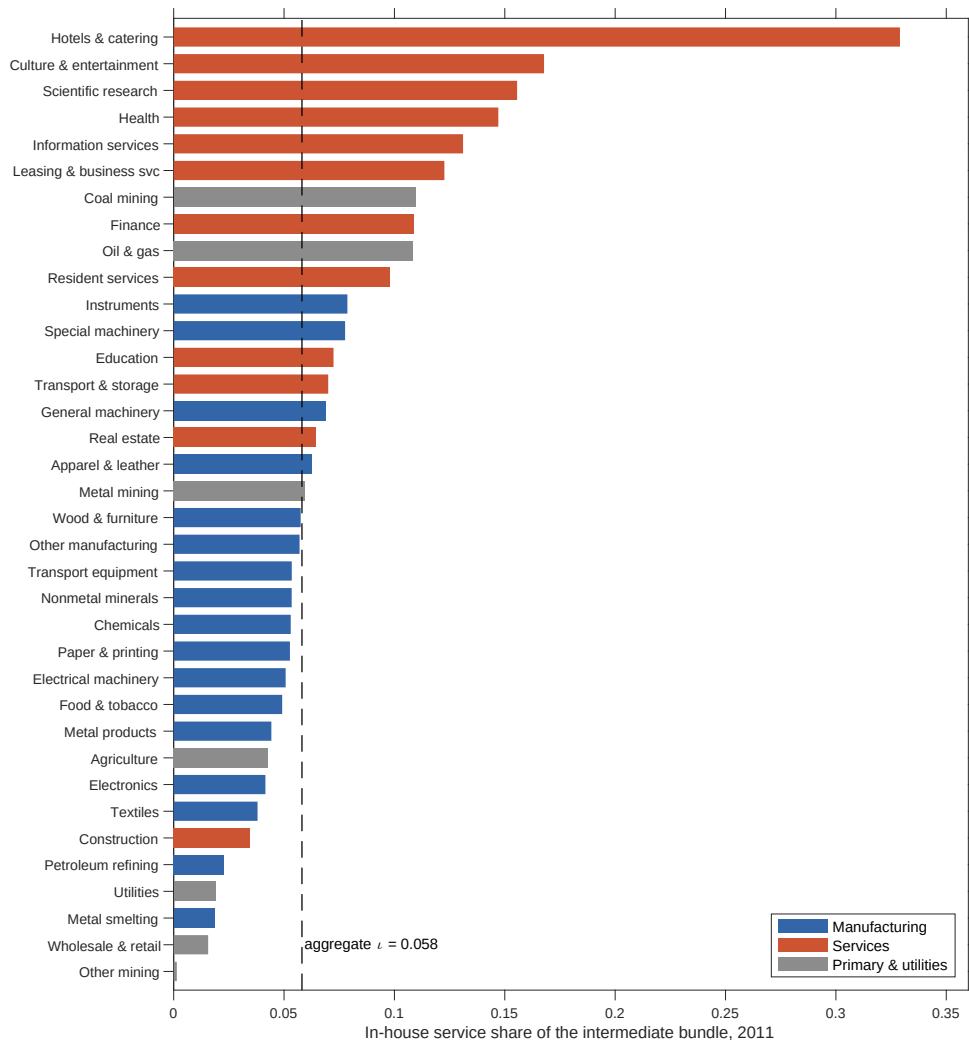
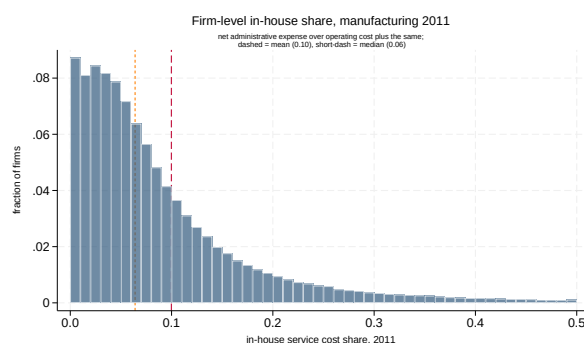
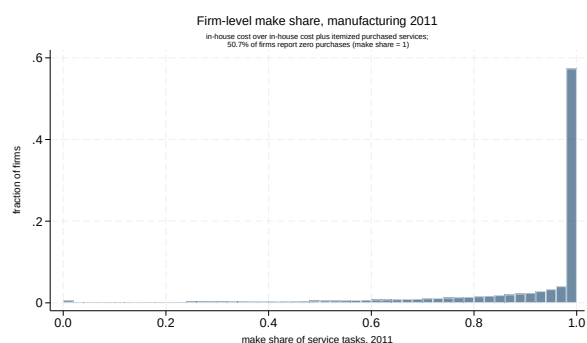


Figure 4: The measured in-house service share, by sector

Notes: In-house service cost as a share of gross cost by sector (the axis label of the figure calls it a share of the intermediate bundle), measured from the 2011 wave of the tax survey: administrative expense net of itemized purchased professional fees, over operating cost plus the same, aggregated within sector at reported values. The dashed line is the aggregate under the model's intermediate-spending weights ($\bar{\iota} = 0.056$). Unmeasured sectors (gas and water utilities, repair, sundry business services) inherit their group's value in the model; the non-market sector is excluded. The underlying variable is defined in Section 5.2.3; the sector vector is `iota_io41_vector_2011.csv` in the replication files.



(a) In-house cost share



(b) Make share of service tasks

Figure 5: The firm-level distributions behind the measured in-house share

Notes: Manufacturing firms in the 2011 tax-survey wave. Panel (a): the firm-level in-house service cost share — net administrative expense over operating cost plus the same, winsorized at $1/99$ — with mean 0.10 (dashed) and median 0.06 (short-dash); $N = 278,948$; the display is truncated at 0.5. Panel (b): the firm-level make share — net administrative expense over the same plus the seven itemized purchased-service categories, defined where the denominator is positive; $N = 275,549$. The distribution has a mass point at one: 50.7 percent of manufacturers report zero purchases in the itemized categories, a mass that mixes genuinely internal provision with unrecorded purchases (the schedule covers seven service categories, and pre-reform filings carried no credit incentive to document purchases); the zero share falls from 75 percent in the smallest size quartile to 31 percent in the largest. The value-weighted aggregate make share is 0.73 and the firm mean 0.89. Generated by 06h_IotaFirm_Distributions.do.

8 Quantification Results

In this section, I evaluate the welfare effects of switching from the turnover tax to the value-added tax system. The model is calibrated to $N = 32$ regions (31 provinces and a constructed rest of the world) and $J = 41$ sectors. Inter-regional and sectoral trade flows come from the Interregional Input-output table, converted to the basic prices as described in Section 6.7. Statutory business-tax and VAT rates by sector and year are compiled from Ministry of Finance and State Council circulars, and adjusted to the rates effectively paid in the sectors where the two differ (Section 7). The sectoral in-house service shares are measured from the 2011 tax survey and the trade elasticity is set to $\theta = 4.5$ (Section 7). I solve the model in exact hat algebra following Section 6. The calibration strategy ensures that the model exactly replicates the 2012 interregional table, including existing trade imbalances across provinces. That table is not quite a pre-reform economy, because Shanghai moved its transport and modern-service sectors to the value-added tax on 1 January 2012. I therefore first solve the model backward from the observed table, returning Shanghai's pilot sectors to the business tax, and take the resulting economy, in which every province is unreformed, as the baseline for all counterfactuals (Section 6.7 and Appendix C.3). The transfers that finance these imbalances are held fixed in the counterfactual analysis, so that changes in tax structure alter the composition and efficiency of trade flows, and the volume of exports, which responds to producer prices with elasticity θ (Section 6.6), but not the transfers themselves.

I quantify the effects of the tax reform by performing five counterfactual exercises that inform the impact of the reform itself and also the structural difference between turnover tax and value-added tax system:

1. First, I introduce into the model the change in tax structure from 2012 to 2017: both the new rate schedule and the extension of input crediting to the reformed sectors, holding the tax treatment of the rest of the world fixed; wages, costs, prices, and trade flows adjust in general equilibrium. This exercise measures the effect of the reform as legislated.
2. In the second exercise, I hold the reform fixed and vary the assignment of tax revenue across provinces. Under the origin-based rule in force, each province keeps what its firms remit and refunds the credits that its buyers claim; I compare this rule with a pooled rebate that holds every province harmless at its baseline collections and shares the aggregate change in proportion to baseline tax revenue. This exercise measures the fiscal incidence of the reform across provinces.

3. In the third exercise, I repeat the first with the make-or-buy margin shut down: I set the in-house service share to zero in every sector, so that firms cannot substitute between purchased and in-house services, and introduce the same change in tax structure. Comparing this exercise with the first measures how much of the effect of the reform operates through vertical integration, and how much would be missed by a model that ignored it.
4. In the fourth exercise, I introduce the two components of the reform separately: the change in crediting holding rates at their 2012 levels, and the change in rates holding crediting at its 2012 coverage. Because the two components interact, I report their contributions averaged over both orderings, and I compare the reform as legislated with a benchmark in which the credit chain of the reformed sectors is complete (full crediting for all sectors). I also separate what the reform owes to raising less revenue from what it owes to how the revenue is raised, by comparing it with a cut in business-tax rates that gives up the same revenue and with a version of the reform whose rates are scaled up until no revenue is lost.
5. In the fifth exercise, I step outside the historical reform and compare two uniform tax systems, a value-added tax with full crediting and a turnover tax without crediting, under the same treatment of exports and at matched levels of fiscal capacity, tracing each system as its rate is scaled up. That a value-added tax dominates a turnover tax is a theorem; this exercise puts a magnitude on it in the calibrated economy, at any given revenue requirement, and shows how that magnitude grows with the revenue the government needs to raise.

Note that welfare in a province is measured by the change in its real income, $\hat{U}_d = \hat{I}_d / \hat{P}_d$. The national measure is aggregate real income: the sum of the provincial gains in yuan over national income, which is the income-share-weighted mean of the provincial changes. It gives every yuan the same weight, it is the standard aggregation in quantitative trade models with immobile groups (Galle et al., 2023), and a transfer between provinces leaves it unchanged, so it separates efficiency from the distribution across provinces, which I report province by province. Fiscal capacity is measured by tax revenue as a share of national income, relative to its 2012 value. Finally, the comparison of tax architectures depends on how the rest of the world is closed. A value-added tax with zero-rated exports and border-adjusted imports is neutral with respect to trade, whereas a turnover tax embeds tax in the cost of exports and therefore acts, in part, as an export tax; when foreign demand is less than perfectly elastic, a small export tax transfers income from foreigners to the home country. I therefore let foreign demand respond to export prices with the trade

elasticity in all exercises (Section 6.6), so that this transfer is finite. Appendix E reports the closed-economy version of the fifth exercise, in which the transfer is absent and the ranking of the two architectures is a theorem, together with the results under the alternative closure that holds export revenue fixed.

8.1 Headline Reform and Fiscal Federalism

The reform raises welfare, measured as aggregate real income, by 1.30 percent. The real wage rises by 3.3 percent, real tax revenue falls by 10.7 percent, and export revenue rises by 5.7 percent. In the aggregate, households gain more from higher wages and lower prices than they lose from the fall in tax revenue.

Figure 6 shows how these gains are distributed across provinces. Welfare rises in 29 of the 31 provinces, which are home to 98.2 percent of the population, and the gains range up to 3.8 percent in Shaanxi. Two provinces lose: Beijing (−3.0 percent) and Tibet (−1.5 percent). This is surprising, because these are among the provinces most specialized in the sectors the reform was designed to help. The reformed sectors account for 61 percent of value added in Beijing and 49 percent in Tibet, against 32 percent nationally. Their producers do benefit: the real wage rises by 3.7 percent in Beijing and 0.9 percent in Tibet, and it rises in every other province as well. Shanghai, where the reformed sectors are 45 percent of value added, does not lose: its real wage rises by 3.8 percent and its welfare by 1.2 percent. The difference between Beijing and Shanghai is the subject of the rest of this subsection.

The explanation is on the revenue side. In the model, the income of a province is the sum of its wage income, the tax revenue collected in the province, which is rebated to its households, and a transfer that is held fixed.¹³ Consider first a province that did not trade with the others. A reform that removes a cascading tax would reduce its revenue and raise its wages. Its households would receive a smaller rebate, but they would gain more as workers and consumers, because the tax that is removed is distorting: when the whole country is treated as one closed economy, the same reform raises welfare by 1.6 percent

¹³In practice the value-added tax is shared between the central and provincial governments. The provincial share was 25 percent until April 2016 and has been 50 percent since May 2016, when the State Council raised it to compensate provinces for the loss of the business tax, which had accrued almost entirely to them (State Council of the People's Republic of China, 2016); the 50 percent share was confirmed in 2019 (State Council of the People's Republic of China, 2019). The provincial share is assigned according to where the tax is paid, so the origin principle studied here governs it directly. The model abstracts from the central share and rebates all revenue collected in a province to that province's households. The provincial revenue losses reported below are therefore the incidence of origin-based assignment on all revenue collected in a province, of which the provincial budget bears its statutory share.

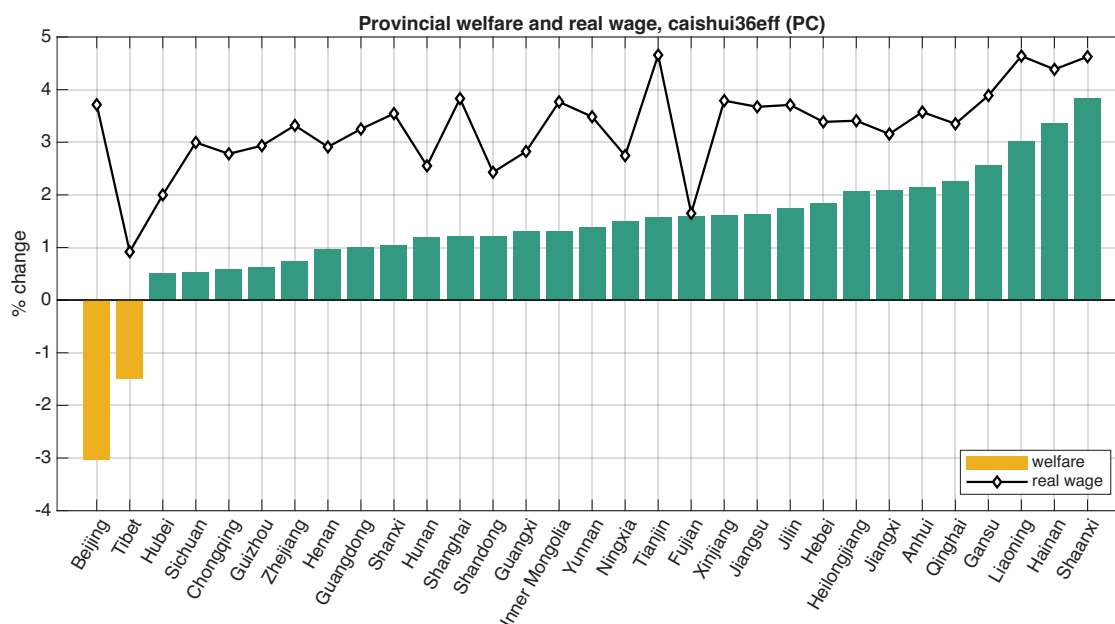


Figure 6: Provincial welfare effects of the reform as legislated

Notes: Bars show each province's welfare change (real income, $\hat{U}_d = \hat{I}_d / \hat{P}_d$) under the reform as legislated, at effective rates, with provinces ordered by gain; the line shows the change in the provincial real wage. Aggregate real income rises by +1.30%; the real wage rises by +3.2% on average (weighted by labor income) and in all 31 provinces. The baseline is the unreformed 2012 economy, with Shanghai's 2012 pilot undone. Trade elasticity $\theta = 4.5$; in-house service shares measured from the 2011 tax survey (Figure 4); destination border ledger; foreign demand priced with elasticity θ .

(Appendix E). The main way a province can lose is therefore that part of the revenue it gives up goes to residents of other provinces. Panel (a) of Figure 7 shows where revenue falls the most. Real tax revenue falls in 30 of 31 provinces, but it falls by 38 percent in Beijing and 76 percent in Tibet, against 10.7 percent nationally and 9 percent in Shanghai.¹⁴ Measured against provincial income, the loss of revenue is 6.1 percent in Beijing, compared with 1.9 percent in Shanghai and about 1.4 percent in the average province.

Why do these provinces lose so much revenue? The answer lies in how the value-added tax assigns revenue across provinces. The tax is collected at origin: when a firm sells, its own province keeps the output tax. The buyer then credits that tax against its own liability, which reduces what the *buyer's* province collects. When seller and buyer are in the same province, the two entries cancel within one budget. When they are in different provinces, they do not: the buyer's province gives up revenue on account of a tax that the seller's province collected and kept. Case 4 of the two-firm example in Section 4 shows this transfer in its simplest form. Before the reform, service and construction firms could not credit the value-added tax on the goods they bought, so their provinces gave up nothing.

¹⁴The figure for Tibet is large because its tax base is small: baseline tax revenue is 3.5 percent of provincial income, the lowest in the country. Tibet accounts for 0.2 percent of national income and does not affect any aggregate.

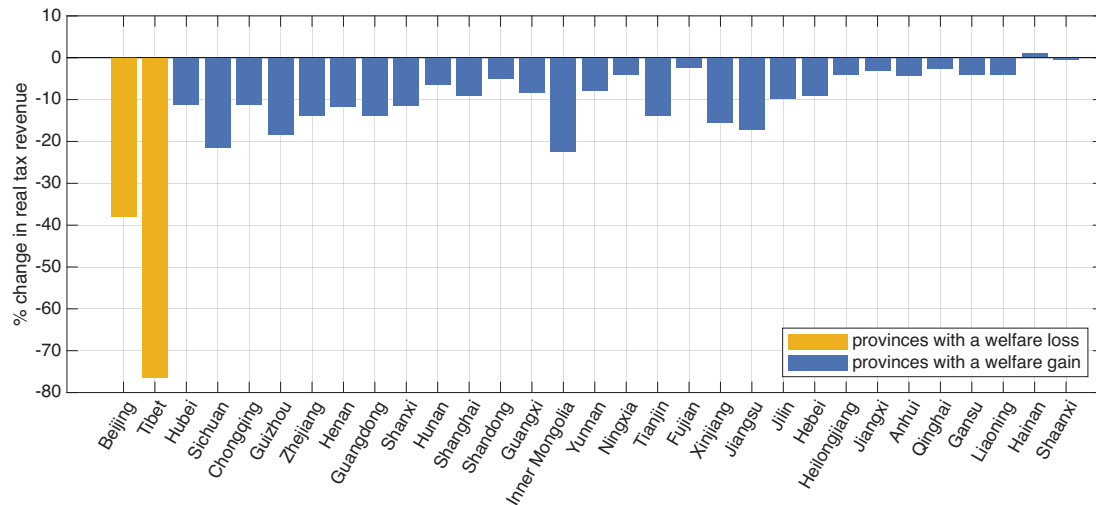
The reform switches these credits on. A province pays more, the larger its reformed sectors and the more of their inputs they buy from other provinces. The gap in rates adds to this. Most reformed services are taxed at 6 or 11 percent, while the goods they buy carry 13 or 17 percent, so the credit is large relative to the output tax that the service firm's province collects; in the limit, the firm is in the net-refund position of Case 2 in [Section 4](#).

Panel (b) of [Figure 7](#) measures these transfers. For each province, I compute the credits that other provinces grant on its sales, minus the credits that it grants on its purchases from other provinces, as a share of its baseline tax revenue. By construction, these transfers sum to zero across provinces in yuan; the shares shown sum to zero once weighted by provincial baseline revenue. In total, 17.3 percent of national baseline tax revenue moves between provincial budgets in this way. Beijing is a net payer of 31 percent of its baseline revenue, Shanghai of 4 percent, and Tibet of more than its entire baseline revenue. The net recipients are provinces that produce goods and sell them to the rest of the country, such as Hebei and Henan (+13 percent each) and Liaoning (+10 percent). Across provinces, the correlation between the change in real revenue in panel (a) and the net transfer in panel (b) is 0.89: the interprovincial transfers account for most of the variation in revenue losses.

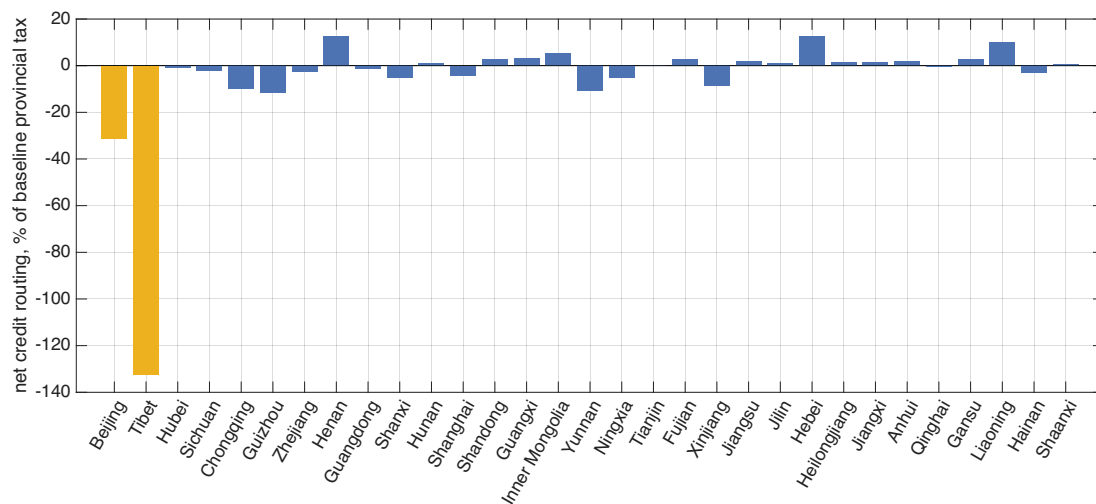
This is why the size of a province's revenue loss does not determine whether it gains. Jiangsu and Zhejiang lose 2.3 and 2.8 percent of income in tax revenue, as much as Tibet (2.7 percent), and real revenue falls by about a fifth in Sichuan and Inner Mongolia. Welfare rises in all four, because the revenue they give up goes mainly to their own firms and households; their net position in panel (b) is small. Beijing and Tibet are the provinces in which a large part of the revenue given up is a transfer to other provinces: 5.0 and 4.7 percent of provincial income. Shanghai shows that specialization in services is not enough to produce a loss. Its reformed sectors buy a much smaller share of their inputs from other provinces than Beijing's do: 36 percent of purchased inputs against 56 percent, and 49 percent of purchased goods against 82 percent, the highest share of any province. Goods bought from other provinces are 26 percent of the output of Beijing's reformed sectors and 12 percent of Shanghai's. Shanghai's net transfer to other provinces is 0.8 percent of income, and the gains to its workers and consumers exceed it.¹⁵

These losses are a matter of incidence and not of efficiency. At the national level the credits cancel, and the reform raises welfare. What differs across provinces is which budget bears the cost of the credits. The benefit of the credits, cheaper services and construction

¹⁵A regression across the 31 provinces confirms this. I split each province's revenue change, as a share of its income, into the net interprovincial transfer of panel (b) and the remainder. The coefficient of the provincial welfare change on the transfer is 1.00 (standard error 0.09), and the province's wage and price gains do not respond to the transfer; the coefficient on the remainder is 0.69 (0.12), and the two components explain 80 percent of the variation in welfare across provinces.



(a) Change in real tax revenue



(b) Net interprovincial credit routing under origin-based collection

Figure 7: Provincial tax revenue and interprovincial credit routing under the reform as legislated

Notes: Provinces are in the same order as in Figure 6 (by welfare gain, lowest to highest) in both panels. Yellow bars are the two provinces whose welfare falls (Beijing and Tibet); blue bars are the provinces whose welfare rises. Panel (a): the change in each province's real tax revenue, that is, collections under the origin-based rule deflated by the provincial consumer price index, relative to 2012. National real revenue falls by 10.5%, and real revenue falls in 30 of 31 provinces; the largest falls are in Tibet (−77%, from a baseline tax take of 3.5% of provincial income) and Beijing (−38%), and Hainan is the only province whose real revenue rises (+1.2%). Panel (b): net interprovincial credit routing as a share of each province's baseline tax collections. A firm remits output VAT to its own province while the input credits its customers claim are refunded by the customers' provinces; the bar is the refunds other provinces give on the province's sales minus the refunds it gives on its purchases from other provinces, so it sums to zero across provinces once weighted by baseline revenue. Under the reform, 17.4% of the national baseline tax base moves between provincial ledgers in this way. The correlation across provinces between panels (a) and (b) is 0.89; the correlation of the provincial welfare change with panel (a) is 0.77 and with panel (b) 0.59. Same calibration and closure as Figure 6.

and the expansion they bring, is shared by buyers in every province. The cost falls on the budget of the province where the credits are claimed. This is the question of origin-based versus destination-based assignment of the value-added tax, and the reform makes it more

important, because it extends the credit chain to the sectors that are most concentrated in space.

To measure how much of the distribution of gains is due to the assignment rule, I hold the reform fixed and replace origin-based assignment with a pooled rebate: every province receives its baseline revenue, and the national change in revenue is shared across provinces in proportion to their baseline tax revenue. Under this rule, revenue changes by the same percentage in every province, so provinces no longer give up revenue to one another through the credit chain. **Figure 8** compares the two rules. Under the pooled rebate, Beijing moves from a loss of 3.0 percent to a gain of 1.6 percent, and Tibet from a loss of 1.5 percent to a gain of 1.0 percent. The provinces that gained the most under the origin rule give up part of their gains: Shaanxi falls from 3.8 to 1.8 percent, and Hainan from 3.4 to 1.8 percent. Welfare rises in 30 of 31 provinces, and the standard deviation of provincial welfare changes falls from 1.27 to 0.66 percentage points.¹⁶ This is the counterpart of the closed-economy argument above: once provinces stop giving up revenue to one another through the credit chain, the reform raises welfare almost everywhere.

Pooling is a redistribution and not an efficiency gain. Measured in yuan, the provincial gains and losses add up to the same total under both rules, 1.30 percent of national income.¹⁷ The redistribution runs, on balance, from poorer to richer provinces: the richer half of provinces, with 46 percent of the population, receives a net transfer of 0.09 percent of national income. How the pooled revenue is shared also matters. If the national change in revenue is shared in proportion to income and not to baseline tax revenue, every province gives up 1.7 percent of its income. Tibet's own tax revenue is only 3.5 percent of its income, so this rule still makes it a net payer to other provinces, and it still loses 0.4 percent; Tibet and Fujian (−0.1 percent) are then the two provinces whose welfare falls, and the standard deviation is 0.72 percentage points.

¹⁶The exception is Fujian (−0.3 percent), the only province in which the consumer price index rises, by 1.7 percent against a fall of 1.2 percent in the average province. The interregional table records 15.5 percent of Fujian's final demand as construction purchased from abroad, all of it in fixed capital formation, against zero in the median province; the tax on these purchases rises from 3 to 11 percent with no offsetting fall in cost, which alone adds 1.1 percentage points to Fujian's price index. Whether this entry reflects foreign contractors or the balancing of the table cannot be determined from the table, and I leave it as published. Under the origin rule Fujian keeps the revenue from this tax, including the tax collected at the border on these imports, and its welfare rises by 1.6 percent; under pooling that revenue is shared with the other provinces, which costs Fujian 1.9 percent of its income. With a complete credit chain (the full-crediting benchmark of **Section 8.3**) and the same pooled rebate, welfare rises in all 31 provinces.

¹⁷With homothetic preferences, the equivalent variation of province d is its baseline income times its welfare change, $I_d(\hat{U}_d - 1)$. Summing over provinces and dividing by national income gives 1.30 percent under the origin rule and under both pooled rules. A transfer between provinces leaves this sum unchanged.

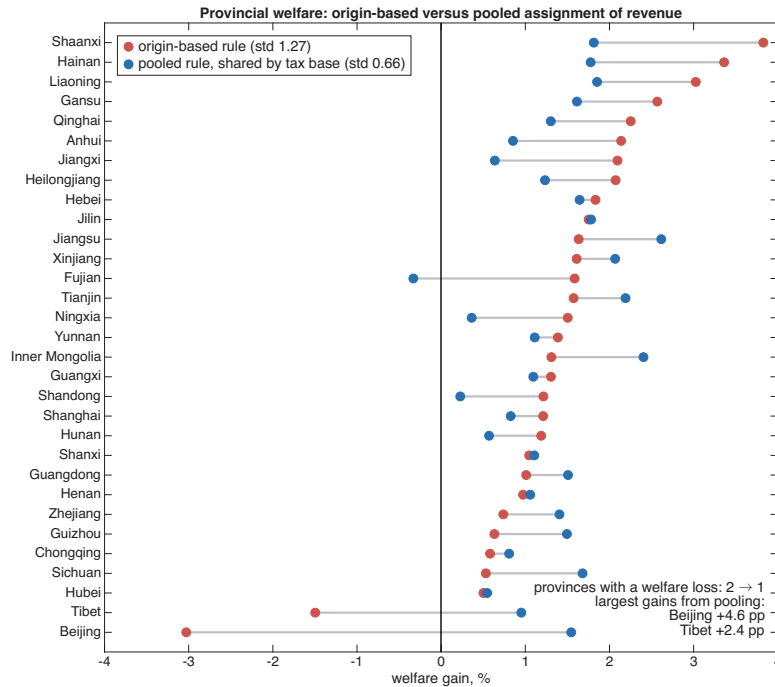


Figure 8: Origin-based versus pooled assignment of revenue, by province

Notes: Each province's welfare change under the origin-based rule (red) and under a pooled rebate (blue) that gives every province its baseline revenue and shares the national change in revenue in proportion to baseline tax revenue, so that revenue changes by the same percentage in every province. Provinces are in the same order as in Figure 6. Pooling is a redistribution: measured in yuan, the provincial gains and losses add up to the same total under both rules. Same calibration and closure as Figure 6.

8.2 Gains from Reducing In-house Production

How much of this gain operates through the make-or-buy margin? The business tax fell on services purchased in the market but not on services produced in-house, so it raised the relative cost of buying and tilted firms' input mix toward making more of their own services; extending input credits removes that wedge, and firms shift part of the bundle back to the market. In the model, the in-house share of the input bundle falls by 6.8 percentage points in the reformed sectors relative to the sectors that were always under the VAT (Figure A.2 in the appendix shows the distribution across region-sector pairs). To measure what this margin contributes, I remove the margin by setting the in-house share to zero in every sector and introduce the same reform. The welfare gain is then 0.92 percent instead of 1.30 percent. A model that ignored vertical integration would therefore understate the gain from the reform by 0.38 percentage points, or 29 percent. The remaining 0.92 percent is the gain from removing cascading alone; it depends only on the input-output structure, the statutory rates, and the trade elasticity, and not on how the in-house share is measured. Figure 9 shows that the gain rises almost linearly in the aggregate in-house share when the measured sector vector is scaled proportionally, by

about 0.53 percentage points for every 10 points of the share: from 1.06 percent at an aggregate share of 0.02 to 1.75 percent at 0.15, around the measured value of 0.056.

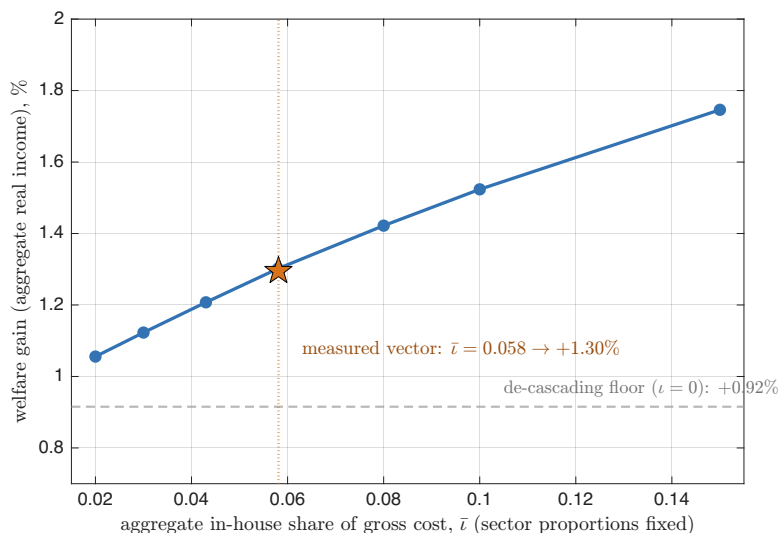


Figure 9: Welfare and the measured in-house share

Notes: Welfare gain (aggregate real income) of the as-legislated reform as a function of the aggregate in-house share $\bar{\iota}$, scaling the measured sector vector proportionally so that sector proportions are held fixed. The starred point is the measured calibration ($\bar{\iota} = 0.056$, welfare +1.30 percent). The dashed line is the welfare gain with the make-or-buy margin removed ($\iota = 0$ in every sector, +0.92 percent): the gain from removing cascading alone, which does not depend on the measurement of the in-house share.

8.3 Crediting, Rates, and Uniformity

The reform changed the tax rate and input crediting condition. It led firms credit the tax on input purchases, and it raised the rate on services themselves, from 3 to 5 percent under the business tax to 6 to 11 percent under the value-added tax. The first removes cascading; the second is a tax rate increase. It also left carve-outs in place, so that some sectors still cannot credit their inputs. Table 7 separates these pieces by holding the economy fixed and changing one at a time. The first row is the reform as legislated. The next three extend input crediting to every sector and differ only in the rates. Completing the credit chain at the legislated rates raises the gain from 1.30 to 1.54 percent at a real revenue ratio one point lower: the carve-outs collect revenue precisely by leaving cascading in place on the excluded links. Holding full crediting and returning the service rates to their 2012 levels lowers the gain to 1.19 percent and costs 21 points of revenue. Once the chain is intact, raising the service rates is therefore not costly; it narrows the gap between the rates on services and on goods. The reform also gave up revenue, so part of its gain is the gain from a lower tax burden. Two experiments separate the two. In the first, I leave the 2012 system in place and cut the business-tax rates by a common factor (to 80 percent of their level) until

real revenue falls by as much as it does under the reform. This cut alone raises welfare by 0.98 percent (Table 9 in Appendix E): revenue falls by 1.57 percent of national income and households gain 0.98 percent net, so each yuan of business-tax revenue given up returns about 1.6 yuan. The business tax is an expensive way to raise revenue, and a large part of what the reform achieves is to raise less revenue with it. The reform delivers 1.30 percent for the same revenue, a third more. In the second experiment, I keep the reform and scale the new service rates up until real revenue is back at its 2012 level. The gain is still 1.1 percent: recovering the same 1.57 percent of income costs 0.17 points of welfare, about 1.1 yuan per yuan raised. The reform is therefore not a tax cut in disguise; it replaces an instrument whose revenue is expensive with one whose revenue is cheap. The same rate increase without crediting, applied while service customers still cannot deduct, lowers welfare by 1.0 percent, because every added point of tax on services cascades through the firms that buy them. What a rate costs depends on the chain beneath it. Averaging the two components over both orderings in which they can be introduced attributes 1.65 points of welfare, and 26 points of real revenue given up, to the crediting change, and -0.36 points of welfare, and 15 points of revenue recovered, to the rate change, with their interaction of 1.21 points shared equally. Per point of real revenue, the crediting change is worth 0.063 points of welfare and the rate change costs 0.023.

The last row shows how far the legislation stops short of a uniform value-added tax. A single rate of 17 percent on every sector, with full crediting and exports zero-rated, raises welfare by 3.9 percent and real revenue by 25 percent, against 1.30 percent and a 10.7 percent revenue loss under the reform as legislated. The distance is the cost of rate dispersion across sectors and of the block of final demand, about a sixth, that the tax does not reach (agriculture, education, health, and public administration). Uniformity within the taxed block alone does not close it: the same 17 percent on only the sectors the tax currently reaches gives 1.5 percent. One reading is that, with part of final demand untaxed, equalizing the rates on the rest is not the second-best structure; the experiment does not separate this from the effect of the higher revenue it raises. This is the bridge to the next section, which asks what the credit chain itself is worth once rates are set aside.

8.4 Tax Architecture at Scale

In a closed economy with full compliance, a value-added tax dominates a turnover tax as a matter of theory: a uniform tax with full crediting falls on final consumption and leaves producer prices alone, so with revenue rebated it is equivalent to a lump sum, whereas a turnover tax falls on every transaction and taxes intermediate links as many times as they

Table 7: Crediting, rates, and uniformity

	Welfare (%)	Real revenue ratio	Provinces gaining
Reform as legislated	+1.30	0.893	29/31
<i>Full crediting in every sector:</i>			
at 2012 rates	+1.19	0.680	24/31
at 2017 statutory rates	+1.54	0.886	29/31
at a uniform 17 percent, every sector	+3.87	1.252	30/31

Notes: Welfare is aggregate real income, the sum of provincial gains in yuan over national income; real revenue is collections deflated by the provincial consumer price index, relative to 2012. “Gaining” counts provinces whose welfare rises. The first row is the reform as legislated, at effective rates and with the statutory carve-outs in place. The next three rows extend input crediting to every sector and differ only in the rates: the 2012 rates (business-tax rates of 3 to 5 percent on the reformed services), the 2017 statutory rates (6 to 11 percent on services, 17 percent on goods), and a single rate of 17 percent on every sector, including agriculture, the exempt sectors, and the nonmarket sector, with exports zero-rated. The same 17 percent applied only to the sectors the value-added tax currently reaches, leaving the exempt and nonmarket block untaxed, gives +1.54 percent at a real revenue ratio of 1.060: with part of consumption outside the tax, uniformity within the taxed block is not the second-best rate structure. For comparison, applying the 2017 rates without extending crediting, that is, raising the rates on services while their customers still cannot credit them, lowers welfare by 0.96 percent and raises real revenue by 12 percent. The baseline is the unreformed 2012 economy (Shanghai’s pilot undone); public administration earns no input credits in any row.

change hands (Diamond & Mirrlees, 1971). Outside those conditions the ranking is not guaranteed. A turnover tax that reaches exports collects rents from foreign buyers when foreign demand is less than perfectly elastic, and with informality a cascading tax can be second-best (Emran & Stiglitz, 2005; Keen, 2014). And even where the sign is settled, theory delivers the magnitude only as a local formula, one half of an elasticity times a squared wedge times a base, summed over links; it does not say what that comes to in an economy with a given input-output structure, trade with other provinces and the rest of the world, and a margin between buying services and producing them in-house, nor how it depends on the revenue the government must raise. This section puts the number on it, in the calibrated Chinese economy with its trade side priced, and shows how it scales.

The reform as legislated is not the right experiment for this, because it changed rates, extended crediting to some sectors and not others, and left carve-outs in place, so its welfare effect mixes the structure of the tax with its level. I therefore compare two clean systems on the same economy. The first is a uniform value-added tax with full crediting in every sector. The second is a uniform turnover tax that falls on every sale and credits nothing; it is the business tax extended to the whole economy. I trace each system by scaling its statutory rate, from 3 to 22 percent for the value-added tax and from 0.8 to 8 percent for the turnover tax, and I compare the two at the same fiscal capacity, defined as tax revenue as a share of national income relative to the 2012 baseline. Two capacities matter for China. A capacity of 1.00 is the 2012 system. A capacity of 0.88 is where the country stands after the reform, which lowered tax revenue as a share of income by 11.8 percent (10.7 percent in real terms). The answer, in one sentence, is that at China’s post-reform capacity the

crediting architecture is worth 1.4 percent of real income, slightly more than the entire reform delivered, that the value is 2 percent at the 2012 capacity and grows more than proportionally with the revenue raised, and that no force in the open economy comes close to reversing it.

Figure 10a plots welfare against fiscal capacity for the two systems. Two features of the figure need to be read correctly. First, revenue is rebated to households as a lump sum, and welfare places no value on public spending beyond that rebate, so a curve slopes down only if the tax carries a wedge that grows with the rate. The turnover tax and the value-added tax that exempts the nonmarket sector both do, and both slope down. The truly uniform value-added tax carries no wedge, and its curve is flat: raising its rate from 3 to 22 percent changes aggregate real income by less than 0.02 percentage points. The level of a curve therefore says nothing about whether taxation is desirable; only the vertical distance between two curves at the same capacity does. Second, the 2012 system, marked by the black square, lies between about 1.6 and 3.9 percentage points below every uniform system at the same capacity. It combined dispersed rates with a broken credit chain, and each feature is costly on its own. That distance is the price of the pre-reform structure, and it is not the object of this section.

The comparison between the two solid curves is the main result. The value-added tax is above the turnover tax at every capacity. At a capacity of 0.44, the lowest point solved, the gap is 0.23 percentage points. It is small for a reason: at low rates every tax is close to a lump sum, and the cost of the cascade is second order in the rate, so there is little yet for the credit chain to remove. As capacity rises the cascade starts to bite, and the advantage of the value-added tax grows more than proportionally with capacity: 0.52 percentage points at a capacity of 0.60, 1.28 at 0.85, 1.97 at 1.00, and 3.8 at 1.27. At China's post-reform capacity of 0.88, the crediting architecture is worth about 1.4 percent of welfare relative to raising the same revenue with a turnover tax; at its 2012 capacity, about 2 percent.

Figure 10b isolates this gap for the truly uniform value-added tax. The gap is positive at every capacity, 0.24 points at the lowest, and it widens more than proportionally, to 1.5 points at a capacity of 0.88 and 2.0 at 1.00. There is no capacity at which the two architectures are equivalent, but there is a capacity below which the difference is too small to matter, and China passed it long before the reform. A country that raised half of what China raised in 2012 would gain about a third of a point from the credit chain; a country at China's scale gains two points, and keeps gaining as it grows. This is the sense in which the choice of architecture is a question of scale.

One version of the turnover tax does better than the other, and it is worth being precise about why. In [Figure 10a](#), the dashed turnover curve taxes exports, whereas the value-added tax zero-rates them, as value-added taxes do in practice. With finite foreign demand, a small tax on exports collects rents from foreign buyers. The rates in question sit far below the inverse of the trade elasticity, about 22 percent, so the levy is on the rising part of its Laffer curve ([Lemma 1](#) in [Appendix E](#)). This levy is worth 0.5 percentage points on average across the range. It is trade policy, not tax structure, which is why the solid turnover curve removes it by giving the turnover tax the same export treatment as the value-added tax, so that the two solid curves differ only in whether purchases are credited. Even with the levy, the taxed-exports turnover tax stays below the value-added tax at every capacity: by 0.22 points at a capacity of 0.60, 0.74 at 0.85, and 1.3 at 1.00.

The closed economy gives the same ranking without the trade side. When I shut the trade leg, the value-added tax curve is exactly flat: raising its rate collects revenue at no welfare cost, because a uniform tax with full crediting and lump-sum rebates is a uniform consumption tax that leaves producer prices unchanged ([Diamond & Mirrlees, 1971](#)). The turnover tax falls below it at every matched capacity, by 0.6 to 3.6 percentage points ([Table 8](#) in [Appendix E](#)). The open economy changes the magnitudes, not the ranking: the export levy narrows the gap for the arm that taxes exports, and the priced foreign demand lets the cascade's cost spill into the terms of trade, but at no capacity does either effect reverse the order. What the comparison adds to the theorem is the scale at which the architecture starts to matter, and for China that scale was passed long before the reform.

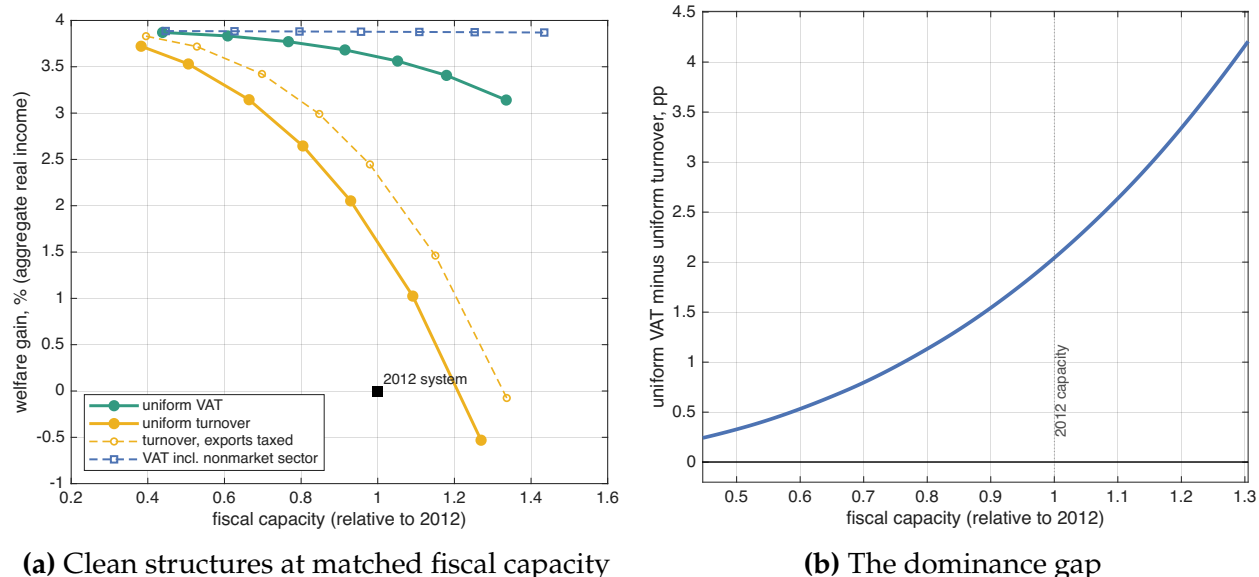


Figure 10: Tax architecture at matched fiscal capacity

Notes: Welfare is aggregate real income, the sum of provincial real-income gains over national income, in percent; fiscal capacity is tax revenue as a share of national income relative to the 2012 baseline. Panel (a): four uniform tax systems under the paper's closure (priced export demand with elasticity θ) and the measured in-house vector: a uniform value-added tax with full crediting and zero-rated exports; a uniform turnover tax with exports zero-rated, so that the two solid curves differ in tax structure only; a uniform turnover tax with exports taxed; and the value-added tax extended to the nonmarket sector. In each comparison the turnover arm has the same sector coverage as its value-added-tax comparator: the solid pair in panel (a) both exempt the nonmarket sector, and both arms in panel (b) include it. Each curve is traced by scaling the system's statutory rate (value-added tax 3–22 percent, turnover tax 0.8–8 percent). The black square is the 2012 system. Revenue is rebated to households as a lump sum, so comparisons are vertical distances at matched capacity. Panel (b): the gap between the value-added tax including the nonmarket sector and the uniform turnover tax with exports zero-rated, at matched capacity. The gap is positive at every capacity, so the curves have no crossing; the dotted line marks the 2012 capacity.

9 Conclusion

This paper asks what a tax on business revenue does to a production network, and what removing it is worth. The business tax fell on every sale, so it taxed intermediate inputs, raised the cost of buying services from the market, and rewarded producing them in-house. China's replacement of the business tax with a value-added tax in the service sectors, rolled out across provinces and sectors between 2012 and 2016, is the laboratory.

The evidence shows the network responding along each margin. The number of firms in reformed service sectors rises by about 21 percent relative to manufacturing, and their revenue by about 22 percent. Intermediate trade flows with a service sector at either end rise by 36 percent relative to flows among manufacturing sectors. Manufacturers in the pilot provinces, whose own tax regime did not change, buy more of the reformed services from the market than manufacturers elsewhere, and the administrative expense that houses their internal service departments falls.

To put a welfare number on these responses, I build a spatial general-equilibrium model of the production network in which every sector can make or buy its service inputs, the credit chain of the value-added tax is modeled transaction by transaction, and revenue is assigned to the province where the tax is paid. The in-house share is measured from firms' administrative expenses before the reform rather than calibrated. The reform raises welfare by 1.30 percent and real wages by 3.3 percent, and lowers real tax revenue by 10.7 percent. A model without the make-or-buy margin misses 30 percent of the gain.

Three lessons follow for the design of taxes. First, what a tax rate costs depends on the credit chain beneath it. A yuan of business-tax revenue costs households about 1.6 yuan, and a yuan raised by higher rates on services once their customers can credit the tax costs about 1.1. Raising the rate on services lowers welfare only while their customers cannot deduct the tax, and a uniform value-added tax with full crediting on every sector would raise welfare by 3.9 percent. Against a turnover tax raising the same revenue, the crediting architecture is worth 1.4 percent of real income at China's post-reform fiscal capacity and 2 percent at its 2012 capacity, and the value grows more than proportionally with the revenue raised, so the choice of architecture is a question of scale. Second, how revenue is assigned across regions decides who gains. Under origin-based collection, more than a sixth of the tax base moves between provinces through interprovincial input credits. Beijing, whose service sectors buy their inputs from the rest of the country, loses although its real wage rises by 3.7 percent, while Shanghai, whose service sectors buy closer to home, gains; pooling the change in revenue turns Beijing's loss into a gain without changing the aggregate. Third, the boundary of the firm is an adjustment margin for tax policy, and one that can be measured: the share of service tasks produced in-house has an address in the firm's accounts, and it moves when the tax on market inputs is removed.

The framework can be used to study how firms redraw their boundaries under other changes in the relative price of market inputs, such as trade policy, contract enforcement, or digital service platforms, and to evaluate destination-based alternatives to the origin-based assignment of value-added tax revenue that the reform made more consequential.

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A Derivations

A.1 Marginal Cost

The marginal cost c_{dj} of the representative producer is obtained by solving the following cost minimization problem subject to the production constraint $y_{dj} \geq 1$:

$$\min_{\ell_{dj}, Y_{dj}^M} w_d \ell_{dj} + P_{dj}^M Y_{dj}^M, \quad (\text{A1})$$

The production technology is Cobb–Douglas:

$$y_{dj} = \ell_{dj}^{\alpha_{dj}} (Y_{dj}^M)^{1-\alpha_{dj}}. \quad (\text{A2})$$

First-order conditions imply the cost share rules:

$$\frac{w_d \ell_{dj}}{P_{dj}^M Y_{dj}^M} = \frac{\alpha_{dj}}{1 - \alpha_{dj}}.$$

Thus the unit cost function is

$$c_{dj} = \mathbf{C}_1(\alpha_{dj}) w_d^{\alpha_{dj}} (P_{dj}^M)^{1-\alpha_{dj}}, \quad \mathbf{C}_1(\alpha) \equiv \alpha^{-\alpha} (1 - \alpha)^{-(1-\alpha)}. \quad (\text{A3})$$

The input demands for one unit of output are:

$$\ell_{dj}^* = \frac{\alpha_{dj} c_{dj}}{w_d} = \alpha_{dj} \mathbf{C}_1(\alpha_{dj}) w_d^{\alpha_{dj}-1} (P_{dj}^M)^{1-\alpha_{dj}}, \quad (\text{A4})$$

$$(Y_{dj}^M)^* = \frac{(1 - \alpha_{dj}) c_{dj}}{P_{dj}^M} = (1 - \alpha_{dj}) \mathbf{C}_1(\alpha_{dj}) w_d^{\alpha_{dj}} (P_{dj}^M)^{-\alpha_{dj}}. \quad (\text{A5})$$

Substituting the effective prices, the marginal cost can be expressed as

$$c_{dj} = \alpha_{dj}^{-\alpha_{dj}} (1 - \alpha_{dj})^{-(1-\alpha_{dj})} (w_d)^{\alpha_{dj}} (P_{dj}^M)^{1-\alpha_{dj}}. \quad (\text{A6})$$

A.2 Intermediate Price Index

The inner optimization problem associated with the choice of demanding intermediate inputs subject to the constraint $Y_{dj}^M \geq 1$ is:

$$\min_{Y_{ok,dj}^m, Y_{k,dj}^H} \sum_{o=1}^N \sum_{k=1}^J p_{ok,dj} Y_{ok,dj}^m + \sum_{k=1}^J \frac{w_d}{z_{dj}^H} Y_{k,dj}^H, \quad (\text{A7})$$

where $p_{ok,dj}$ is the tax-inclusive (purchasers') price and $Y_{k,dj}^H$ is the in-house alternative, priced at the wage.

This gives the following demand function:

$$Y_{ok,dj}^m = \iota_{ok,dj} \left[\frac{p_{ok,dj}}{P_{dj}^M} \right]^{-(1+\theta)} Y_{dj}^M \quad (\text{A8})$$

and the following price index:

$$P_{dj}^M = \left[\sum_{o=1}^N \sum_{k=1}^J \iota_{ok,dj} [p_{ok,dj}]^{-\theta} + \sum_{k=1}^J \iota_{Hk,dj} \left(\frac{w_d}{z_{dj}^H} \right)^{-\theta} \right]^{-\frac{1}{\theta}}, \quad (\text{A9})$$

A.3 Competitive Prices, Shares, and Gravity

Under perfect competition with constant returns, the good of (o, k) is delivered to producers (d, j) at $p_{ok,dj} = (1 + t_{ok,dj})c_{ok}\kappa_{ok,dj}$ and to consumers at $p_{ok,d} = (1 + t_{ok,d})c_{ok}\kappa_{ok,d}$ (zero profits). Substituting into the CES demands yields the shares

$$\gamma_{ok,dj}^P = \iota_{ok,dj} \left[\frac{(1 + t_{ok,dj})c_{ok}\kappa_{ok,dj}}{P_{dj}^M} \right]^{-\theta}, \quad \gamma_{Hk,dj}^P = \iota_{Hk,dj} \left[\frac{w_d/z_{dj}^H}{P_{dj}^M} \right]^{-\theta},$$

with the consumer analogue on P_d , and expenditures $X_{ok,dj}^P = \gamma_{ok,dj}^P E_{dj}^M$, $X_{ok,d}^C = \gamma_{ok,d}^C I_d$. The in-house row enters the same nest priced at the wage: the make-or-buy margin is an ordinary CES margin against an untaxed inside option. The wedge elasticity of every bilateral flow is $-\theta$, the trade elasticity.

A.4 Second-Order Welfare and Wedge Dispersion

For a buyer cell with expenditure weights s_m and effective-cost wedges w_m across its CES substitution margins, a second-order expansion of the welfare cost around undistorted prices, with revenue rebated lump-sum, gives

$$\Delta W \approx -\frac{1+\theta}{2} \sum_m s_m (\ln w_m - \overline{\ln w})^2, \quad \overline{\ln w} = \sum_m s_m \ln w_m,$$

so only the *dispersion* of wedges across margins matters: the common component is a price-level shift returned by the rebate. This is the formula behind the bundle decomposition of the certification appendix (a uniform full-credit VAT has zero wedge variance at every rate; a cascade has variance proportional to τ^2).

A.5 Numerical Certification

The counterfactual system is verified computationally rather than only algebraically: with no change in policy the published table is reproduced at 10^{-15} and the rebuilt unreformed baseline at 10^{-7} (TEST_REBASE), the provincial fiscal ledger reproduces the equilibrium decomposition at 10^{-14} (DIAG_ROUTING_CLOSURE), and on a closed-economy version of the calibrated data the framework reproduces lump-sum equivalence of a uniform consumption tax to 2×10^{-5} per capacity point and the dominance of the crediting architecture at every revenue level. The certification, the trade-closure diagnostics, and their measured magnitudes are collected in Appendix E.

A.6 Circular Flow and the Trade Balance

This subsection derives the trade (im)balance condition (6.28) from the household and firm budget constraints, and shows that the ad valorem tax enters the circular flow only through the locally rebated revenue T_d , never through the goods balance. It is essential here to work with the statutory *remittance* $T_d = \sum_k T_{dk}^P$ of (6.26)–(6.27)—the output tax collected on a sector’s sales, net of the input credits it reclaims—and *not* with the cascading wedge $t_{ok,dj}$. As Case 1 of the numerical illustration showed, the remittance is generically positive even when every wedge $t_{ok,dj} = 0$, because the output-tax term collects the statutory rate t_k on a value-added base; the two objects coincide only after the credit-invoice cancellation nets pairwise across the whole economy, established in the Walras step below. Throughout, R denotes net-of-tax sales and X^P, X^C the bilateral expenditures at purchaser (tax-inclusive) prices (6.21), so a seller keeps $X/(1+t)$ and the price wedge $\frac{t}{1+t}X$ is the object in (6.23)—distinct from the remittance.

Building blocks. Households in region d spend all income on final goods at purchaser prices:

$$\sum_{o=1}^N \sum_{k=1}^J X_{ok,d}^C = I_d = w_d L_d + T_d + D_d, \quad (\text{A10})$$

with T_d the rebated tax revenue (6.27) and D_d the exogenous transfer. On the production side, profits are zero (so net revenue equals total expenditure, as established above), and the net-of-tax revenue of each sector (d, j) is paid out fully to value added and to market intermediate purchases at purchaser prices:

$$R_{dj} = \text{VA}_{dj} + \sum_{o=1}^N \sum_{k=1}^J X_{ok,dj}^P, \quad (\text{A11})$$

where VA_{dj} is sector value added, i.e. labor income (in-house labor included). Summing over j and using $\sum_j VA_{dj} = w_d L_d$,

$$\sum_{j=1}^J R_{dj} = w_d L_d + \sum_{j=1}^J \sum_{o=1}^N \sum_{k=1}^J X_{ok,dj}^P. \quad (\text{A12})$$

Throughout this derivation T_d is the export-adjusted remittance of (6.26): output tax accrues on domestic sales only, while credits on export-associated inputs remain reclaimable, so export-oriented sectors can hold $T_{dk}^P < 0$ (a net refund position), which the derivation accommodates without modification since no step requires $T_d \geq 0$.

Regional resource constraint. Solving (A10) for $w_d L_d$ and substituting into (A12), labor income cancels and leaves

$$\underbrace{\sum_{o,k} X_{ok,d}^C + \sum_j \sum_{o,k} X_{ok,dj}^P}_{\text{absorption } G_d^{\text{purch}} \text{ (purchaser prices)}} - \underbrace{\sum_j R_{dj}}_{\text{net sales}} = T_d + D_d. \quad (\text{A13})$$

That is, purchases valued at purchaser prices, net of the value of local output, equal the remittance collected plus the transfer. Note the two sides of the tax here refer to *different* transactions: the left-hand difference is the tax embedded in the prices of goods *purchased by* region d , whereas T_d on the right is the tax *remitted by* region d 's sellers (output tax net of credits). Under the origin basis these are not equal region by region—that inequality is the origin-versus-destination wedge—and (A13) holds because T_d is defined as the remittance and enters household income.

Remittance decomposition. Under the origin rule, region d 's remittance splits by the location of the counterparty,

$$T_d = \underbrace{W_d^{\text{dom}}}_{\text{tax embedded in within-region transactions}} + \underbrace{\text{Tax}_d^{x,n} + \text{Tax}_d^{x,c}}_{\text{output tax on sales to buyers in } o \neq d: \text{ noncreditable, creditable}} - \underbrace{\text{Cr}_d^m}_{\text{credits } d \text{ refunds on purchases from } o \neq d}. \quad (\text{A14})$$

On a within-region link the seller's output tax and, if the link is creditable, the buyer's credit are entries in the same budget, so the net remittance on within-region transactions equals the tax embedded in their purchaser prices: the consumer wedge and the noncreditable wedge, W_d^{dom} . On cross-region sales d 's sellers remit the output tax whether or not the buyer can credit it; on cross-region purchases d 's government refunds the credits its buyers claim. The last two terms are the two legs of interprovincial credit routing.

From the resource constraint to the trade balance. Split each sum in (A13) into within-region ($o = d$) and cross-region ($o \neq d$) parts. Within-region purchases at purchaser prices

less within-region net sales equal W_d^{dom} , which cancels against the same term in T_d , leaving

$$\text{Import}_d^P - \text{Export}_d^{\text{net}} = D_d + \text{Tax}_d^{x,n} + \text{Tax}_d^{x,c} - \text{Cr}_d^m, \quad (\text{A15})$$

where Import_d^P is d 's purchases from other regions at the model's purchaser prices, inclusive of noncreditable tax only, and $\text{Export}_d^{\text{net}}$ its sales to other regions at net-of-tax prices. Adding the noncreditable tax on d 's cross-region sales to both sides values exports in the same purchaser-price convention, $\text{Export}_d^P = \text{Export}_d^{\text{net}} + \text{Tax}_d^{x,n}$, and gives the balance in the model's own valuation,

$$\text{Import}_d^P - \text{Export}_d^P = D_d + N_d, \quad N_d \equiv \text{Tax}_d^{x,c} - \text{Cr}_d^m, \quad (\text{A16})$$

where N_d is the net interprovincial credit routing of Figure 7(b): the output tax d collects on creditable sales to other provinces, less the credits it refunds on creditable purchases from them. Valued at invoice prices instead, that is, adding the creditable VAT invoiced on cross-region sales and purchases to both flows, the routing term cancels and

$$\text{Import}_d^{\text{inv}} - \text{Export}_d^{\text{inv}} = D_d, \quad (\text{A17})$$

which is the valuation in which the published interregional table records trade and the form stated in the main text, (6.28). The baseline transfers D_d are calibrated from the household budget (A10), with the published trade table as a check: it reproduces them to within 0.13 percent of provincial income, the discrepancy being production taxes outside the VAT and the business tax that the model carries in a separate residual account (Section 6.7). The tax enters the circular flow only through T_d , collected and rebated locally, and through the routing term, which is a transfer between provincial budgets.

Walras consistency. Summing (A13) over the Chinese provinces,

$$\sum_{d \leq N-1} \left(G_d^{\text{purch}} - \sum_j R_{dj} \right) = \sum_{d \leq N-1} T_d + \sum_{d \leq N-1} D_d. \quad (\text{A18})$$

The routing terms cancel in the sum ($\sum_d N_d = 0$), and on every creditable link the seller's output tax, the buyer's credit, and the tax in the purchase price cancel identically. No further term arises from rate differences along a chain: as Case 1 of Section 4 shows, a creditable chain remits in total the downstream rate on cumulative value added, however that total is split across stages, so the split moves collections between budgets without adding to them. What survives on both sides is the set of terminal charges: tax on final consumers, noncreditable tax on intermediate links, and the border collections on consumer imports,

$$\sum_{d \leq N-1} \left(G_d^{\text{purch}} - \sum_j R_{dj} \right) = \sum_{d \leq N-1} T_d + (\text{ROW-side legs}), \quad (\text{A19})$$

where the ROW-side legs are the business tax on exports, the price-internal production taxes on sales to the rest of the world, and the residual lumps, none of which has a domestic

counterparty. The rest of the world has no budget in the model; its implicit transfer is

$$D_N \equiv - \sum_{d \leq N-1} D_d, \quad (\text{A20})$$

which in the 2012 data equals 2.1 percent of national income, the counterpart of the external surplus of 4.1 percent net of the ROW-side legs (Appendix E). With D_N so defined, $\sum_{d=1}^N D_d = 0$ and the closure satisfies Walras' law. The argument runs on remittances and the credit-invoice cancellation; it does not assert that the per-link tax is zero on creditable transactions (the wedge is zero there, the remittance is not), and it is consistent with strictly positive statutory collections throughout.

B Land Transactions after the Reform

The results from the quantitative model presented in the paper provide insights into the effects of the Business to Value-Added Tax (B2V) reform. However, one limitation of the baseline analysis is the omission of land, a crucial factor in China's economy and a central source of fiscal revenue for local governments. This omission may bias the measured welfare effects, particularly in regions where land finance plays an outsized role.

In the model, service-intensive provinces such as Beijing experience a large decline in tax revenue following the reform. Yet when we examine land market outcomes, a complementary dynamic emerges that mitigates some of these fiscal pressures. Using a similar event-study specification as in the firm-level analysis, I document in Figure A.1 that the reform is associated with a differential increase in both the unit price of land and the total area of land transacted by service firms compared to manufacturing firms.

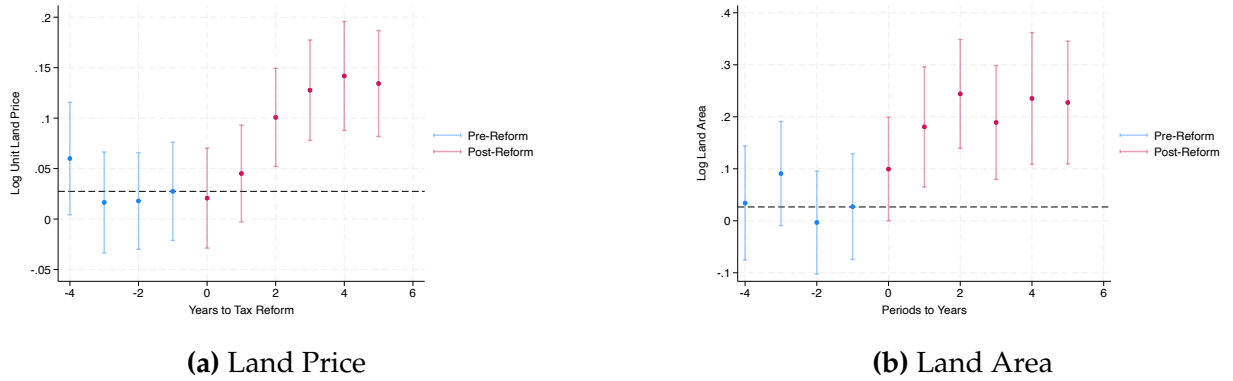


Figure A.1: Effects of the B2V Reform on Land Finance

The estimates show that the average growth in land prices and land area transacted attributable to the B2V reform are 9.3% and 20.3% higher, respectively, in the service sectors compared to manufacturing counterparts. These findings are significant when considering that, as of 2012, land use rights transactions accounted for roughly 40% of local government revenue nationwide. Thus, while the direct removal of the business tax reduced fiscal

inflows, the reform indirectly stimulated land-based revenue through higher demand from expanding service firms. Provinces with large service sectors and heavy reliance on land finance, such as Beijing, may have had part of the lost revenue offset through this channel; the model does not include land and the offset is not quantified.

Robustness of firm-level analysis. The land finance results also provide an independent validation of the main firm-level evidence. The increase in land demand and prices is consistent with service firms expanding operations after the reform, mirroring the observed surge in entry and revenue growth in the tax survey and registry data. In particular, land acquisition is a costly and strategic investment decision, typically undertaken only when firms anticipate long-run expansion and profitability. The fact that land market outcomes exhibit similar reform-induced dynamics strengthens the interpretation that the B2V reform not only raised firm revenues but also shifted investment behavior and resource allocation. Taken together, these complementary results reinforce the conclusion that the reform stimulated service sector expansion through both output and investment margins, with important implications for local fiscal capacity and welfare.

C Counterfactual In-house Share

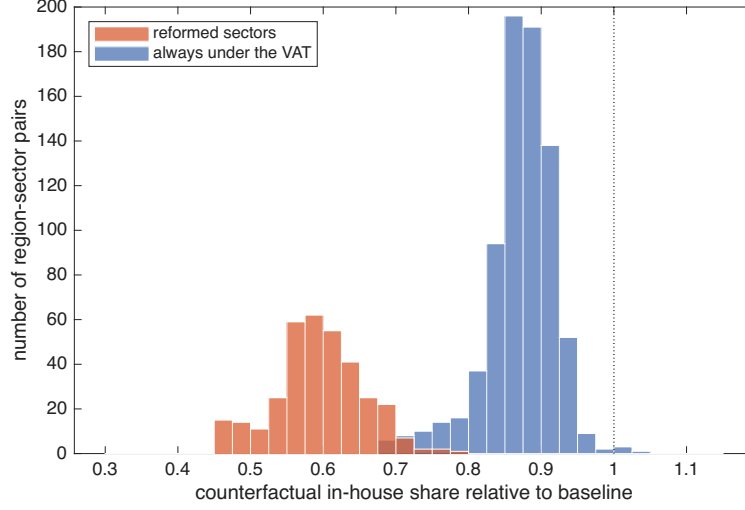


Figure A.2: Counterfactual In-house Share

Notes: The figure plots, for each region–sector pair, the in-house share of the input bundle in the reform equilibrium relative to its baseline value; a value of 0.6 means the pair keeps 60% of its baseline in-house share. Pairs in the reformed service sectors ($n = 341$) have a median of 0.59 (10th to 90th percentile 0.51 to 0.68); pairs in sectors that were under the value-added tax throughout ($n = 777$) have a median of 0.88 (0.82 to 0.92). Weighted by the size of the input bundle, the in-house share falls from 13.3% to 7.9% in the reformed sectors and from 5.8% to 5.1% in the others, a difference in differences of -4.7 percentage points.

C.1 The Trade-Wedge Specification: Short-Run Estimates

The reform’s wedge structure implies a continuous counterpart to the difference-in-differences of the main text. Reconstructing bilateral flows from the raw 2012 and 2017 MRIO tables at basic prices, and assigning each sector pair (k, j) its effective-cost wedge $w_{kj} = 1 + (1 - D_{kj})t_k$ under the pre- and post-reform crediting masks and effective rates, the structural gravity equation in relative changes (Section 6) is

$$\hat{X}_{ok,dj} = \hat{\gamma}_{ok,dj}^P \hat{E}_{dj}^M = \left(\frac{\hat{t}_{ok,dj} \hat{c}_{ok}}{\hat{p}_{dj}^M} \right)^{-\theta} \hat{E}_{dj}^M, \quad (\text{A1})$$

so that, in logs, the wedge change enters with coefficient $-\theta$ while the cost and destination terms are absorbed by origin- and destination-time fixed effects. Its estimable counterpart by PPML is

$$X_{ok,dj,t} = \exp \left[\beta (\text{Post}_t \times \Delta \ln w_{kj}) + \mu_{ok,dj} + \eta_{o,t} + \eta_{d,t} \right] \varepsilon_{ok,dj,t}, \quad (\text{A2})$$

with pair, origin-province-year, and destination-province-year fixed effects, two-way clustered by origin and destination cell. The estimate is $\hat{\beta} = -0.43$ (s.e. 0.24): flows rose where wedges fell. Splitting treatment into wedge quadrants gives $+0.07$ (0.04) for always-

VAT sellers to newly-deducting buyers, +0.16 (0.06) for formerly-BT sellers to VAT buyers, and +0.64 (0.06) for service-to-service pairs. Two features discipline the interpretation. First, the 2017 table sits roughly one year after the reform’s completion for the largest cohorts, so $\hat{\beta}$ is a *short-run* response — an order of magnitude below the calibrated long-run trade elasticity, consistent with the standard finding that short-run trade elasticities are a fraction of long-run ones. Second, the quadrant pattern is not proportional to the wedge changes: the service-to-service quadrant dominates despite a moderate wedge cut, the signature of secular service-input deepening loading onto treated pairs. For both reasons the wedge specification is reported as directional evidence only; the trade elasticity θ is calibrated from gravity estimates, and the long-run counterfactual is supplied by the model.

C.2 Policy Configurations: As-Legislated, Effective-Rate, and Full-Crediting Experiments

The quantification evaluates the reform under three policy configurations that differ along two dimensions: how completely the post-reform credit chain operates, and which vintage of tax rates applies. This appendix records the exact content of each, since the distinction carries the interpretation of the headline comparisons (Table 7).

The credit carve-outs. Caishui [2016] No. 36 did not create a complete credit chain: four service categories remained statutorily non-creditable for downstream buyers even after their sellers joined the VAT. The model implements these as fractional creditability δ_k applied to the seller-sector’s links, with the fractions set to the creditable subsector’s share of 2012 output: accommodation and catering $\delta = 0.35$ (catering purchases non-creditable; accommodation creditable), finance $\delta = 0.30$ (loan interest non-creditable; fee-based services creditable), resident services, repair, and other services $\delta = 0.83$ (resident daily services non-creditable; repair and other services creditable), and culture, sports, and entertainment $\delta = 0.55$ (entertainment non-creditable). These weights are output-share approximations and each is a disclosed sensitivity dial.

The rate vintage. The 2012 business-tax wedges are calibrated to $\min(\text{statutory}, \text{effective})$ rates, where effective rates are identified directly from sector-level remittances relative to the tax base — for a cascading tax the mapping requires no inference, and in six sectors the effective rate binds well below statute (threshold exemptions, item exemptions, informality). Applying statutory 2017 rates against effective 2012 wedges would overstate the reform’s burden increase in exactly those six sectors. The central configuration therefore scales each binding sector’s 2017 rate by its 2012 effective-to-statutory ratio — a constant-compliance-wedge assumption.

The three experiments.

- **caishui36** (as-legislated upper bound): statutory 2017 rates, credit carve-outs as enacted. Because VAT's self-enforcement mechanics more plausibly *shrank* the compliance wedge after the reform, this configuration bounds the burden increase from above.
- **caishui36eff** (central): the same carve-outs plus the effective-rate scaling. This is the configuration behind every headline number. On the rebuilt baseline the statutory-rate alternative gives a welfare gain of +1.35% at a real revenue ratio of 0.910, against +1.30% and 0.893 at effective rates, with the same two losing provinces (Beijing −2.8 and Tibet −1.6 percent; DIAG_REVIEW_D). The effective-rate choice is therefore the more conservative of the two on welfare and on revenue, consistent with the rate-dispersion result of Section 8.3. The scaling ratios are 0.84, 0.77, 0.73, 0.48, 0.52, and 0.72 in sectors 30, 34, 35, 36, 37, and 40.
- **ideal** (full crediting): every statutorily creditable link credits at $\delta = 1$ — the four carve-outs removed, loan interest included — at statutory rates with no overlay. This is the frictionless-VAT benchmark: what the credit architecture would deliver if the chain were completed.

What the comparison measures. At the calibrated parameters ($\theta = 4.5$, the measured in-house vector), the central reform delivers a welfare gain of +1.30% (aggregate real income) with a real revenue ratio of 0.893; the full-crediting benchmark delivers +1.54% at 0.886. The gap — roughly a fifth of the reform's total gain, forgone — prices the statutory carve-outs: completing the credit chain would raise welfare further while costing about one percent of revenue, because the carve-outs currently collect revenue precisely by re-introducing cascading on the excluded links. Two caveats discipline this reading. First, the two configurations differ in rate vintage as well as crediting, so the eff-to-ideal gap is not a pure crediting effect; a Shapley decomposition that toggles the credit act and the rate act separately and averages over both orderings (MAIN_6) provides the clean separation: the credit act accounts for +1.65 points of welfare and −26.1 points of real revenue, the rate act for −0.36 and +15.4, with an interaction of +1.21 points of welfare shared equally; in effectiveness terms, +0.063 welfare points per point of real revenue given up by the credit act against −0.023 per point recovered by the rate act. The carve-out gap is small next to either act. Second, the carve-out weights are output-share approximations; the loan-interest exclusion (finance, $\delta = 0.30$) is the economically largest single item, consistent with its prominence in the policy debate; the 2026 VAT Law, in force since January 2026, drops loan services from the statutory list of noncreditable purchases (Article 22), which now names catering, resident daily services, and entertainment; the Implementation Regulations (State Council Order No. 826, Article 21) keep input tax on loan interest noncreditable on a provisional basis and mandate a review, so the exclusion remains in force.

Robustness to the trade elasticity. Section 7 calibrates $\theta = 4.5$; the counterfactuals were rerun at the ends of the literature range (DIAG_REVIEW_OCT; export-demand elasticity set equal to θ , measured in-house vector). The welfare gain of the reform as legislated is +0.89% at $\theta = 3$, +1.30% at 4.5, and +1.71% at 6, with the real revenue ratio at 0.885, 0.893,

and 0.903. The gain with the make-or-buy margin removed is +0.70, +0.92, and +1.12%, so the margin's share of the gain rises from 22 to 29 to 34 percent: a higher elasticity makes both the cascade and the make-or-buy response more costly to leave in place. The provincial pattern is unchanged: Beijing and Tibet are the only losers at every θ (−3.0 and −1.4 percent at $\theta = 3$; −3.2 and −1.7 at $\theta = 6$), and Shanghai gains at every θ (+0.5, +1.2, and +1.9 percent).

Boundaries held fixed versus the margin removed. Section 8 measures the make-or-buy contribution by setting the in-house share to zero. One may prefer a counterfactual in which the margin is present but frozen, so that the baseline technology and allocation are unchanged and only the substitution is switched off. In this model the two are the same experiment. With in-house tasks produced by own labor at constant returns and the value-added nest Cobb–Douglas, holding the in-house expenditure share of the bundle at its baseline value γ_H (a Cobb–Douglas outer nest between own-labor tasks and the market bundle, implemented as `CFG.freezeIH` in `FUN_SOLVE_PC`) gives the unit cost $w^{\alpha+\gamma_H(1-\alpha)}(p_{\text{mkt}})^{(1-\alpha)(1-\gamma_H)}$ and market input demands $(1-\alpha)(1-\gamma_H)R$ times the market CES shares — exactly the cost function and demands of the model in which the in-house labor is relabeled as value added. The solver confirms the identity: the frozen-margin reform gives +0.915%, a real revenue ratio of 0.8762, and Beijing −2.42, Shanghai +0.96, Tibet −1.49 percent, digit for digit the $\iota = 0$ counterfactual (`DIAG_FREEZE_IH`). The 0.38-point contribution is therefore the value of letting firm boundaries move, not an artifact of removing in-house labor from the technology.

C.3 Zero-Rated and Exempt Exports

A value-added tax can relieve an export in two ways, and they are not equivalent. A *zero-rated* export carries no output tax and the exporter is refunded the tax paid on its inputs, so the export leaves the country free of tax. An *exempt* export also carries no output tax, but the input tax attributable to it cannot be credited or refunded; that tax stays in the exporter's cost and is passed into the export price. Exemption therefore removes the tax on the exporter's own value added only, and it leaves a cascading wedge on the inputs.

Let c_{dj} denote the creditable input tax of cell (d, j) per unit of its net output. The tax that remains in one unit of export revenue at seller-net value is

$$x_{dj} = \begin{cases} 0 & \text{zero-rated,} \\ c_{dj} & \text{exempt,} \\ t_j & \text{not relieved (business tax),} \end{cases}$$

and the model carries the export treatment of a cell as the intensity $z_{dj} = 1 - x_{dj}/t_{dj} \in [0, 1]$, where t_{dj} is the cell's output rate: the free-on-board price includes the factor $1 + (1 - z_{dj})t_{dj}$, and the remittance on exports is $(1 - z_{dj})t_{dj}$ per unit of seller-net revenue. Zero-rating is $z = 1$, the business tax is $z = 0$, and exemption lies between them at $z = 1 - c_{dj}/t_{dj}$. The credit intensity c_{dj} is held at its baseline value.

The distinction matters in the baseline for Shanghai. Under the 2012 pilot, Caishui [2011] No. 131 zero-rated international transport and research and design services supplied to foreign entities, and exempted the other exported pilot services: software and information services, consulting and attestation, technology transfer, logistics support, and the leasing of movable property used abroad. Shanghai's pilot cells in the observed 2012 table are read accordingly. For a pilot sector with pilot share ω^x of exports and zero-rated share ζ of pilot exports,

$$x_{dj} = \omega_j^x (1 - \zeta_j) c_{dj} + (1 - \omega_j^x) t_j^{BT},$$

with $(\omega^x, \zeta) = (0.918, 0.941)$ for transport, storage, and post (railway and post were outside the pilot; logistics support was exempt), $(0.601, 0)$ for information services (telecommunications were outside the pilot), $(1, 0)$ for leasing and business services, and $(1, 0.208)$ for research and technical services, the shares taken from the exports of the corresponding subsectors in the 2012 national table with 139 sectors. The resulting intensities are $z = 0.94, 0.61, 0.42$, and 0.36 . The design share of professional technical services cannot be separated, so ζ for research and technical services is a lower bound. In the unreformed baseline that the backward solve produces, these cells return to the business tax and $z = 0$.

D Numerical Solution Algorithm

The counterfactual equilibrium in relative changes is a fixed point in the wage vector $\hat{w}$ and income vector $\hat{I}$, solved by a damped Gauss-Seidel sweep over four blocks. Given the baseline shares $(\gamma^P, \gamma^C, \gamma_H^P, \alpha)$, the wedge hats $\hat{t}$, and the trade elasticity θ :

1. **Costs and price indices.** Given $\hat{w}$, solve the inner fixed point for $(\hat{c}_{dj}, \hat{P}_{dj}^M)$ from $\hat{c} = \hat{w}^\alpha (\hat{P}^M)^{1-\alpha}$ and the CES index over market inputs and the in-house row (priced at $\hat{w}$); compute $\hat{P}_d$.
2. **Trade shares.** Update $\hat{\gamma}^P = (\hat{t}\hat{c}/\hat{P}^M)^{-\theta}$ and $\hat{\gamma}^C = (\hat{t}\hat{c}/\hat{P}_d)^{-\theta}$.
3. **Goods market and fiscal ledger.** Solve the linear expenditure system for counterfactual revenues R' and incomes I' : net revenues equal expenditures net of wedges, remittances follow the statutory ledger (output tax on sales less reclaimed credits, exports relieved according to their zero-rating intensity, destination rule at the border), and collections are rebated to the collecting region. The system is solved either by a nested revenue-income iteration or by a direct sparse solve; both deliver the same fixed point (they agree to 10^{-12}).
4. **Labor market.** Compute market-clearing wages from labor demand $[\alpha + \gamma_H^P(1 - \alpha)]R'$ and update $\hat{w}$ by a damped geometric step with a per-sweep cap; update $\hat{I}$ likewise.

The sweep converges in roughly forty to ninety iterations at a tolerance of 10^{-8} on wages and incomes. The baseline is itself the outcome of a solve: the model is first solved backward from the published 2012 table to the economy in which Shanghai's pilot sectors are under

the business tax, with an outer loop on the in-house shares so that the unreformed economy carries the measured 2011 vector, and that equilibrium is written out in levels and used as the baseline (Section 6.7). Convergence is certified rather than assumed: with no change in policy the published table is reproduced at 10^{-15} and the rebuilt baseline at 10^{-7} ; applying the pilot to the rebuilt baseline returns the published table to 10^{-6} ; the provincial fiscal ledger reproduces the equilibrium decomposition at 10^{-14} , and on a closed-economy version of the data the algorithm reproduces the lump-sum equivalence of a uniform consumption tax to numerical precision (see Appendix E).

E A Closed-Economy Benchmark for the Comparison of Tax Systems

Section 8.4 compares a value-added tax with a turnover tax at equal revenue in the open economy. A uniform tax on all final consumption with full crediting is equivalent to a lump-sum tax, but theory ranks it against a turnover tax without qualification in one environment only: a closed economy, in which the turnover tax has no trade margin on which to collect rents from foreign buyers. This appendix constructs that environment from the calibrated economy, shows that the model reproduces the theoretical ranking there, and uses the benchmark to read the open-economy results. On the closed economy, the welfare curve of the uniform value-added tax is flat in the tax rate, and the value-added tax is ahead of the turnover tax at every matched revenue, by 0.5 to 3.6 percentage points (Table 8). The open-economy comparison differs from this benchmark only through trade, which Sections E.5 and E.6 take up. The scripts that produce every number are listed in Section E.8.

E.1 What theory implies

Two statements are involved.

(T1) Lump-sum equivalence. In a closed economy with inelastic factor supply, a uniform ad-valorem tax τ on all final consumption, with full crediting of taxed inputs and the revenue rebated lump-sum to households, changes no real allocation. Producer relative prices are undistorted, consumer prices rise by $(1 + \tau)$, nominal income rises by the rebate, and welfare is unchanged. Traced against revenue as τ varies, welfare is a horizontal line.

(T2) Dominance over the turnover tax. A uniform turnover tax at rate τ falls on every transaction. It taxes intermediate links, distorts relative producer prices according to the length of the chain behind each good, and taxes purchased inputs against the untaxed in-house alternative. Its welfare cost is of second order in the embedded wedges and positive at any positive revenue. Because the uniform value-added tax raises the same revenue at no cost by T1, it dominates the turnover tax at every revenue level.

Remark 1. Production efficiency in the sense of Diamond and Mirrlees (1971) does not by itself rank a uniform value-added tax against a uniform turnover tax: both are restricted instruments, and comparisons between restricted instruments are in general ambiguous.

The ranking is a theorem in the closed economy because of T1: there the uniform value-added tax is not merely less distorting but costless. The same reasoning identifies what can differ in the open economy, namely the treatment of trade.

E.2 Constructing the closed economy

The closed economy keeps the interprovincial production network of the calibrated model and removes its trade with the rest of the world. It is built from the baseline in four steps.

1. **Consumption shares.** The import rows of the consumption share matrix are set to zero and each destination's shares are rescaled to sum to one, so that demand formerly met by imports is spread over the destination's domestic suppliers in proportion to their existing shares.
2. **Input shares.** The import rows of the input share matrix are set to zero and the remaining market rows of each column are rescaled so that the column's total market share is preserved. Each sector's split between value added and purchased inputs is therefore unchanged; only the sourcing of the market bundle changes. Columns with essentially no domestic supplier (a domestic market share below 10^{-6}) keep their original shares; they account for less than 0.01 percent of baseline revenue.
3. **Exports.** Export revenue is set to zero in every province and sector.
4. **Reference point.** The closed economy is a counterfactual economy and is not recalibrated to data. All results are reported relative to a reference solution at 2012 tax rates on the same closed economy. Both statements above are invariant to this choice: flatness does not depend on the level of the curve, and dominance compares two curves that share the reference.

The two uniform systems. Both systems apply one statutory rate to all 41 sectors, including the nonmarket sector, so that no dispersion from exemptions remains. They differ only in crediting. Under the *uniform value-added tax*, at rates of 5, 10, 15, and 22 percent, every market input is credited. Under the *uniform turnover tax*, at rates of 1.5, 3, 4.5, and 6.5 percent, no input is credited. Revenue is summarized by fiscal capacity, the ratio of tax revenue to income relative to its 2012 value, $\text{cap} = (\sum_d T'_d / \sum_d I'_d) / (\sum_d T_d^{12} / \sum_d I_d^{12})$. The rate grids are chosen so that the two systems overlap in capacity, and the systems are compared at matched capacity by interpolation, not at matched rates.

E.3 The balancing transfer

Closing the economy requires one additional object. In the calibrated open economy, the provincial transfers D_d are the financing counterpart of the external position: as shares of national income,

$$\underbrace{\sum_d D_d}_{-2.13\%} + \underbrace{(E - M)}_{+4.08\%} = +1.95\%,$$

where $E = 24.34$ percent and $M = 20.26$ percent are exports and imports, and the remaining 1.95 percent is the tax collected on transactions with the rest of the world (Appendix A). Removing exports and imports while keeping the transfers at their nominal values leaves an injection with no counterpart.

Proposition E.1 (No equilibrium with an unbacked nominal injection). In the closed economy, with all household income spent, labor supplied inelastically, and a nonzero net nominal injection $S = \sum_d D_d + (\text{other legs fixed in nominal terms}) \neq 0$, there is no wage vector at which all provincial labor markets clear. Aggregate expenditure and aggregate factor income differ by S at every wage level, and because S is fixed in nominal units while every endogenous magnitude scales with wages, the difference cannot be absorbed.

The closed economy is therefore completed with a lump-sum transfer proportional to labor income, $B_d = b \hat{w}_d w_d L_d$, with the scalar b chosen so that the aggregate labor market clears. The transfer has a closed form. Write the goods-market system as $r = Mr + c_0 + b c_1$, where c_1 is the final demand generated by one unit of the transfer. Its solution is affine in b ,

$$r(b) = r_0 + b r_1, \quad r_0 = (I - M)^{-1} c_0, \quad r_1 = (I - M)^{-1} c_1,$$

and with $\ell_{dj} = \alpha_{dj} + \gamma_{dj}^H (1 - \alpha_{dj})$ the total labor share of each province and sector (direct labor plus in-house tasks), aggregate labor-market clearing gives

$$\ell^\top r(b) = \sum_d \hat{w}_d w_d L_d \quad \implies \quad b^* = \frac{\sum_d \hat{w}_d w_d L_d - \ell^\top r_0}{\ell^\top r_1}.$$

The transfer enters household budgets only. It changes no rate, no wedge, and no margin, so it cannot affect producer prices, the make-or-buy decision, or the ranking of the two systems: T1 and T2 hold for any lump-sum transfer. Welfare is measured against baseline income, so the transfer counts as counterfactual income. In the reference solution the transfer is 4.4 percent of labor income; across the experiments of Table 8 it lies between 2.9 and 3.3 percent and is nearly constant in the tax rate within each system.

Transfers fixed in nominal terms. The provincial transfers D_d remain fixed in nominal terms, so their real value falls when a higher rate raises the consumer price level, while the balancing transfer is paid in proportion to labor income. This redistributes income across provinces and has no efficiency content. It therefore leaves aggregate real income, $\sum_d (I'_d / \hat{P}_d) / \sum_d I_d^0$, unchanged, which is the welfare measure of the paper and the measure on which the two statements are evaluated.

E.4 Results

Both statements hold. The value-added tax curve is flat: over a range of revenue from 57 to 144 percent of the 2012 level, welfare moves by 0.002 percentage points. The level of the curve, 2.7 percent above the reference, is the gain from replacing the 2012 system, with its dispersed rates and incomplete crediting, by any uniform tax with full crediting. The turnover tax curve falls at an increasing rate, from +2.3 to -0.8 percent as the rate rises

Table 8: Value-added tax and turnover tax in the closed economy

System	Rate	Capacity	Welfare (%)
Uniform value-added tax	0.050	0.572	+2.734
	0.100	0.854	+2.735
	0.150	1.112	+2.735
	0.220	1.438	+2.736
Uniform turnover tax	0.015	0.521	+2.338
	0.030	0.763	+1.611
	0.045	0.966	+0.667
	0.065	1.189	−0.822

Slope of the value-added tax curve: 0.0000 points of welfare per point of capacity.
Value-added tax minus turnover tax at matched capacity: minimum +0.55, mean +1.81, maximum +3.56.

Welfare is aggregate real income, in percent deviation from the reference solution of the closed economy at 2012 rates. Capacity is tax revenue over income relative to 2012. Trade elasticity $\theta = 4.5$ and the measured in-house vector of Section 7. All solutions converge in 42 to 54 iterations at a tolerance of 10^{-7} .

from 1.5 to 6.5 percent, the quadratic cost of cascading. The value-added tax is ahead at every matched capacity, by 0.55 points at the lowest revenue and 3.56 at the highest, and the gap is smallest where revenue is lowest, as the second-order nature of the cascading cost implies.

E.5 The trade side

Imports. Imports are Armington varieties whose shares respond to relative costs with elasticity θ , in the baseline and in every counterfactual. The value-added tax follows the destination principle: tax is collected on imports at the border and credited to registered buyers, so that an imported input carries no net tax, and imports for final consumption are taxed. A turnover tax grants no credit and therefore taxes imported inputs.

Exports. Foreign demand for each province and sector's exports has constant elasticity (Section 6.6):

$$R_{dj}^{x'} = R_{dj}^x (\hat{p}_{dj}^x)^{-\theta}, \quad \hat{p}_{dj}^x = \hat{c}_{dj} \frac{1 + \tau_j^{x'}}{1 + \tau_j^x},$$

so that export revenue falls with elasticity θ when the cost or the tax on exports rises. The value-added tax zero-rates exports. A turnover tax, unless exports are explicitly relieved, taxes them, and a tax on exports facing a finite demand elasticity collects rents from foreign buyers.

Lemma 1 (Export taxes below the optimum raise home welfare). *Let foreign expenditure on a home export good be fixed and the home share small, so that export revenue at FOB price p is $R^x(p) = \bar{R} p^{-\theta}$ and the quantity shipped is $q(p) = \bar{R} p^{-(\theta+1)}$, $\theta > 0$. Let the good be supplied at*

constant marginal cost c , and let an export tax $\tau \geq 0$ be levied on the FOB price, so that $p = c(1 + \tau)$, with the proceeds rebated lump-sum at home. Home welfare relative to free trade is

$$W(\tau) = (p - c) q(p) = \bar{R} c^{-\theta} \tau (1 + \tau)^{-(\theta+1)},$$

which is single-peaked with its maximum at $\tau^* = 1/\theta$, and $W'(\tau) > 0$ for every $\tau \in (0, 1/\theta)$. To first order $W(\tau) = \bar{R} c^{-\theta} \tau + O(\tau^2)$: the gain is linear in the tax. By Lerner symmetry (Costinot & Werning, 2019; Lerner, 1936) the same statement holds for a tax on imported inputs, and for a turnover tax that passes through to export costs with elasticity $\eta \in (0, 1)$ it holds with τ replaced by $\eta\tau$.

Proof. Differentiating, $W'(\tau) = \bar{R} c^{-\theta} (1 + \tau)^{-(\theta+2)} [(1 + \tau) - (\theta + 1)\tau] = \bar{R} c^{-\theta} (1 + \tau)^{-(\theta+2)} (1 - \theta\tau)$, which is positive for $\tau < 1/\theta$, zero at $\tau = 1/\theta$, and negative beyond; $W(0) = 0$ and $W'(0) = \bar{R} c^{-\theta} > 0$. For a tax on imported inputs, Lerner symmetry maps the import tax into an export tax of the same rate; for a turnover tax, the FOB price is $c(1 + \eta\tau)$ and the argument applies to $\eta\tau$. $\square$

With $\theta = 4.5$ the optimum is about 22 percent, and every turnover rate considered here (0.8 to 8 percent) lies on the rising branch. This is the optimal-tariff result of Johnson (1953), in the CES form of Gros (1987) and Costinot and Rodríguez-Clare (2014). A destination-based value-added tax, with zero-rated exports and border-adjusted imports, is trade-neutral in the sense of Feldstein and Krugman (1990) and G. M. Grossman (1980) and sets this margin to zero by design.

The comparison of Section 8.4 therefore gives the turnover tax the same export treatment as the value-added tax, so that the two systems differ only in crediting, and reports the turnover tax with taxed exports separately (Figure 10a). The export levy is worth 0.11 percentage points of welfare at a turnover rate of 0.8 percent and 0.40 at 4.5 percent, and 0.5 points on average over the range of capacities compared.

Sensitivity to export demand. The gain from the reform as legislated is +1.30 percent at an export-demand elasticity of 4.5, +1.22 percent at 3, and +1.36 percent at 6. If export revenue is instead held fixed in foreign currency, so that foreign buyers do not respond to the fall in Chinese costs, the gain is +0.97 percent. Beijing and Tibet are the two provinces that lose in each case.

E.6 Reading the open-economy comparison

In the open economy the uniform value-added tax is ahead of the turnover tax with the same export treatment at every capacity computed (Figure 10b): by 0.24 percentage points at a capacity of 0.45, 0.53 at 0.60, 1.5 at China's post-reform capacity of 0.88, 2.0 at capacity 1, and 4.2 at 1.31. Lump-sum equivalence (T1) carries over to the open economy, because a destination-based value-added tax with zero-rated exports and border-adjusted imports is a uniform tax on consumption: the welfare curve of the uniform value-added tax moves from +3.886 to +3.870 percent as capacity rises from 0.45 to 1.43, a change of

0.016 percentage points. What does not carry over is the ranking as a theorem, since the turnover tax now also taxes trade. The gaps are smaller than in the closed economy at the same capacity, and the shape is the same. A second-order decomposition shows why.

The domestic term. Consider a buyer that allocates spending across substitution margins m , the origin–sector pairs of its market bundle and the in-house alternative, with expenditure weights s_m and effective-cost wedges w_m . To second order around undistorted prices, with revenue rebated lump-sum, the welfare cost of the wedges is

$$\Delta W^{\text{dom}} \approx -\frac{1+\theta}{2} \sum_m s_m (\ln w_m - \overline{\ln w})^2, \quad \overline{\ln w} = \sum_m s_m \ln w_m.$$

Only the dispersion of wedges across margins matters; the common component is a shift in the price level that the rebate returns, which is T1 in local form. Under a uniform value-added tax with full crediting, every market margin has $w_m = 1$ and the in-house margin is untaxed, so the dispersion is zero at every rate. Under a uniform turnover tax the wedge is zero on the in-house margin, about τ on a direct purchase, and a multiple of τ along longer chains. Every wedge is proportional to τ , so

$$\Delta W_{\text{turnover}}^{\text{dom}} \approx -A \tau^2, \quad A = \frac{1+\theta}{2} \left[\underbrace{s_H(1-s_H)}_{\text{make-or-buy margin}} + \underbrace{\sum_m s_m (\delta_m - \bar{\delta})^2}_{\text{chain-depth dispersion}} \right] > 0,$$

where δ_m is the number of taxed stages embedded in margin m and s_H is the in-house share. Both components of A are properties of the production network, not of the rate. This is the term that Table 8 isolates.

The trade term. In the open economy a turnover tax also taxes imported inputs, which by Lemma 1 acts as a tax on exports at rate $\eta\tau$ even when exports themselves are relieved. To first order its contribution to home welfare is

$$\Delta W_{\text{turnover}}^{\text{trade}} \approx +B \tau, \quad B = \frac{E}{Y} \cdot \eta,$$

where E/Y is the export share of income. The term is linear in the rate, it is bounded by the size of trade, and it is revenue collected from foreigners, not domestic efficiency. A value-added tax sets $B = 0$.

The gap. At matched revenue the gap between the two systems is therefore, to second order,

$$G(\tau) \equiv W^{\text{VAT}} - W^{\text{turnover}} \approx A \tau^2 - B \tau.$$

The trade term lowers the gap at every rate, which is why the open-economy gaps are below their closed-economy counterparts. The domestic term grows with the square of the rate while the trade term is linear and bounded, so the gap widens more than proportionally

with capacity. In principle the linear term could dominate at rates close to zero; in the calibrated economy the gap is positive at the lowest capacity computed, and it remains positive when the turnover tax keeps its export levy (0.22 points at a capacity of 0.60, 0.74 at 0.85, and 1.3 at 1.00). At low revenue both systems are close to costless and the choice between them matters little; at China's revenue it is worth 1.4 to 2 percent of real income.

A value-added tax with an exempt sector. The value-added tax of Figure 10a that leaves the nonmarket sector untaxed is a restricted instrument, and theory does not rank it against the turnover tax even in the closed economy: at matched revenue it carries one wedge, between the exempt sector and the rest, against the many small wedges of the turnover tax. In the calibrated economy it is ahead at every capacity computed, by 0.23 points at a capacity of 0.44 and 1.97 at 1.00, slightly less than the uniform value-added tax.

E.7 Level and structure of the reform

The same two terms account for the difference between the open and the closed economy in the decomposition of the reform into the level of taxation and its structure (Section 8.3).

Table 9: Level and structure of the reform at matched real revenue

	Total (%)	Level (%)	Structure (%)	Scaling factor
Open economy	+1.30	+0.98	+0.32	0.794
Closed economy	+1.62	+0.88	+0.74	0.786

Total is the reform as legislated. Level is the 2012 system with the business-tax rates multiplied by the scaling factor, chosen so that real revenue equals its value under the reform. Structure is the difference. Welfare is aggregate real income. The closed-economy row is computed on the economy of Section E.2, relative to its reference solution at 2012 rates.

The level component is similar in the two economies. The structure component is 0.74 points in the closed economy and 0.32 in the open economy. The difference is the trade term: the 2012 system embedded a tax on exports in exporters' costs, and the reform, which is trade-neutral, removes it. Export revenue rises by 5.7 percent, and the rents that the embedded tax collected from foreign buyers (Lemma 1) are given up. This is a loss of rents, not of efficiency, and it does not arise in the closed economy.

E.8 Scripts

Table 10: Scripts

Script	Output
DIAG_AUTARKY_THEOREM.m	Table 8
MAIN_5_DECOMPOSE_STRUCTURE.m	Table 9, open economy
DIAG_AUTARKY_LEVELSTRUCT.m	Table 9, closed economy
MAIN_11_CLEAN_AND_HARMONIZE.m, DIAG_F11_AGG_GAP.m	Figures 10a and 10b; open-economy gaps
DIAG_REVIEW_OCT.m	export levy; sensitivity to export demand
FUN_EXPEND_PRIME.m (balanceLabor)	balancing transfer of Section E.3